

Predistribution at Work

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Abstract

An occupation-blind income tax distorts workers toward higher-amenity jobs by taxing compensating differentials. Using a multidimensional Mirrleesian framework with job amenities, I derive two results. First, predistribution that improves conditions in disamenable jobs is desirable due to a fiscal externality: it induces workers to reallocate toward low-amenity occupations, earning compensating differentials that are taxed. The inability to condition taxes on occupation breaks the Atkinson-Stiglitz theorem and justifies labor market interventions on fiscal grounds alone. Second, accounting for this distortion lowers optimal taxes. A calibration to U.S. data finds that compensating differentials lower optimal marginal tax rates by one to four percentage points. Even coarsely applied to the worst jobs, predistribution acts as implicit occupational targeting and internalizes the fiscal externality far more effectively than tax reductions alone.

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1 Introduction

Not all jobs are created equal. Jobs differ in the physical toll they exact, the autonomy they afford, the risks they impose, and the satisfaction they deliver—differences that are reflected in wages as compensating differentials. Despite this, modern income tax systems are largely occupation-blind, treating a dollar earned in a coal mine the same as one earned behind a desk. By taxing compensating differentials, the income tax implicitly favours high-amenity work and penalizes unpleasant jobs, distorting occupational choice. Workers, too, are not created equal: their talents lie in different domains and they sort into occupations according to their comparative advantage.

This paper studies optimal policy in the presence of compensating differentials arising from job amenities and comparative advantage. I do so using a Mirrlees-Roy framework in which workers with multidimensional skills sort into occupations along both margins (Rothschild and Scheuer, 2013; Lockwood et al., 2017).

I establish two results. First, the government can increase welfare through predistribution: improving amenities at jobs with undesirable non-wage characteristics (henceforth, ‘bad jobs’) beyond what the private net benefits of those improvements justify. Doing so leads workers to pursue their comparative advantage in bad jobs, allowing the government to capture a fiscal externality from the compensating differentials they earn. Making bad jobs better compresses amenity differentials across occupations, counteracting the wage compression from income taxation that leads workers to favour amenities over pay. This contributes to more efficient occupational sorting. These gains persist even when predistribution is applied uniformly across bad jobs rather than customized to each, reducing the informational demands on the planner.

This first finding represents a departure from the celebrated result of Atkinson and Stiglitz (1976) that, under certain assumptions, redistribution can be accomplished solely through a nonlinear income tax, rendering commodity taxation and labor market interventions redundant. Prior work has identified conditions under which this result breaks down: non-separable or heterogeneous preferences (Atkinson and Stiglitz, 1976; Nichols and Zeckhauser, 1982; Saez, 2002a), general equilibrium wage effects (Naito, 1999; Costinot and Werning, 2023), and dual-purpose expenditures (Goldin et al., 2024). A separate literature rationalizes labor market regulation on grounds of externalities (Summers, 1989), monopsony power (Naidu and Posner, 2022), asymmetric information (Viscusi, 1979), and behavioral biases (Akerlof and Dickens, 1982). I connect these literatures by showing that multidimensional skills and an occupation-blind income tax are sufficient to overturn the Atkinson-Stiglitz theorem and justify labor market regulation on fiscal grounds alone. Intuitively, predistribution through job quality serves as a second-best substitute for occupation-specific taxation.

The second finding is that taxes should be lower than a pure intensive-margin analysis recommends, since they distort not only the intensive margin of labor supply but also the extensive margin of occupational choice. Crucially, this force for lower tax rates is not uniform: it is strongest at income levels where a tax cut would induce the most switching into jobs with the largest fiscal externalities.

Horizontal equity considerations reinforce the efficiency case for predistribution and lower taxation. Since incomes overstate welfare when they contain compensating differentials, a planner with redistributive preferences over utilities — through a concave social welfare function or Pareto weights tilted toward workers with comparative advantage in bad jobs — has an independent motive for both policies. Concretely, predistribution should be more aggressive the larger the utility difference between workers in bad and good jobs, and marginal tax rates should be lower at income levels where disamenable work is prevalent. The well-documented empirical regularity that bad jobs tend to be low-paid (Sockin, 2022; Maestas et al., 2023; Caldwell et al., 2025; Humlum et al., 2026) makes this case for predistribution especially compelling.

The fiscal externality from compensating differentials cannot be captured by a simple sufficient statistic,

because its two key components—the tax wedge and the density of occupation switchers—are themselves endogenous to the tax schedule and the extent of predistribution.¹ Consequently, solving for optimal policy requires two policy-invariant inputs beyond the standard Mirrleesian framework: the multidimensional skill distribution and occupational amenities. Both compensating differentials and the generalized Roy model are difficult to estimate for well-documented reasons. Thus, I follow [Lockwood et al. \(2017\)](#) in imposing structure—a Gaussian copula governing skill correlation across occupations and a logit model of occupational choice—to back out these primitives. As a key requirement of this framework is that it accurately predict the responsiveness of occupation choice to taxation, I calibrate preferences to match the occupation-switching elasticity from [Powell and Shan \(2012\)](#).

Internalizing the fiscal externality from compensating differentials in the tax schedule lowers optimal marginal rates by 1–4 percentage points, with larger reductions when workers have comparative advantage rather than perfect rank correlation of skills, or are more sensitive to income relative to amenities. Even coarsely applied, predistribution does far more to shift workers into higher-paying jobs and internalize the fiscal externality than tax reductions alone.

Related literature The idea that wages compensate for undesirable job characteristics traces to Adam Smith and was formalized by [Rosen \(1986\)](#). Within this tradition, prior work has noted that income taxation distorts workers toward lower-paying, higher-amenity occupations ([Powell and Shan, 2012](#); [Lockwood et al., 2017](#)),² yet the policy implications of this distortion remain unresolved. This paper addresses this gap by deriving and quantifying optimal tax and predistribution policy under compensating differentials.

My theoretical framework builds on the Mirrleesian literature on optimal taxation with occupational choice and multidimensional skills. [Lockwood et al. \(2017\)](#) show that when high-paying professions impose negative externalities and low-paying ones generate positive externalities, progressive taxation acts as an implicit Pigouvian tax, reallocating workers toward socially valuable occupations. My analysis extends theirs by equipping the government with an additional policy lever of predistribution through job quality, and replacing their formula with a [Diamond \(1998\)](#)-style ABC formula that makes explicit the determinants of the fiscal externality. I set aside occupational externalities, which are theoretically well-studied but empirically elusive, and show that compensating differentials have first-order policy implications even in their absence.

[Rothschild and Scheuer \(2013\)](#) explore how the additional margin of occupational choice affects optimal taxation when general equilibrium wage effects à la [Stiglitz \(1982\)](#) enable indirect redistribution. This paper, by contrast, emphasizes a different — and, as recent empirical work suggests, quantitatively important — source of endogeneity in the wage distribution: workers sort into jobs on the basis of both comparative advantage ([Kirkeboen et al., 2016](#); [Taber and Vejlín, 2020](#)) and amenities ([Sockin, 2022](#); [Humlum et al., 2026](#)).

A growing body of work studies predistribution³ — policies that shape the pre-tax distribution of income such as labor market regulations, wage-setting institutions, and public employment guarantees ([Rodrik and Stantcheva, 2021](#); [Kuziemko et al., 2023](#); [Bozio et al., 2024](#)) — but provides limited guidance on whether

¹This echoes [Jacquet and Lehmann \(2021\)](#), who emphasize that sufficient statistics are endogenous to policy.

²A similar argument applies to location choice: [Albouy \(2009\)](#) shows that federal income taxes distort workers away from high-wage, low-amenity cities, generating substantial locational inefficiencies. My framework extends naturally to this setting, with cities replacing occupations. While [Gaubert et al. \(2021\)](#) do not emphasize this distortion, they show that comparative advantage can provide a rationale for place-based redistribution. My optimal tax formulas nest this component of their paper, since they restrict skill to a single dimension, whereas I allow for multidimensional skills.

³This definition of predistribution differs from usage in some optimal taxation papers, where predistribution refers to indirect redistribution through general equilibrium incidence effects.

such interventions are desirable when a nonlinear income tax is available. I study the optimality of public investments in job quality, adding to a literature that explores redistribution beyond the income tax schedule via policies such as commodity taxes,⁴ social insurance,⁵ and in-kind transfers.⁶ I also contribute to the literature on the optimal tax implications of discrete choice margins such as whether to work (Saez, 2002b), which occupation to enter (Rothschild and Scheuer, 2013; Lockwood et al., 2017; Gomes et al., 2018; Ales and Sleet, 2022), where to live (Gaubert et al., 2021), and “jumping effects” more generally (Bergstrom and Dodds, 2021).

The tax code already implicitly subsidizes fringe benefits whenever they are deductible to employers and excluded from the individual income tax base, most prominently, employer-provided health insurance (Gruber, 2011).⁷ A substantial literature finds that the fringe-benefit share of compensation responds to tax incentives (Woodbury, 1983; Long and Scott Jr, 1984; Gruber and Poterba, 1994). One might think this undermines the case for predistribution — if amenities are already subsidized, further intervention seems unnecessary — but I show it does not. Subsidies on fringe benefits consumed uniformly across occupations, like health insurance, drop out of the fiscal externality from occupational switching, leaving the case for predistribution intact.

This paper also connects to a literature on compensation inequality arguing that assessments of inequality and welfare must account for non-wage amenities (Pierce, 2001; Sockin, 2022; Lamadon et al., 2022; Dube et al., 2022; Ouimet and Tate, 2023; Maestas et al., 2023). This literature quantifies how amenities contribute to compensation and inequality, but stops short of deriving policy implications. I show that non-wage amenities generate horizontal inequality in welfare among workers at the same income level and that leaving them untaxed distorts occupational sorting, motivating predistribution as a corrective.

The paper proceeds as follows. Section 2 presents the baseline model and characterizes optimal policy under occupation-blind (2.2) and occupation-specific income taxation (2.3). Section 3 extends the model to heterogeneous preferences (3.1), an arbitrary number of occupations (3.2), endogenous amenities, possibly subject to uniform preexisting subsidies (3.3), and general equilibrium wage effects (3.4). Section 4 describes the empirical strategy and data, Section 5 presents the results, and Section 6 concludes.

2 Model

2.1 Setup

I begin with a simple model of an economy with a unit mass of individuals, each of whom chooses to work in one of two occupations, A or B . Each individual has a skill type $(\theta_A, \theta_B) \in \mathcal{A} \times \mathcal{B}$ with $\mathcal{A} = \mathcal{B} = [\underline{w}, \bar{w}]$, where θ_A and θ_B denote productivity in each occupation. Worker types are distributed with cdf $F(\theta_A, \theta_B)$ and density $f(\theta_A, \theta_B)$. Worker preferences are represented by

$$U(c, l, d) = \begin{cases} c - \frac{l^{1+1/\varepsilon}}{1+1/\varepsilon} & \text{if } o(\theta_A, \theta_B, d) = A, \\ c - \frac{l^{1+1/\varepsilon}}{1+1/\varepsilon} - d & \text{if } o(\theta_A, \theta_B, d) = B \end{cases} \quad (1)$$

⁴e.g., Saez (2002a); Costinot and Werning (2023) and Ferey et al. (2024).

⁵e.g., Rochet (1991); Cremer and Pestieau (1996); Boadway et al. (2006) and Ferey (2022).

⁶e.g., Nichols and Zeckhauser (1982) and Blackorby and Donaldson (1988).

⁷A separate literature explores the role of closely-held firms as tax shelters (Alstadsæter et al., 2014; Leite, 2024). I focus on the tax exclusion of amenities and fringe benefits for employees, rather than tax avoidance strategies available only to owners of closely-held firms.

where $o(\theta_A, \theta_B, d)$ denotes the occupational choice of a worker of type (θ_A, θ_B) given disutility $d > 0$ from working in B . Wages are

$$w(\theta_A, \theta_B) = \begin{cases} \theta_A & \text{if } o(\theta_A, \theta_B, d) = A \\ \theta_B & \text{if } o(\theta_A, \theta_B, d) = B. \end{cases} \quad (2)$$

Income is $y(\theta_A, \theta_B) \equiv w(\theta_A, \theta_B)l(\theta_A, \theta_B)$.

This model differs from [Rosen \(1986\)](#) in two respects. First, the Rosen framework assumes homogeneous skills, leaving no scope for comparative advantage. Compensating differentials from switching jobs are identical across workers, and determined in equilibrium by firms' costs of amenity provision and workers' preferences. Second, it abstracts from the intensive margin of labor supply, which is central to the Mirrlees problem.

My model instead features workers with heterogeneous, multidimensional skills, so wage differentials are worker-specific and exogenous — each worker's gain from switching occupations depends on their skill vector rather than equilibrium prices. Thus, all wage-relevant heterogeneity is captured by the multidimensional skill distribution, and workers who choose unpleasant jobs do so because they have a comparative advantage there. This enables two features central to the analysis: an intensive labor supply margin, and compensating differentials that vary with skills and policy. I return to endogenous wages in [Section 3.4](#).

2.2 Occupation-blind taxation

Suppose that occupation-specific income taxation is infeasible: both occupations face a common nonlinear tax schedule $T(\cdot)$. Occupation B imposes a baseline disamenity of $\bar{d}$. The government can reduce the disamenity to d at a cost $\gamma(d)$ per B -worker, where $\gamma(\bar{d}) = 0, \gamma'(d) < 0$. This operationalizes predistribution as public expenditures on job quality; [Section 3.3](#) considers amenity subsidies. Each worker chooses occupation $o \in \{A, B\}$ to maximize their utility,

$$o^*(\theta_A, \theta_B, d) = \arg \max_{o \in \{A, B\}} V^o(\theta_o) \quad (3)$$

where

$$V^o(\theta_o) = \max_y y - T(y) - \frac{(y/\theta_o)^{1+\frac{1}{\varepsilon}}}{1 + \frac{1}{\varepsilon}} - d \cdot \mathbb{1}[o = B]. \quad (4)$$

Suppose that the planner can observe workers' income y and consumption $c = y - T(y)$, but not their type (θ_A, θ_B) , effort e , or occupation o . It can observe and set d , implementing a uniform disamenity level across all B -workers. Under these constraints, any incentive-compatible allocation $\{c(\theta_A, \theta_B), y(\theta_A, \theta_B), o(\theta_A, \theta_B)\}$ must satisfy two properties. First, occupation choice is governed by a threshold rule $\check{\theta}_B(\cdot, d)$ with inverse $\check{\theta}_A(\cdot, d)$: for any θ_A ,

$$o^*(\theta_A, \theta_B, d) = \begin{cases} A & \text{if } \theta_B < \check{\theta}_B(\theta_A, d), \\ B & \text{otherwise} \end{cases} \quad (5)$$

where $\check{\theta}_B(\theta_A, d)$ is defined as the θ_B that leaves a worker with skill θ_A in A indifferent between the two occupations, given disamenity d ,

$$c(\theta_A, \check{\theta}_B(\theta_A, d)) - \frac{(y(\theta_A, \check{\theta}_B(\theta_A, d))/\theta_A)^{1+\frac{1}{\varepsilon}}}{1 + \frac{1}{\varepsilon}} = c(\theta_A, \check{\theta}_B(\theta_A, d)) - \frac{(y(\theta_A, \check{\theta}_B(\theta_A, d))/\check{\theta}_B(\theta_A, d))^{1+\frac{1}{\varepsilon}}}{1 + \frac{1}{\varepsilon}} - d. \quad (6)$$

Through (6), $\check{\theta}_B(\theta_A, d)$ depends on both the tax schedule and predistribution spending. Endogenous occupational choice generates a compensating differential: a marginal worker must earn a higher wage in B than in A to offset the disamenity,

$$\check{\theta}_B(\theta_A, d) > \theta_A. \quad (7)$$

Since the tax schedule compresses the return to selecting a higher-wage but lower-amenity occupation, occupation choice is endogenous to the tax schedule. In particular, higher marginal tax rates raise the compensating differential—that is, the threshold wage in occupation B , $\check{\theta}_B(\theta_A, d)$.

Second, workers earning the same wage must be assigned the same consumption and income. For all $(\theta_A, \theta_B), (\theta'_A, \theta'_B)$ such that $w(\theta_A, \theta_B) = w(\theta'_A, \theta'_B)$,

$$c(\theta_A, \theta_B) = c(\theta'_A, \theta'_B), \quad y(\theta_A, \theta_B) = y(\theta'_A, \theta'_B). \quad (8)$$

Among individuals with the same wage, utility is d units lower for those in B than in A . Since taxes cannot condition on occupation, workers earning the same income pay the same taxes and consume the same amounts, regardless of occupation. Behind the veil of ignorance, all individuals prefer to be more skilled in A . That is, for $\theta_1 > \theta_2$, a worker is better off if they are endowed with skills (θ_1, θ_2) over (θ_2, θ_1) . A concave social welfare function, or Pareto weights tilted toward workers with comparative advantage in bad jobs, motivates predistribution because occupation-blind income taxation cannot equalize welfare across workers who earn the same wage in jobs with different amenities.

Planner's problem. I assume the single-crossing condition holds: the marginal rate of substitution between income and consumption, $\frac{-U_l(c, y/w)}{wU_c(c, y/w)}$, is decreasing in w . Since workers earning the same wage must be assigned the same bundle, the planner's problem reduces to choosing $\{l(w), V^A(w)\}$. A B -worker earning wage w receives utility $V^A(w) - d$. The threshold rule for occupation choice (6) becomes

$$V^A(\theta_A) = V^A(\check{\theta}_B(\theta_A, d)) - d, \quad (9)$$

so the distribution of workers across occupations and wages is endogenous to policy. The skill distribution $F(\theta_A, \theta_B)$ and (9) induce occupation-specific wage densities

$$h_A(w) = \int_{\underline{w}}^{\check{\theta}_B(w, d)} f(w, \theta_B) d\theta_B, \quad h_B(w) = \int_{\underline{w}}^{\check{\theta}_A(w, d)} f(\theta_A, w) d\theta_A \quad (10)$$

with total wage density $h(w) = h_A(w) + h_B(w)$.

The social planner has redistributive preferences summarized by the concave social welfare function $W(\cdot)$. They maximize

$$\max_{l(w), V^A(w), d} \int_{\underline{w}}^{\bar{w}} W(V^A(w)) h_A(w) + W(V^A(w) - d) h_B(w) dw \quad (11)$$

subject to local incentive compatibility constraints

$$V^{A'}(w) - \frac{l(w)^{1+1/\varepsilon}}{w} = 0 \quad \forall w \in [\underline{w}, \bar{w}] \quad (12)$$

and the resource constraint

$$\int_{\underline{w}}^{\bar{w}} \left(wl(w) - c(V^A(w), l(w)) \right) h(w) - \gamma(d) h_B(w) dw \geq 0. \quad (13)$$

Occupation choice is embedded implicitly as h_A, h_B are defined by (9) and (10).

Proposition 1 *Optimal marginal tax rates (MTRs) are characterized by*

$$\frac{T'(y(w))}{1 - T'(y(w))} = \frac{1 + \frac{1}{\varepsilon}}{wh(w)} \left(\int_{\underline{w}}^{\bar{w}} h(z) - \frac{1}{\mu} [W'(V^A(z))h_A(z) + W'(V^A(z) - d)h_B(z)] dz - \Xi^t(w, d) \right) \quad (14)$$

$$\Xi^t(w, d) = \int_{\underline{w}}^{\check{\theta}_B(w, d)} \Delta T(\theta_B, d) f(\check{\theta}_A(\theta_B, d), \theta_B) \frac{\check{\theta}_A(\theta_B, d)}{l(\check{\theta}_A(\theta_B, d))^{1 + \frac{1}{\varepsilon}}} d\theta_B \quad (15)$$

where $\Delta T(\theta_B, d) = T(y(\theta_B)) - \gamma(d) - T(y(\check{\theta}_A(\theta_B, d)))$ is the tax revenue generated when a marginal worker earning wage θ_B in B switches from A to B .

Optimal predistribution through job quality satisfies

$$-\gamma'(d) = \frac{1}{\mu} \mathbb{E}_B [W'(V^B(\theta_B))] + \Xi^p(d) \quad (16)$$

$$\Xi^p(d) = \mathbb{E}_B \left[\Delta T(\theta_B, d) \frac{f(\check{\theta}_A(\theta_B, d), \theta_B)}{h_B(\theta_B)} \frac{\check{\theta}_A(\theta_B, d)}{l(\check{\theta}_A(\theta_B, d))^{1 + \frac{1}{\varepsilon}}} \right] \quad (17)$$

where $\mathbb{E}_B[x(\theta_B, d)] = \frac{\int_{\underline{w}}^{\bar{w}} x(\theta_B, d) h_B(\theta_B) d\theta_B}{\int_{\underline{w}}^{\bar{w}} h_B(\theta_B) d\theta_B}$ is the weighted average of $x(\theta_B, d)$ among B -workers.

Proof. Appendix A.1.

Corollary 1 *At the no-predistribution baseline $\gamma(\bar{d}) = 0$,*

- (i) *the fiscal externality from compensating differentials is positive, $\Delta T(w, \bar{d}) > 0 \forall w \in [\underline{w}, \bar{w}]$,*
- (ii) *compensating differentials are a force for lower MTRs, $\Xi^t(w, \bar{d}) \geq 0 \forall w \in [\underline{w}, \bar{w}]$, and*
- (iii) *the social return to predistribution weakly exceeds its private return, $\Xi^p(\bar{d}) \geq 0$.*

The inequalities in (ii) and (iii) are strict whenever marginal switchers have positive density, $f(\check{\theta}_A(\theta_B, \bar{d}), \theta_B) > 0$, on some subinterval of $(w, \check{\theta}_B(w, \bar{d}))$ and $[\underline{w}, \bar{w}]$, respectively.

I discuss these results in two parts: optimal income taxes, then predistribution through job quality.

Optimal income taxes Consider first the case with no predistribution, $\gamma(\bar{d}) = 0$. Optimal MTRs are lower than in a naïve benchmark that sets $\Xi^t(w, d) = 0 \forall w, d$ — which obtains either when the planner ignores the fiscal externality, or when occupations are identical. This reflects both allocative efficiency and horizontal equity considerations.

First, income taxation blunts workers' incentives to pursue their comparative advantage by compressing wage differentials while leaving amenities unchanged. Thus, taxes distort not only the intensive margin of effort but also the extensive margin of occupational choice: some individuals who would have taken higher-paying but lower-amenity jobs in B under lump-sum taxation instead choose lower-paying, higher-amenity jobs in A . Intuitively, a one-dollar tax cut for all workers with wages exceeding w generates $\Xi^t(w, \bar{d}) \geq 0$ cents in fiscal externalities as workers shift into B and pay taxes on their compensating differentials.

Figure 1 illustrates why $\Xi^t(w, \bar{d})$ can be signed, while Appendix A.1 contains the formal argument. A reduction in the MTR at $y(w)$ by ϵ only leads to occupation switching by workers who are doubly marginal:

on the margin of occupation choice and for whom $y(w)$ lies between their incomes in the two occupations. The dashed line depicts these workers, who lie on the locus of indifferent types $(\theta_A, \check{\theta}_B(\theta_A, d))$ and have $\theta_A < w < \check{\theta}_B(\theta_A, d)$, so $y(\theta_A) < y(w) < y(\check{\theta}_B(\theta_A, d))$. All other workers are inframarginal. Either they have strict comparative advantage ($\theta_B \neq \check{\theta}_B(\theta_A, d)$), or their incomes in both occupations fall on the same side of $y(w)$, so the tax variation changes their tax burden equally across both occupations and leaves their occupation choice unchanged. The fiscal externality $\Xi^t(w, d)$ is driven entirely by doubly marginal switchers, who move into B and earn more there.

While the occupation-switching fiscal externality is always weakly positive and pushes toward lower optimal MTRs, its magnitude depends critically on the density of marginal workers, $f(\check{\theta}_A(\theta_B, d), \theta_B)$ and the size of the tax wedge $\Delta T(\theta_B, d)$. Thus, the downward pressure on MTRs is greatest where the density of marginal workers is greatest and the tax wedge from compensating differentials is largest.

Second, the presence of non-wage amenities weakens the link between income and welfare. At every income level, B -workers' welfare is lower than A -workers' despite them earning identical incomes. A redistributive planner accounts for this disparity, recognizing equal income does not imply equal welfare when compensating differentials are present. This is a second force for lower progressivity, formally captured by the term $W'(V^A(z))h_A(z) + W'(V^A(z) - \bar{d})h_B(z)$ in equation (14): a concave SWF assigns B -workers higher welfare weights than their A counterparts at the same wage, reducing optimal MTRs at income levels where B -workers are prevalent. This logic relies on welfare weights being endogenous and decreasing in utility, so redistribution responds not only to marginal utilities of consumption but also utility levels. A similar result obtains under exogenous Pareto weights, provided the planner places greater weight on workers with a comparative advantage in the bad job.

Optimal predistribution through job quality Suppose now the planner can choose d . Optimal predistribution equates the marginal cost of reducing d to its marginal benefit, which has two parts. The first, $\frac{1}{\mu} \mathbb{E}_B [W'(V^B(\theta_B))]$, is the average social welfare weight on B -workers relative to the population average μ , capturing the planner's valuation of the direct gain to B -workers. The second, $\Xi^p(d)$, is the fiscal externality from occupation switching.

Predistribution is desirable for two reasons. First, it addresses horizontal inequity. Workers with comparative advantage in B bear a disamenity that those in A do not, despite earning the same wage. From an ex ante perspective, comparative advantage is a matter of luck, so the planner may view predistribution as insurance against being suited to the worse occupation — analogous to how the tax-transfer system insures against a low skill draw. This motive is captured by $\frac{1}{\mu} \mathbb{E}_B [W'(V^B(\theta_B))]$, which exceeds 1 under a concave SWF whenever B -workers are worse off on average than A -workers. This is not automatic but depends on how the realized wage distributions h^A and h^B compare. A sufficient condition is that h^A first-order stochastically dominates h^B , so that B -workers earn no more than A -workers yet bear the disamenity, and thus have lower utilities on average. This is the empirically relevant case, as a large literature finds that poor working conditions and low pay tend to coincide (e.g., [Maestas et al., 2023](#)).

Second, it improves allocative efficiency: by making B more attractive, it compresses amenity differentials across occupations and counteracts the distortionary effect of income taxation on the wage-amenity tradeoff. Formally, $\Xi^p(d) \geq 0$ as long as predistribution is not so extensive that it reverses the sign of the fiscal externality, so the fiscal motive reinforces the horizontal equity motive.

Allowing for predistribution has an ambiguous effect on optimal MTRs through three channels. The direct cost of predistribution reduces the net revenue gain from switchers, lowering $\Xi(w, d)$ and pushing

toward greater progressivity. It also alters the threshold wages $\check{\theta}_B(\theta_A, d)$ and the mass of indifferent workers, with indeterminate effects on $\Xi(w, d)$ that depend on the skill distribution. Finally, since workers who switch to B earn more than they did in A , predistribution shifts the wage distribution rightward, tending to raise optimal MTRs at the bottom and lower them at the top.

2.3 Occupation-specific taxation

Proposition 2 *Suppose the government can observe occupation and thus set occupation-specific taxes $T^A(\cdot)$ and $T^B(\cdot)$. Optimal MTRs satisfy*

$$\frac{T^{A'}(y^A(w))}{1 - T^{A'}(y^A(w))} = \frac{1 + \frac{1}{\varepsilon}}{w h_A(w)} \int_w^{\bar{w}} \left[h_A(\theta_A) - \frac{1}{\mu} W'(V^A(\theta_A)) h_A(\theta_A) \right. \\ \left. + \Delta T(\check{\theta}_B(\theta_A, d), d) f(\theta_A, \check{\theta}_B(\theta_A, d)) \frac{\check{\theta}_B(\theta_A, d)}{l^B(\check{\theta}_B(\theta_A, d))^{1 + \frac{1}{\varepsilon}}} \right] d\theta_A, \quad (18)$$

$$\frac{T^{B'}(y^B(w))}{1 - T^{B'}(y^B(w))} = \frac{1 + \frac{1}{\varepsilon}}{w h_B(w)} \int_w^{\bar{w}} \left[h_B(\theta_B) - \frac{1}{\mu} W'(V^B(\theta_B)) h_B(\theta_B) \right. \\ \left. - \Delta T(\theta_B, d) f(\check{\theta}_A(\theta_B, d), \theta_B) \frac{\check{\theta}_A(\theta_B, d)}{l^A(\check{\theta}_A(\theta_B, d))^{1 + \frac{1}{\varepsilon}}} \right] d\theta_B \quad (19)$$

where $y^o(w)$ is the income that a worker earning w in occupation o chooses to earn, $l^o(w)$ is their labor supply, $V^o(w)$ is their indirect utility, and $\Delta T(\theta_B, d) = T^B(y^B(\theta_B)) - \gamma(d) - T^A(y^A(\check{\theta}_A(\theta_B, d)))$ is the tax revenue generated when a marginal worker earning θ_B in B switches from A to B .

Optimal predistribution through job quality satisfies

$$-\gamma'(d) = 1 \quad (20)$$

Proof. Appendix A.2.

Equations (18)-(19) characterize the optimal occupation-specific MTRs. Occupation-specific taxes are differentiated to internalize the fiscal externality from compensating differentials and promote horizontal equity, with both forces tending to push T^A above and T^B below the occupation-blind benchmark. Differentiating taxes in this manner erodes the fiscal externality that motivates the differentiation: the net fiscal gain ΔT from marginal workers who move from A into B is diminished as T^A rises and T^B falls.

Once occupation-specific income taxes are available, fiscal externalities from compensating differentials no longer motivate predistribution — the ΔT term in (17) drops out of (20). The only rationale for predistribution is that d is a non-market good: workers cannot adjust it continuously, only choose between fixed occupation-amenity bundles.

Corollary 2 *Suppose workers can individually purchase job quality improvements at cost $\gamma(d)$ and occupation-specific taxation is feasible. The optimal subsidy s on workers' purchases of job quality is zero.*

$$s^* = 0 \quad (21)$$

Proof. Appendix A.3.

Predistribution is redundant once occupation-specific taxes are available, unless amenity markets are missing — predistribution is a second-best substitute for differentiated income taxation. The Atkinson-Stiglitz theorem is restored when taxes can be conditioned on occupation.

3 Model extensions

This section extends the baseline model in four directions. Two extensions add empirical content: heterogeneous preferences over occupations and an arbitrary number of occupations. The other two show that the rationale for predistribution is attenuated, but not eliminated, by a uniform preexisting subsidy on amenities or by general equilibrium wage effects.

3.1 Heterogeneous preferences

Workers have intrinsic preferences over occupations: some prefer the physical engagement of manual labor over cognitive tasks, while others prefer the reverse. I model this with a taste parameter ξ for occupation B . Worker types $(\theta_A, \theta_B, \xi)$ are distributed according to $F(\theta_A, \theta_B, \xi)$.

The utility function (1) is replaced by

$$U(c, l, d) = \begin{cases} c - \frac{l^{1+1/\varepsilon}}{1+1/\varepsilon} & \text{if } o(\theta_A, \theta_B, d, \xi) = A, \\ c - \frac{l^{1+1/\varepsilon}}{1+1/\varepsilon} - d + \xi & \text{if } o(\theta_A, \theta_B, d, \xi) = B. \end{cases} \quad (22)$$

The threshold functions for occupation choice depend on both skills and preferences. Let $\check{\theta}_B(\theta_A, d, \xi)$ be the value of θ_B at which a worker is indifferent between occupations, given A -skill θ_A , disamenity d , and preference parameter ξ . A worker with a stronger preference for occupation B will have a lower threshold θ_B ,

$$\frac{\partial \check{\theta}_B(\theta_A, d, \xi)}{\partial \xi} < 0. \quad (23)$$

This generates sorting on taste, which dampens both the efficiency and redistributive rationales for predistribution. Marginal workers with stronger preferences for B require smaller compensating differentials, and thus generate a smaller fiscal externality when switching occupations. Sorting also narrows the average utility gap between A - and B -workers, weakening the redistributive motive for predistribution.

While taste heterogeneity adds a dimension to the type distribution, it does not alter the approach to solving for optimal policy. Occupation-wage densities simply involve an additional integral over preference shocks. Specifically, (10) is replaced by (24).

$$h_A(w) = \int_{\underline{\xi}}^{\bar{\xi}} \int_{\underline{w}}^{\check{\theta}_B(w, d, \xi)} f(w, \theta_B, \xi) d\theta_B d\xi, \quad h_B(w) = \int_{\underline{\xi}}^{\bar{\xi}} \int_{\underline{w}}^{\check{\theta}_A(w, d, \xi)} f(\theta_A, w, \xi) d\theta_A d\xi \quad (24)$$

3.2 > 2 occupations

I extend the model to an arbitrary number of occupations, where choice is governed by a collection of threshold functions rather than a single one, following [Gomes et al. \(2018\)](#). Even in this richer setting, the fiscal externality from coarse predistribution—a uniform improvement amenities across the worst jobs—is unambiguously positive.

Suppose there are J occupations, ranked so that $j = 1$ is least amenable and $j = J$ is most. Workers draw a skill vector $\theta \equiv (\theta_1, \dots, \theta_J)$, where θ_j is productivity in occupation j . Worker types are distributed according to a continuous J -dimensional CDF F . Let F_j be the marginal of F with respect to the j -dimension (with bounded density f_j). Denote by $F_{-j|j}(\theta_{-j}|\theta_j)$ the conditional distribution of $\theta_{-j} \equiv (\theta_1, \dots, \theta_{j-1}, \theta_{j+1}, \dots, \theta_J)$ given θ_j (with bounded density $f_{-j|j}$).

Throughout this subsection, I work with amenities ψ_j rather than disamenities, so a higher ψ_j corresponds to a more pleasant occupation. The planner chooses the amenity level in each occupation j ,

$$\psi_j = \bar{\psi}_j + \chi_{j|k},$$

where

$$\chi_{j|k} = \begin{cases} \chi & \text{if } j \leq k \\ 0 & \text{else.} \end{cases} \quad (25)$$

I consider two policy scenarios that differ in the informational demands they place on the planner. The first is coarse predistribution, where the government can uniformly raise the amenity level in the k least amenable occupations by χ at a cost-per-worker of $\gamma_{\chi|k}(\chi)$, for fixed k and $\bar{\psi}_j$. The second is granular predistribution, where the government sets $\chi = 0$ and chooses a distinct amenity level $\bar{\psi}_j$ for each occupation j at a cost-per-worker of $\gamma_j(\bar{\psi}_j)$.

Under occupation-blind income taxes, workers choose their occupation to maximize utility:

$$u^j(\theta_j) \equiv \max_y y - T(y) - \frac{(y/\theta_j)^{1+1/\varepsilon}}{1+1/\varepsilon} + \bar{\psi}_j + \chi_{j|k} \quad (26)$$

$$U(\theta) \equiv \max_{j \in \{1, \dots, J\}} u^j(\theta_j) \quad (27)$$

Occupational choice can be described by a collection of threshold functions $\check{\theta} \equiv (\check{\theta}_{j\ell})_{j \neq \ell}$, where $j, \ell \in \{1, \dots, J\}$ throughout, such that a worker of type θ joins occupation j if and only if, for all $\ell \neq j$, $\theta_\ell < \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)$. As in the two-occupation case, these threshold functions are defined by the indifference condition and are strictly increasing. Appendix A.4 provides further details.

Together, the skill distribution F and threshold functions $\{\check{\theta}_{j\ell}\}_{j \neq \ell}$ induce occupation-specific wage densities. The density of workers earning θ_j in occupation j in equilibrium is

$$h_j(\theta_j) \equiv f_j(\theta_j) F_{-j|j}((\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))_{j \neq \ell} | \theta_j). \quad (28)$$

Define cumulative densities within occupations

$$H_j(x) = \int_{\underline{w}}^x h_j(z) dz, \quad (29)$$

the endogenous wage density

$$h(w) = \sum_{j=1}^J h_j(w), \quad (30)$$

and the density of workers at the (j, ℓ) occupation margin,

$$h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) = f_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) F_{-(j,\ell)|(j,\ell)}((\check{\theta}_{jk}(\theta_j, \psi_j, \psi_k))_{k \neq j,\ell} \mid \theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)). \quad (31)$$

Proposition 3 Suppose there are J occupations.

(i) (Optimal marginal tax rates.) Optimal MTRs are characterized by

$$\frac{T'(y(w))}{1 - T'(y(w))} = \frac{1 + \frac{1}{\varepsilon}}{wh(w)} \left[\sum_{j=1}^J \int_w^{\bar{w}} h_j(z) \left(1 - \frac{1}{\mu} W'(V(z) + \psi_j) \right) dz - \Xi^t(w, \bar{\psi}, \chi) \right] \quad (32)$$

$$\Xi^t(w, \bar{\psi}, \chi) = \sum_{j=1}^J \sum_{\ell=j+1}^J \int_w^{\check{\theta}_{\ell j}(w, \psi_\ell, \psi_j)} \Delta T_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell, \chi) h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) \frac{\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{l(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))^{1+1/\varepsilon}} d\theta_j \quad (33)$$

where $\Delta T_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell, \chi)$ is the tax revenue generated when a marginal worker earning θ_j in j switches from ℓ to j . Its precise form depends on the predistribution policy in place (see (ii) and (iii) below).

(ii) (Coarse predistribution.) Suppose the planner can uniformly reduce disamenities by χ in all occupations $j \leq k$ at a per-worker cost $\gamma_{\chi|k}(\chi)$, with $\gamma_{\chi|k}(0) = 0, \gamma'_{\chi|k}(\chi) > 0$. The optimal χ satisfies

$$\gamma'_{\chi|k}(\chi) = \frac{1}{\mu} \mathbb{E}_{j \leq k} [W'(V(\theta_j) + \psi_j)] + \Xi^x(\chi) \quad (34)$$

$$\Xi^x(\chi) = \mathbb{E}_{j \leq k} \left[\sum_{\ell > k} \Delta T_{j\ell}^x(\theta_j, \chi) \frac{h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))}{h_j(\theta_j)} \frac{\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{l(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))^{1+1/\varepsilon}} \right] \quad (35)$$

where $\Delta T_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell, \chi) = \Delta T_{j\ell}^x(\theta_j, \chi) = T(y(\theta_j)) - \mathbb{1}[j \leq k] \gamma_{\chi|k}(\chi) - T(y(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))) + \mathbb{1}[\ell \leq k] \gamma_{\chi|k}(\chi)$ is the tax revenue generated when a worker earning θ_j in j moves from ℓ to j , and $\mathbb{E}_{j \leq k} [x(\theta_j, \psi_j)] = \frac{\sum_{j \leq k} \int_w^{\bar{w}} x(\theta_j, \psi_j) h_j(\theta_j) d\theta_j}{\sum_{j \leq k} \int_w^{\bar{w}} h_j(\theta_j) d\theta_j}$ is the weighted average of $x(\theta_j, \psi_j)$ over workers in treated occupations $j \leq k$.

(iii) (Granular predistribution.) Suppose the planner can set a distinct amenity level $\bar{\psi}_j$ in each occupation at per-worker cost $\gamma_j(\bar{\psi}_j)$. Optimal predistribution in occupation j satisfies

$$\gamma'_j(\bar{\psi}_j) = \frac{1}{\mu} \mathbb{E}_j [W'(V(\theta_j) + \bar{\psi}_j)] + \Xi_j^p(\bar{\psi}) \quad (36)$$

$$\Xi_j^p(\bar{\psi}) = \mathbb{E}_j \left[\sum_{\ell \neq j} \Delta T_{j\ell}^{\bar{\psi}}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell) \frac{h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell))}{h_j(\theta_j)} \frac{\check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell)}{l(\check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell))^{1+1/\varepsilon}} \right] \quad (37)$$

where $\Delta T_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell, \chi) = \Delta T_{j\ell}^{\bar{\psi}}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell) = T(y(\theta_j)) - \gamma_j(\bar{\psi}_j) - T(y(\check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell))) + \gamma_\ell(\bar{\psi}_\ell)$, and $\mathbb{E}_j [x(\theta_j, \psi_j)] = \frac{\int_w^{\bar{w}} x(\theta_j, \psi_j) h_j(\theta_j) d\theta_j}{\int_w^{\bar{w}} h_j(\theta_j) d\theta_j}$ is the weighted average of $x(\theta_j, \psi_j)$ over workers in occupation j .

Proof. Appendix A.4.

Corollary 3 At the no-predistribution baseline $\chi = 0, \gamma_j(\bar{\psi}_j) = 0$,

- (i) compensating differentials are a force for lower MTRs, $\Xi^t(w, \bar{\psi}, 0) \geq 0 \forall w$;
- (ii) the fiscal externality from coarse predistribution $\Xi^x(0) \geq 0$, so the social return to coarse predistribution weakly exceeds its private return; but
- (iii) the fiscal externality from granular predistribution $\Xi_j^p(\bar{\psi})$ cannot be signed.

The core intuition carries through: income taxation distorts workers toward higher-amenity jobs, and this extensive-margin distortion pushes toward a flatter optimal tax schedule. The fiscal externality term (33) mirrors the two-occupation case (15), but sums over all $J(J-1)/2$ occupation pairs, and can be signed by the same cancellation argument (Appendix A.4).

Under granular predistribution in part (iii), optimal amenity improvements need not be monotone in job quality. Raising $\bar{\psi}_j$ draws workers from every other occupation — both better and worse — so the net fiscal externality $\Xi_j^p(\bar{\psi})$ depends on the composition of inflows: how many switchers come from higher- versus lower-amenity occupations. If a moderately bad occupation draws disproportionately from higher-amenity jobs, the planner may invest more there than in an even worse occupation whose inflows are predominantly from lower-amenity jobs. Thus, implementing granular predistribution requires detailed knowledge of the full matrix of cross-occupation flows, making it informationally demanding.

Coarse predistribution in part (ii) avoids this problem. By raising amenities uniformly across all occupations below the cutoff k , it leaves the relative attractiveness of jobs within the treated group unchanged, and likewise within the untreated group — so no switching occurs on either side of the cutoff. The only induced switches are from untreated occupations ($\ell > k$) into treated ones ($j \leq k$), i.e., from more to less amenable jobs. Since compensating differentials imply these switchers earn strictly more in their new, less amenable occupation, $\Delta T_{j\ell}^x(w, 0) \geq 0$: every switcher contributes a positive fiscal externality. In contrast to granular predistribution, the planner needs only a cutoff k and a uniform χ , not the full matrix of cross-occupation flows.

3.3 Endogenous amenities

In practice, amenities are endogenous: compensation consists of both wages and non-wage benefits, and workers can choose their preferred mix. Some amenities are occupation-specific (e.g., safer conditions in risky jobs at the cost of lower wages), while others are not tied to a particular job (e.g., health insurance). From the firm’s perspective, what matters is that total compensation equals marginal product, regardless of its composition. I therefore model amenities as part of the worker’s consumption bundle, even if they are provided by firms.

A potential objection to Proposition 1’s prescription of predistribution toward bad jobs is that amenities already benefit from implicit tax subsidies, as they are deductible against firms’ taxable income and excluded from workers’ taxable income. If the tax code already favors amenity provision, adding differential subsidies on top may be redundant or even distortionary. However, the major tax-favored amenities like health insurance are typically consumed regardless of occupation, so they enter the worker’s bundle symmetrically across jobs and do not enter the fiscal externality that motivates differential subsidies. For amenities that are occupation-specific, the relevant question is whether existing subsidies are sufficiently large to reverse the sign of the fiscal externality. In practice, the case for predistribution survives unless preexisting subsidies are already implausibly more generous in the disamenable occupation.

I extend the baseline model to incorporate endogenous general and occupation-specific amenities, potentially subject to a uniform subsidy. Workers derive utility from consumption c and general amenities k , with increasing and concave subutility $\zeta(k)$, and disutility from both labor supply and occupation-specific disamenities, d_A and d_B . Workers can mitigate these disamenities through expenditures captured

by occupation-specific cost functions $\gamma_A(d_A)$ and $\gamma_B(d_B)$.

$$U(c, l, h, d) = \begin{cases} c + \zeta(k) - \frac{l^{1+1/\varepsilon}}{1+1/\varepsilon} - d_A & \text{if } o(\theta_A, \theta_B) = A, \\ c + \zeta(k) - \frac{l^{1+1/\varepsilon}}{1+1/\varepsilon} - d_B & \text{if } o(\theta_A, \theta_B) = B. \end{cases} \quad (38)$$

Workers maximize utility subject to their budget constraint,

$$wl - T(wl) = \begin{cases} c + \frac{k}{1+\bar{s}} + \frac{\gamma_A(d_A)}{1+\bar{s}} & \text{if } o(\theta_A, \theta_B) = A, \\ c + \frac{k}{1+\bar{s}} + \frac{\gamma_B(d_B)}{1+\bar{s}+s^B} & \text{if } o(\theta_A, \theta_B) = B. \end{cases} \quad (39)$$

A preexisting, irremovable distortion $\bar{s}$ implicitly subsidizes expenditures on all amenities, general (k) and specific (d_A, d_B). The government's policy instruments are an occupation-blind income tax $T(\cdot)$ and a targeted subsidy on job-quality expenditures in occupation B , denoted s_B . I do not allow the government to tax or subsidize the general amenity k or job-quality expenditures in occupation A , γ_A ; if it could, the planner would simply neutralize the preexisting distortion $\bar{s}$ entirely. The question is whether, despite the universal implicit subsidy $\bar{s}$, the planner still finds it optimal to impose a differential subsidy s_B on job quality in occupation B .

Proposition 4 *Suppose all amenities (k, d_A, d_B) face a preexisting irremovable subsidy $\bar{s}$, on top of which the government levies an occupation-blind income tax and a targeted subsidy s^B on amenity expenditures in occupation B . The optimal subsidy s^B is characterized by*

$$1 + \bar{s} + s^B = \frac{1}{\mu} \mathbb{E}_B [W'(V^B(\theta_B))] + \mathbb{E}_B \left[\Delta T(\theta_B, d_A, d_B) \frac{f(\check{\theta}_A(\theta_B), \theta_B)}{h^B(\theta_B)} \frac{\check{\theta}_A(\theta_B)}{l(\check{\theta}_A(\theta_B))^{1+\frac{1}{\varepsilon}}} \right] \quad (40)$$

where the occupation-switching fiscal externality

$$\Delta T(\theta_B, d_A, d_B) = (T(y(\theta_B)) - T(y(\check{\theta}_A(\theta_B)))) - \left(\frac{(\bar{s} + s^B)\gamma_B(d_B)}{1 + \bar{s} + s^B} - \frac{\bar{s}\gamma_A(d_A)}{1 + \bar{s}} \right)$$

captures the net change in tax revenue and subsidy expenditure when a worker switches occupations and $\mathbb{E}_B[x(\theta_B, d_A, d_B)] = \frac{\int_{\underline{w}}^{\bar{w}} x(\theta_B, d_A, d_B) h_B(\theta_B) d\theta_B}{\int_{\underline{w}}^{\bar{w}} h_B(\theta_B) d\theta_B}$ is the weighted average of $x(\theta_B, d_A, d_B)$ for B -workers.

Proof. Appendix A.5.

First, general amenities k do not enter the fiscal externality. Major tax-favored fringe benefits such as employer-provided health insurance and retirement contributions are best understood as general amenities, and are immaterial for the optimal differential subsidy.

Second, the bias toward amenities over income reappears here in amplified form: income is taxed while occupation-specific amenity expenditures $\gamma_A(d_A)$ and $\gamma_B(d_B)$ are actively subsidized — pushing workers further toward amenities over income. This distortion operates not only on the extensive margin of occupation choice but also on the intensive margin: workers spend excessively on amenities within their occupation, or equivalently, their firms do so on their behalf.

Third, even after endogenous amenity spending, some jobs remain more disamenable than others. The fiscal externality from switching out of an amenable occupation into a disamenable one therefore remains positive, unless workers in disamenable jobs are so much more heavily subsidized that the subsidy differential outweighs the tax differential — an implausible scenario. Thus, although a preexisting uniform subsidy $\bar{s} > 0$

makes the conditions for differential predistribution more stringent, it remains desirable under reasonable conditions.

3.4 General equilibrium wage effects

The linear production technology assumed so far precludes the endogenous adjustment of pretax wages to taxation. Yet, when taxes fall, workers keep more of every dollar, including compensating differentials. This makes bad jobs more attractive at the margin, bidding down their pretax wages relative to good ones. [Powell \(2012\)](#) finds evidence of exactly this: a 10 percent increase in the net-of-tax rate decreases pretax wages in dangerous industries by 1–3 percent more than in safe industries.

I characterize optimal policy in the presence of general equilibrium wage effects by replacing the linear technology with a constant returns to scale production function ([Rothschild and Scheuer, 2013](#)). These effects enter through a single new term whose sign is an empirical question: if bad jobs are also low-paid in the cross-section, they attenuate the desirability of predistribution.

Let $V(w)$ be the indirect utility a worker earning wage w obtains, excluding amenities. The threshold function $\check{\theta}_{B,x}(\theta_A, d)$ is implicitly defined by

$$V(Y_A(x)\theta_A) = V(Y_B(x)\check{\theta}_{B,x}(\theta_A, d)) - d. \quad (41)$$

Technology is described by a constant returns to scale aggregate production function $Y(E_A, E_B)$ that combines skill-weighted aggregate hours in the two sectors, E_A and E_B , defined as

$$E_A = \int_{s_A} \theta_A l(\theta_A, \theta_B) dF(\theta_A, \theta_B), \quad (42)$$

$$E_B = \int_{s_B} \theta_B l(\theta_A, \theta_B) dF(\theta_A, \theta_B) \quad (43)$$

where $s_A = \{(\theta_A, \theta_B) : \theta_A \geq \check{\theta}_{A,x}(\theta_B, d)\}$ and $s_B = \{(\theta_A, \theta_B) : \theta_A < \check{\theta}_{A,x}(\theta_B, d)\}$. As marginal products depend only on the ratio of aggregate hours, $x = \frac{E_A}{E_B}$, I denote them by $Y_A(x)$ and $Y_B(x)$. Each worker's wage is

$$w(\theta_A, \theta_B, d) = \begin{cases} Y_A(x)\theta_A & \text{if } \theta_A \geq \check{\theta}_{A,x}(\theta_B, d) \\ Y_B(x)\theta_B & \text{otherwise} \end{cases} \quad (44)$$

Given x and policy, workers choose their occupation optimally, yielding endogenous wage distributions

$$H_x(w) \equiv F\left(\frac{w}{Y_A(x)}, \frac{w}{Y_B(x)}\right) \quad (45)$$

$$h_x^A(w) = \frac{1}{Y_A(x)} \int_{\underline{\theta}_B}^{\check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right)} f\left(\frac{w}{Y_A(x)}, \theta_B\right) d\theta_B, \quad h_x^B(w) = \frac{1}{Y_B(x)} \int_{\underline{\theta}_A}^{\check{\theta}_{A,x}\left(\frac{w}{Y_B(x)}, d\right)} f\left(\theta_A, \frac{w}{Y_B(x)}\right) d\theta_A \quad (46)$$

with associated CDFs $H_x^A(w)$ and $H_x^B(w)$, and total density $h_x(w) = h_x^A(w) + h_x^B(w)$.

The planner's problem mirrors that of Proposition 1, augmented with a consistency constraint that ensures labor-market clearing:

$$(1 - \alpha(x)) \int_{\underline{w}_x}^{\bar{w}_x} w l(w) h_x^A(w) dw = \alpha(x) \int_{\underline{w}_x}^{\bar{w}_x} w l(w) h_x^B(w) dw \quad (47)$$

where $\alpha(x) = \frac{Y_A(x)E_A}{Y(E_A, E_B)}$ is the aggregate income share of occupation A .

Proposition 5 *Defining μ as the multiplier on the resource constraint and $\mu\xi$ as the multiplier on the consistency constraint (47), optimal predistribution satisfies*

$$-\gamma'(d) = \frac{1}{\mu} \mathbb{E}_B [W'(V(w) - d)] + \mathbb{E}_B \left[(\Delta T(w, d) - \xi \Delta C(w, d)) \frac{f(\check{\theta}_{A,x}(w, d), w)}{h_x^B(w)} \frac{Y_A(x) \check{\theta}_{A,x}(w, d)}{l(Y_A(x) \check{\theta}_{A,x}(w, d))^{1+\frac{1}{\varepsilon}}} \right] \quad (48)$$

where $\mathbb{E}_B[x(w, d)] = \frac{\int_w^{\bar{w}} x(w, d) h_x^B(w) dw}{\int_w^{\bar{w}} h_x^B(w) dw}$ is the weighted average of $x(w, d)$ among B -workers, $\Delta T(w, d) = T(y(w)) - \gamma(d) - T\left(y\left(Y_A(x) \check{\theta}_{A,x}\left(\frac{w}{Y_B(x)}, d\right)\right)\right)$ is the fiscal externality, and $\Delta C(w, d) = \alpha(x) Y_A(x) \check{\theta}_{A,x}\left(\frac{w}{Y_B(x)}, d\right) l\left(Y_A(x) \check{\theta}_{A,x}\left(\frac{w}{Y_B(x)}, d\right)\right)$ is the change in the consistency constraint arising when a marginal worker switches from A to B .

Proof: Appendix A.6.2. The appendix also characterizes optimal MTRs, which closely follow Rothschild and Scheuer (2013).

Predistribution raises labor supply in the disamenable occupation B , depressing wages there. The $\xi \Delta C(w, d)$ term in (48) captures this general equilibrium wage channel. $\Delta C(w, d) > 0$ is the change in the consistency constraint when a worker earning w in B switches from A , and ξ is the shadow price of that constraint. The sign of ξ reflects whether lowering B -wages relative to A -wages is welfare-reducing ($\xi > 0$), attenuating the case for predistribution, or welfare-improving ($\xi < 0$), strengthening it. Appendix A.6.3 provides an expression for ξ evaluated at the optimal x . Unlike Rothschild and Scheuer (2013), I cannot derive a general sufficient condition for signing ξ because the discrete wage jumps associated with occupation switching under compensating differentials introduce additional terms of ambiguous sign.

When $d = 0$, compensating differentials vanish and my expression for ξ coincides with Rothschild and Scheuer (2013), so by continuity ξ retains this sign for small $d > 0$: it is positive, and the wage channel works against predistribution, whenever B -workers are concentrated at low wages in the cross-section ($h_x^B(w)/h_x(w)$ decreasing in w), so that depressing B -wages harms the already low-paid.⁸ This is an empirical question, but the evidence is suggestive of $\xi > 0$: better-paid workers also tend to enjoy better working conditions (e.g., Maestas et al., 2023), so worse jobs are disproportionately low-wage and this wage channel attenuates predistribution.

4 Empirical strategy

Setup Unlike the Mirrleesian benchmark with quasilinear isoelastic utility, which requires only a unidimensional skill distribution and the labor supply elasticity, solving for optimal policy in the presence of compensating differentials requires two additional policy-invariant primitives: the multidimensional skill distribution and occupational amenities. These cannot be replaced by an occupation-switching elasticity: any reduced-form estimate, such as that of Powell and Shan (2012), is endogenous to the tax schedule, since both the fiscal externality from occupation switching and the density of marginal workers change with policy.

These primitives are precisely the objects of the generalized Roy model—in which workers sort on comparative advantage and amenities—but their estimation poses identification challenges. The multidimensional

⁸Strictly, Rothschild and Scheuer (2013) use exogenous Pareto weights, and their sufficient condition for $\xi > 0$ also requires that the welfare weight on B -workers exceed that on A -workers at every wage. A concave social welfare function satisfies this automatically: at a given wage, B -workers attain lower utility than A -workers because they bear the disamenity d , so a concave $W(\cdot)$ assigns them a higher marginal welfare weight.

skill distribution and amenities cannot be nonparametrically identified from cross-sectional wage and occupation data alone, because each worker is observed in only one occupation (French and Taber, 2011). Following Lockwood et al. (2017), I impose parametric assumptions to infer the skill distribution and amenities. The estimates and resulting optimal policy should therefore be read as illustrative, elucidating the model’s mechanisms rather than providing precise quantitative prescriptions.

The empirical implementation combines the extensions from Sections 3.1 and 3.2: workers sort across J occupations and have idiosyncratic tastes for each. Each worker has a $2J$ -dimensional type $\nu = (\theta, \psi)$, where $\theta = (\theta_1, \dots, \theta_J)$ is their skill vector and $\psi = (\psi_1, \dots, \psi_J)$ is their amenity vector. Worker ν ’s indirect utility in occupation j is given by

$$u_\nu^j(\theta_j) = V(\theta_j) + \psi_j$$

Each worker’s productivity θ_j in occupation $j = 1, \dots, J$ is drawn from an occupation-specific distribution F_j^θ , with the correlation structure of productivity draws given by a Gaussian copula

$$f(\theta_1, \dots, \theta_n) = f_{0,\Sigma}^\mathcal{N}(\Phi^{-1}(F_1^\theta(\theta_1)), \dots, \Phi^{-1}(F_n^\theta(\theta_n))), \quad (49)$$

where Φ denotes the standard normal CDF and $f_{\mu,\Sigma}^\mathcal{N}$ is the PDF of a multivariate normal with mean μ and covariance matrix Σ . Σ has unit diagonal and a common off-diagonal entry ρ , which controls cross-occupation skill correlation. At $\rho = 1$, a worker occupies the same skill percentile across all occupations, while at $\rho < 1$ there is scope for comparative advantage.

Conditional on the productivity vector θ , each occupation-specific amenity ψ_j is drawn independently from

$$\psi_j \sim \beta^{-1} \bar{\theta}(\theta) (\psi_j^0 + \chi_{j|k} + F_\psi), \quad (50)$$

where ψ_j^0 is the baseline amenity level in occupation j , F_ψ is the standard Gumbel distribution, and $\bar{\theta}(\theta) = \frac{1}{J} \sum_{j=1}^J \theta_j^{1+\varepsilon}$ normalizes occupational choice to be scale-invariant with respect to income. This delivers a logit model of occupational choice. The allocative sensitivity parameter β governs the importance of income relative to amenities in occupational choice: larger values make occupational choice more responsive to income differences, and thus more sensitive to tax-induced changes in relative incomes. $\chi_{j|k}$ parameterizes the extent of coarse predistribution in the k most disamenable occupations, defined in (25). The per-worker cost of predistribution takes the power form

$$\gamma_{\chi|k}(\chi) = \frac{c_p \cdot \chi^{1+1/\varepsilon_p}}{1 + 1/\varepsilon_p}. \quad (51)$$

The model has four primitives: the skill distributions f_j^θ , amenities ψ_j^0 , allocative sensitivity parameter β , and skill correlation parameter ρ . Since cross-sectional data offers no information on how skills covary across occupations for the same worker, I fix ρ by assumption. Given ρ , the Gaussian copula structure and logit model of occupational choice yield a system of functional equations that jointly identifies f_j^θ , ψ_j^0 and β . The inputs are the empirical wage distributions $\tilde{f}_j^\theta$, occupation shares s_j , and the occupation-switching elasticity from Powell and Shan (2012), who find that a 10 percent increase in the net-of-tax rate induces workers to switch to jobs with 0.3 percent higher wages. Appendix B.1 describes this system. Baseline predistribution is normalized to zero ($\chi = 0$) in the inference procedure, so optimal predistribution is relative to the status quo.

Data and calibration I use the 2015 American Community Survey (ACS) 1-year public-use microdata sample, restricted to civilians aged 17–65 who are employed full-time (at least 35 hours per week) and full-year (50 or more weeks), report positive wage and salary income, work in nonfarm occupations, and are not unpaid family workers. Workers are grouped into 12 occupations using 2010 Census occupation codes (Table B1).

I infer the empirical skill marginals by fitting a Pareto-lognormal distribution to each occupation’s wage distribution via maximum likelihood and inverting the labor supply first-order condition. I set $\rho = 0.75$ as my baseline, which allows for comparative advantage in skills. This yields $\hat{\beta} = 0.33$. In robustness checks, I consider two alternative parameterizations. First, I shut down comparative advantage by setting $\rho = 1$, as maintained in Lockwood et al. (2017) and Gaubert et al. (2021), which requires reinferring allocative sensitivity, skill distributions, and amenities. Second, I raise the relative weight on earnings in occupational choice by doubling β to 0.66 instead identifying it from the occupation-switching elasticity, and backing out the skill distributions and amenities given this value.

Optimal policy I modify the planner’s objective in two ways. First, since amenities are scaled by $\bar{\theta}(\theta)$, a worker’s cardinal utility varies with their skill vector, making welfare comparisons under a concave SWF sensitive to this normalization. I therefore replace the concave SWF with exogenous Pareto weights $g(\theta)$. Second, with logit preference shocks, the appropriate measure of individual welfare is the inclusive value — the expected maximum utility under logit preferences. Social welfare is

$$\int_{[\underline{w}, \bar{w}]^J} (IV(\theta) + \gamma) g(\theta) d\theta,$$

where $IV(\theta) = \mathbb{E}[\max_j u_v^j(\theta)] = \ln(\sum_{\ell=1}^J \exp(\frac{\beta}{\bar{\theta}(\theta)} V(\theta_\ell) + \psi_\ell))$ is the inclusive value and γ is the Euler–Mascheroni constant. I use Pareto weights that encode two redistributive motives: a preference for lower-skilled workers, and a preference for those with a comparative advantage in disamenable occupations,

$$g(\theta) = f(\theta) \underbrace{\prod_{j=1}^J \phi (1 - F_j^\theta(\theta_j))^\phi}_{\text{redistribution across skill ranks}}^{-1} \cdot \underbrace{\exp(\lambda CA(\theta))}_{\text{redistribution for comp. adv. in bad jobs}}$$

$$CA(\theta) = \sum_{j=1}^J (\bar{\psi}^0 - \psi_j^0) (F_j^\theta(\theta_j) - \bar{F}(\theta))$$

where $F_j^\theta(\theta_j)$ is the worker’s percentile rank in occupation j ’s skill distribution, $\bar{F}(\theta) = \frac{1}{J} \sum_j F_j^\theta(\theta_j)$ is their average rank, $\bar{\psi}^0 = \frac{1}{J} \sum_j \psi_j^0$ is the average baseline amenity. The first factor captures redistributive preferences for lower-skilled workers: a higher ϕ assigns higher weight to lower-ranked workers, with $\phi = 1$ as the utilitarian benchmark. The second factor governs horizontal equity: the index $CA(\theta)$ measures the extent to which a worker’s comparative advantage lies in disamenable occupations. It is the covariance across occupations between skill rank and disamenity $-\psi_j^0$, so $CA(\theta) > 0$ for workers whose skills are tilted toward bad jobs. A higher λ assigns higher weight to workers with comparative advantage in bad jobs; $\lambda = 0$ is the benchmark with no redistribution based on comparative advantage.

Optimal policy formulas are in Appendix B.2. I use the following parameter values. A Pareto tail with $\alpha = 2$ (Mankiw et al., 2009) is appended above the 99th percentile of each occupation’s skill distribution to ensure well-behaved upper tails. The labor supply elasticity is $\varepsilon = 0.5$ (Kleven et al., 2025). The

predistribution cost parameters are $c_p = 7,500,000$ and $\varepsilon_p = 0.5$. I set $k = 5$, so the five most disamenable occupations receive predistribution. The planner has a redistributive preference $\phi = 1.5$, and no horizontal equity motive, $\lambda = 0$. The latter allows a clean comparison with the $\rho = 1$ case, where the absence of comparative advantage renders λ irrelevant. I report results under a higher predistribution cost c_p and a horizontal equity motive $\lambda > 0$ in Appendix B.3.

5 Results

I compare three policy scenarios: a naïve optimum that ignores the fiscal externality from occupation switching (the standard Mirrleesian benchmark), an externality-corrected optimum that accounts for this externality but restricts the planner to income taxation alone, and a full optimum that adds predistribution as a second instrument alongside income taxation.

Figure 2 presents optimal MTRs and their determinants across the three scenarios. Figure 2a shows that internalizing the fiscal externality from compensating differentials lowers optimal MTRs by approximately 3.1 percentage points across the income distribution relative to the naïve case. Adding predistribution as a second instrument leaves optimal MTRs nearly unchanged relative to the externality-corrected case.⁹

Figure 2b shows how the corrected tax schedule and predistribution affect the skill density $h(w)$ and fiscal externality $\Xi(y)$ relative to the naïve case. The left panel shows that correcting for the fiscal externality shifts the skill density $h(w)$ rightward as workers reallocate toward higher-paying disamenable occupations in response to lower taxes, but the effect is small, consistent with the low occupation-switching elasticity of Powell and Shan (2012). Predistribution to the five least amenable occupations shifts $h(w)$ rightward by substantially more: as a targeted instrument that directly raises the attractiveness of disamenable jobs, it internalizes the fiscal externality from compensating differentials more effectively than a broad-based tax cut. The right panel shows that the fiscal externality correction worsens $\Xi(y)$ at lower earnings levels while internalizing it at higher levels, whereas adding predistribution consistently internalizes $\Xi(y)$ across the earnings distribution, and by a larger magnitude.

Figure 3 presents occupation amenities at baseline and after coarse predistribution that improves amenities equally for the five lowest-amenity occupations: trades & repair, construction, business, transportation, and science & engineering. Baseline amenity values ψ^0 range from 0 to 1.26. The optimal predistribution level of $\chi = 0.10$ —approximately 8% of the baseline amenity range—is achieved at a cost of \$2,490 per worker.

Figure 4 doubles of the allocative sensitivity parameter $\hat{\beta}$, making workers more responsive to earnings relative to amenities. I re-infer skill distributions and amenities at this higher value. When occupational choice is more responsive to income differences, the fiscal externality from compensating differentials is larger, and correcting for it has a correspondingly larger effect on optimal policy. Figure 4a shows that the externality correction lowers optimal MTRs by approximately 6.4 percentage points relative to the naïve case, compared to 3.1 percentage points in the baseline. At the same time, workers are less responsive to amenity improvements, reducing the fiscal return to predistribution. Optimal predistribution falls to $\chi = 0.07$, compared to 0.10 in the baseline.

Figure 5 eliminates comparative advantage, setting $\rho = 1.0$ and re-estimating skill distributions, ameni-

⁹Optimal MTRs depend on both the skill density $h(w)$ and the fiscal externality $\Xi(y)$, and Figure 2b illustrates how predistribution moves each in opposing directions. The rightward shift in $h(w)$ is a force for less progressivity, while the internalization of $\Xi(y)$ is a force for greater progressivity. In absolute terms, however, both effects are small, which explains why optimal MTRs are nearly identical across the externality-corrected and full-optimum cases.

ties, and $\hat{\beta}$. Figure 5a shows that the externality correction reduces MTRs by a similar magnitude to the baseline below \$200,000, but by less above. This reflects a smaller fiscal externality at the top once comparative advantage is shut down: under the baseline $\rho = 0.75$, workers in low-paying occupations could have meaningfully different skill draws in other occupations—placing them, for instance, on the margin between their current job and a much higher-paying one. Under perfect skill correlation, a worker highly skilled in one occupation is highly skilled in all, so skill differentials across occupations are limited by construction. This shrinks the pool of workers marginal to high-paying occupations, reducing the fiscal externality at the top and the corresponding MTR reduction. Optimal predistribution also falls to $\chi = 0.063$ (from 0.100 in the baseline), reflecting this smaller pool of workers marginal to high-paying occupations and thus a lower fiscal return to predistribution.

The left panel of Figure 5b shows a smaller rightward shift in the wage distribution, despite a tax cut of similar magnitude at the lower end. This follows from the absence of comparative advantage: fewer marginal workers means the same tax cut induces less reallocation toward disamenable occupations. The smaller tax cut at the top has negligible effect on density there, as the initial density is near zero.

Appendix Figure B2 shows that introducing a horizontal equity motive, $\lambda > 0$, increases the optimal level of predistribution, while Figure B3 shows that increasing the cost of predistribution, c_p , reduces it.

Taken together, the results point to two conclusions. First, internalizing the fiscal externality from compensating differentials warrants a meaningful reduction in MTRs across the income distribution. In specifications internally consistent with the occupation-switching elasticity of Powell and Shan (2012), these reductions are on the order of 1–4 percentage points (Figures 2 and 5). Reductions are larger when workers are more sensitive to income relative to amenities, or comparative advantage increases the density of marginal workers. Second, even coarse predistribution is more effective at internalizing this externality than a broad-based tax cut. By directly targeting the attractiveness of disamenable occupations, it more squarely addresses the missing-instrument problem, shifting more workers into higher-paying jobs despite being applied uniformly rather than optimally calibrated to each occupation. This makes predistribution a potentially robust policy prescription — one that justifies broad interventions like workplace safety regulation — and suggests it complements rather than substitutes lower MTRs.

6 Conclusion

Although it has long been recognized that an occupation-blind income tax distorts occupational choice by eroding the compensating differentials that induce workers to accept disagreeable jobs, the policy implications of this observation have remained unresolved. This paper fills this gap by extending the canonical Mirrleesian framework to incorporate multidimensional skills and occupational amenities. The central theoretical contribution is to show that the fiscal externality from compensating differentials overturns the Atkinson-Stiglitz theorem by making predistribution through job quality desirable, and calls for lower MTRs. Labor market regulation can thus be justified on purely fiscal grounds, without recourse to the externality, monopsony, asymmetric information and behavioral bias arguments developed in the existing literature.

The fiscal externality from compensating differentials calls for lower MTRs — most forcefully where many workers are on the margin between occupations and the revenue gains from their switching are large. Lower MTRs induce workers to sort more on comparative advantage and less on amenities, improving allocative efficiency and expanding the tax base. Horizontal equity concerns also push toward lower MTRs at income levels where disamenable work is prevalent, whenever the social planner redistributes based on workers’

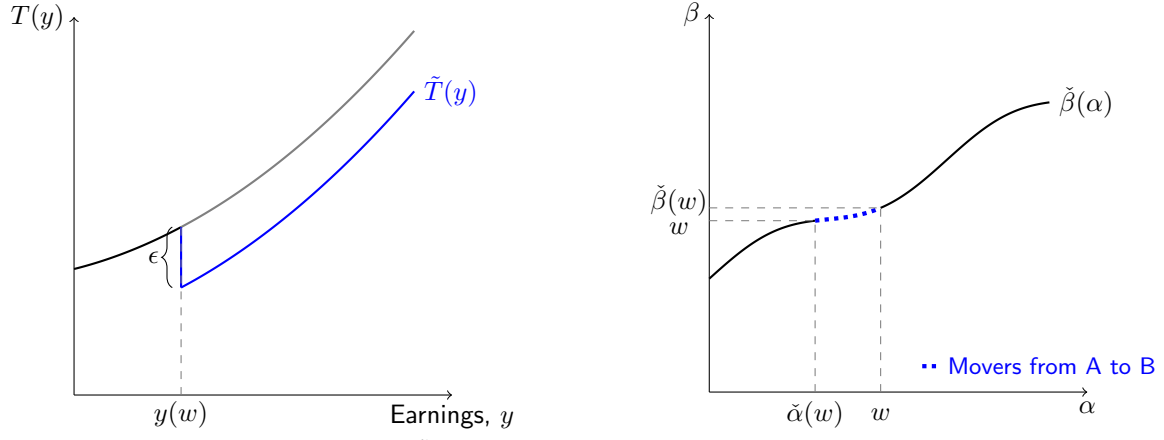
utilities rather than incomes — as under a concave SWF or Pareto weights tilted toward workers with comparative advantage in disamenable jobs.

Relative to a broad-based tax cut, predistribution better approximates occupation-specific taxation. It again serves two ends, counteracting the distortionary effect of income taxation on the wage-amenity tradeoff and correcting the horizontal inequity between equally-paid workers who enjoy different amenities. When predistribution can be differentiated across occupations, job-quality investments should be directed not simply to the worst jobs, but to those with the greatest potential to draw workers away from more amenable ones, thereby maximizing the revenue gains from occupational switching. Yet coarse predistribution—uniform improvements to the least amenable occupations—requires less information and still internalizes the fiscal externality, providing a robust fiscal rationale for interventions such as workplace safety regulation.

The fiscal externality from compensating differentials resists a sufficient statistic representation because its main components—the tax wedge and density of occupation switchers—are endogenous to taxes and predistribution. Thus, implementing the optimal policy formulas requires two policy-invariant primitives beyond the standard Mirrleesian benchmark: the multidimensional skill distribution and occupational amenities. I back out these primitives by imposing structure on the skill distribution and job preferences, and matching the occupation-switching elasticity from [Powell and Shan \(2012\)](#). Internalizing the fiscal externality from compensating differentials lowers optimal MTRs by roughly 1–4 percentage points across the income distribution, with larger reductions when workers have comparative advantage rather than perfectly rank correlated skills, and when their occupational choice responds more to income than to amenities. Even in coarse form, predistribution does far more to shift workers into higher-paying jobs than these tax reductions alone.

Several directions remain for future work. First, while this paper focuses on job quality as the lever of predistribution, the fiscal externality logic extends to any policy that reallocates workers across occupations or places, such as place-based subsidies, training programs, and occupational licensing reform. This suggests a broader agenda for situating these policies within optimal tax theory. Second, the quantitative results should be read as illustrative rather than definitive, as they rest on strong assumptions about workers’ preferences and skill correlation over occupations. Better ways of recovering the multidimensional skill distribution and job amenities, or measuring compensating differentials, would translate into sharper policy guidance. Finally, predistribution is modeled here as a broad stand-in for the many real-world policies that improve job quality, but little is known about how specific policies affect occupational choice, or their costs and incidence. Until such evidence is available, the model speaks more to whether to invest in job quality than to how much. Progress on any of these fronts would move the framework from a qualitative case for predistribution toward more precise quantitative guidance.

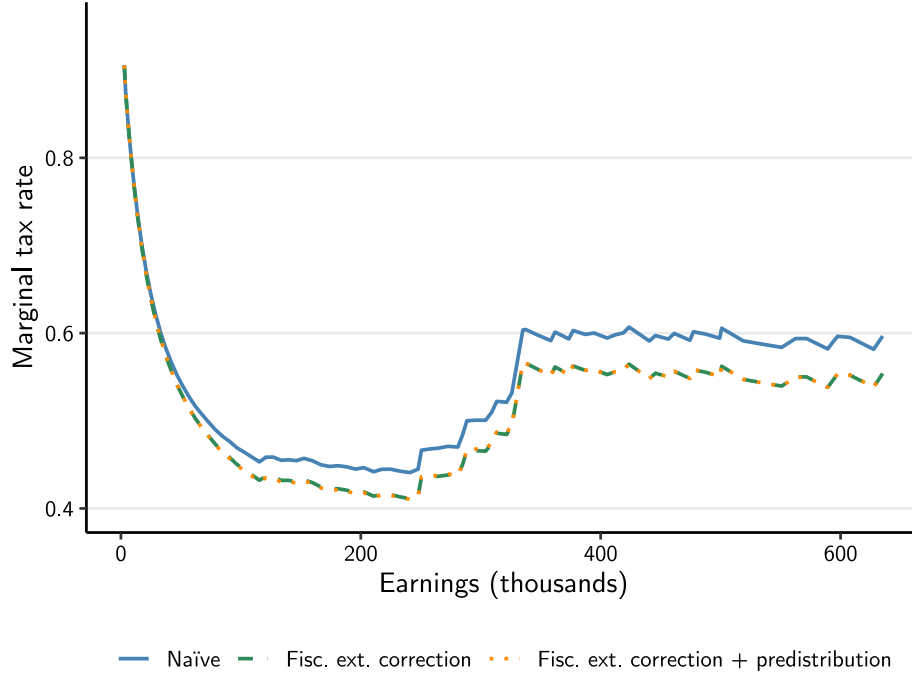
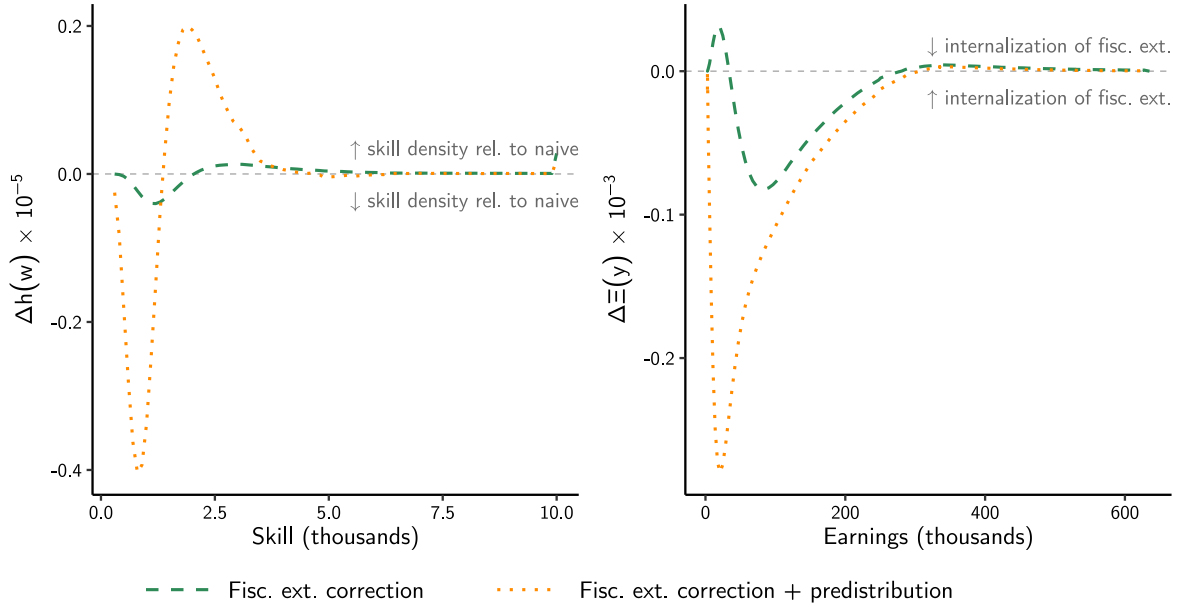
Figure 1: Movers from A to B after a marginal tax cut at $y(w)$



Note: *Left:* The perturbed schedule $\tilde{T}(y)$ reflects a reduction in the marginal tax rate at $y(w)$, lowering the tax burden of all workers with earnings above $y(w)$ by ϵ . *Right:* Only doubly marginal workers on the locus $\check{\beta}(\alpha)$ between $\check{\alpha}(w)$ and w switch from A to B in response.

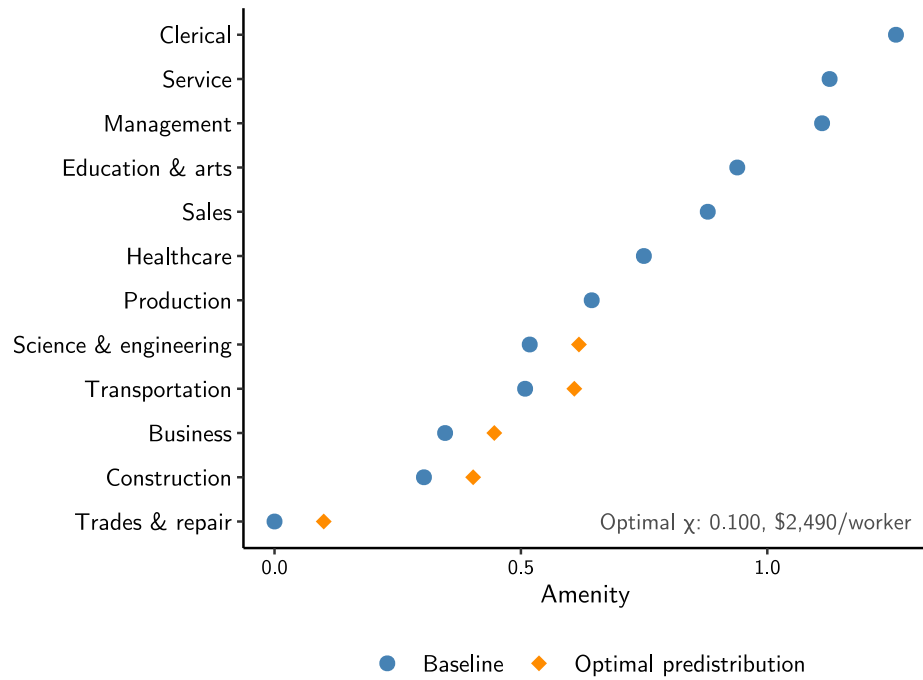
Figure 2

(a) Optimal marginal tax rates

(b) Changes in skill density $h(w)$ and fiscal externality $\Xi(y)$ relative to naïve case

Note: The naïve case sets the fiscal externality from compensating differentials $\Xi(y(w))$ to zero in the optimal tax formula; the remaining two cases set $\Xi(y(w))$ to its true value. The “Fisc. ext. correction + redistribution” case additionally adds optimal redistribution to the five least-amenable occupations (trades & repair, construction, business, transportation, and science & engineering). Panel (b) reports changes relative to the naïve case: the left panel shows the change in the endogenous skill density $h(w)$, and the right panel shows the change in the fiscal externality $\Xi(y(w))$, where the naïve-case fiscal externality is evaluated ex post at the naïve optimal tax schedule.

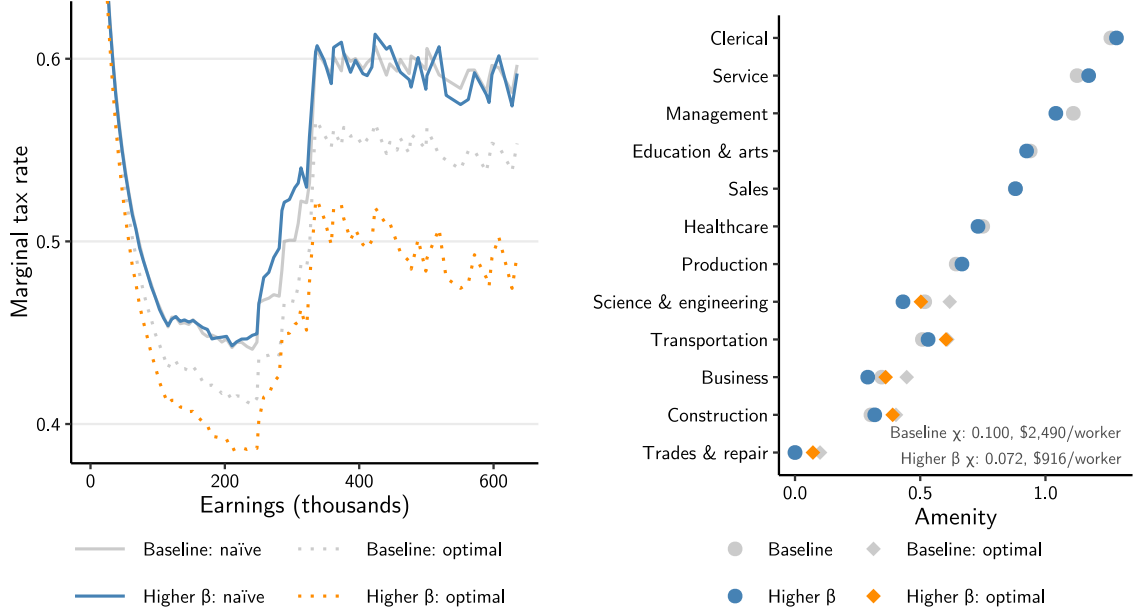
Figure 3: Occupation amenities before and after predistribution



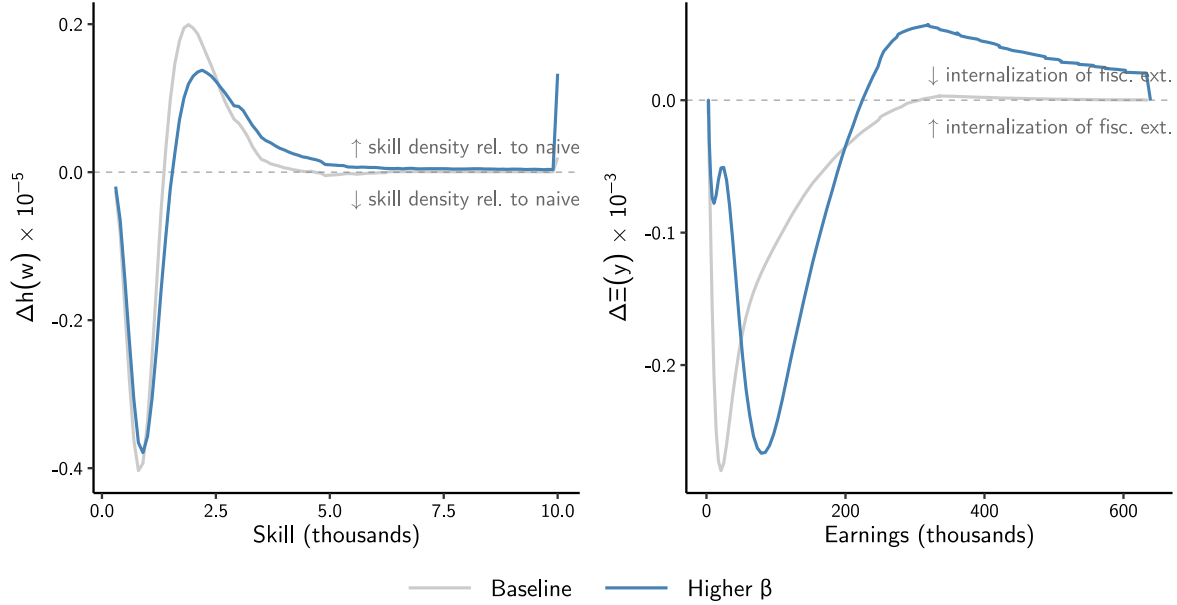
Note: Baseline amenity values ψ^0 are inferred from the system of functional equations described in Section 4. Under optimal predistribution, amenities are $\psi^0 + \chi$ for the five least-amenable occupations and unchanged for the remaining occupations.

Figure 4: Robustness to increased earnings sensitivity of occupation choice, β

(a) Optimal MTRs and predistribution



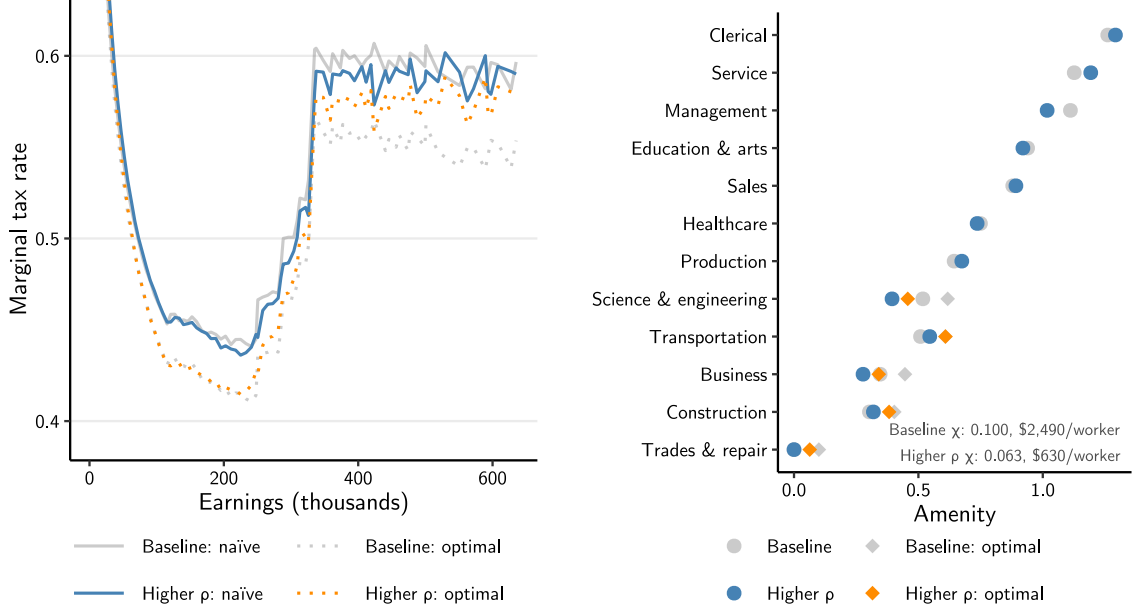
(b) Changes in skill density $h(w)$ and fiscal externality $\Xi(y)$ relative to naïve case



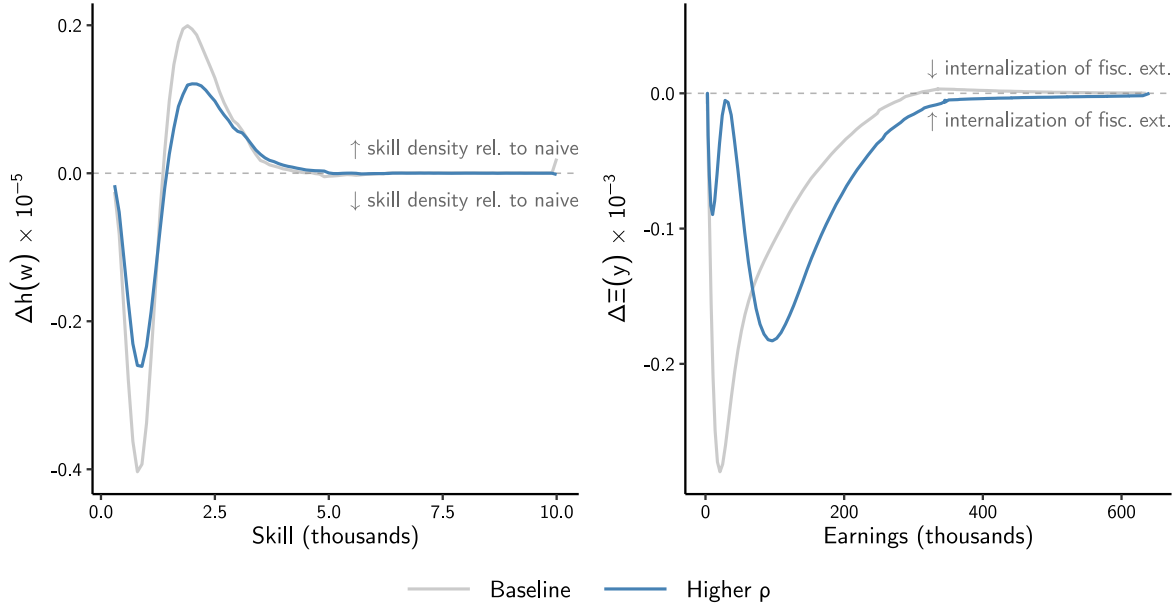
Note: The allocative sensitivity parameter $\hat{\beta}$ governs the importance of earnings relative to amenities in occupational choice: higher values make occupational choice more responsive to earnings differences and less to amenity differences. The baseline specification, corresponding to Figures 2–3, calibrates $\hat{\beta} = 0.33$ from the occupation-switching elasticity of [Powell and Shan \(2012\)](#). The “Higher β ” specification doubles β to 0.66 and re-estimates skill distributions f^θ and amenities ψ^0 given this value. The naïve and optimal cases follow the naïve and “Fisc. ext. correction + predistribution” specifications in Figure 2, respectively. Panel (b) follows the variable definitions therein.

Figure 5: Robustness to higher skill correlation, ρ

(a) Optimal MTRs and predistribution



(b) Changes in skill density $h(w)$ and fiscal externality $\Xi(y)$ relative to naïve case



Note: The skill correlation parameter ρ governs the rank correlation of skills across occupations. The baseline specification, corresponding to Figures 2–3, sets $\rho = 0.75$. The “Higher ρ ” specification sets $\rho = 1$, which implies perfect rank correlation, and reinfers skill distributions f^θ , amenities ψ^0 , and allocative sensitivity β . The naïve and optimal cases follow the naïve and “Fisc. ext. correction + predistribution” specifications in Figure 2, respectively. Panel (b) follows the variable definitions therein.

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A Proofs

A.1 Proposition 1

Placing multipliers μ on the resource constraint (13), and $\eta\mu$ on the local ICCs (12), the Lagrangian is

$$\begin{aligned}\mathcal{L} = & \int_{\underline{w}}^{\bar{w}} W(V^A(w))h_A(w) + W(V^A(w) - d)h_B(w)dw \\ & + \mu \int_{\underline{w}}^{\bar{w}} [wl(w) - c(V^A(w), l(w))] h(w) - \gamma(d)h_B(w)dw \\ & - \mu \int_{\underline{w}}^{\bar{w}} V^A(w)\eta'(w)dw - \mu \int_{\underline{w}}^{\bar{w}} \frac{l(w)^{1+1/\varepsilon}}{w}\eta(w)dw\end{aligned}$$

where the last term uses integration by parts: $-\int_{\underline{w}}^{\bar{w}} V^A(w)\eta'(w)dw = \int_{\underline{w}}^{\bar{w}} V^{A'}(w)\eta(w)dw$.

Using $\frac{\partial c}{\partial V^A} = 1$, the first-order condition with respect to $V^A(w)$ is

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial V^A(w)} = & W'(V^A(w))h_A(w) + W'(V^A(w) - d)h_B(w) - \eta'(w)\mu - \mu h(w) \\ & + \mu \int_{\underline{w}}^{\bar{w}} (w'l(w') - c(V^A(w'), l(w')))\frac{\partial h(w')}{\partial V^A(w)}dw' - \mu \int_{\underline{w}}^{\bar{w}} \gamma(d)\frac{\partial h_B(w')}{\partial V^A(w)}dw' = 0\end{aligned}\tag{A1}$$

Only marginal workers earning w in one occupation change occupations in response to a change in $V^A(w)$. Therefore, the derivatives of A -occupation wage densities are zero everywhere except $w' = w$ and $w' = \check{\theta}_A(w, d)$. Similarly, the derivatives of B -occupation wage densities are zero everywhere except $w' = w$ and $w' = \check{\theta}_B(w, d)$.

$$\begin{aligned}\frac{\partial h_A(w')}{\partial V^A(w)} = 0 \quad \forall \quad w' \in [\underline{w}, \bar{w}] \setminus \{w, \check{\theta}_A(w, d)\} \\ \frac{\partial h_B(w')}{\partial V^A(w)} = 0 \quad \forall \quad w' \in [\underline{w}, \bar{w}] \setminus \{w, \check{\theta}_B(w, d)\}\end{aligned}$$

Where the density derivatives are nonzero, the Leibniz integral rule implies that

$$\frac{\partial h_A(w)}{\partial V^A(w)} = -\frac{\partial h_B(\check{\theta}_B(w, d))}{\partial V^A(w)}, \quad \text{and} \quad \frac{\partial h_A(\check{\theta}_A(w, d))}{\partial V^A(w)} = -\frac{\partial h_B(w)}{\partial V^A(w)}.$$

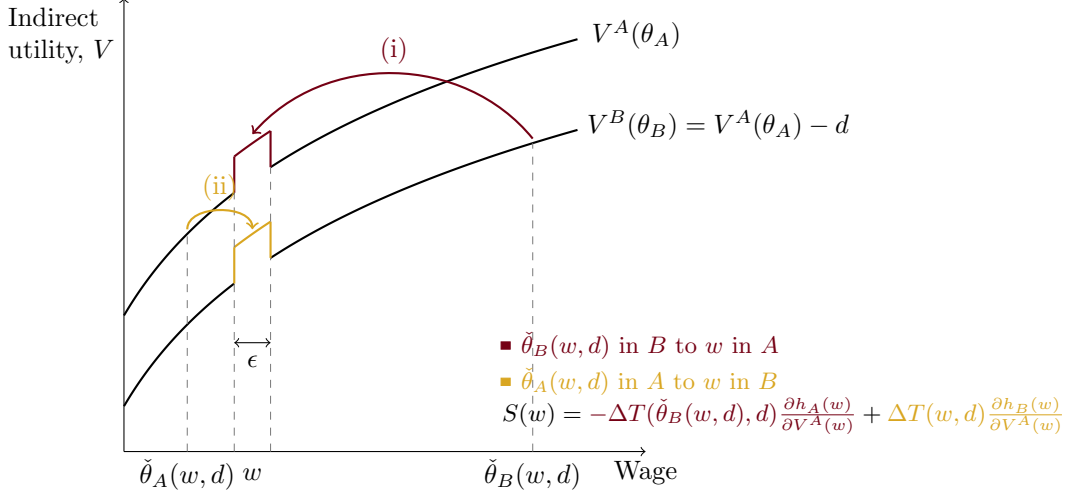
The fiscal externality from occupation switching simplifies to

$$\begin{aligned}S(w) = & \int_{\underline{w}}^{\bar{w}} T(y(w'))\frac{\partial h(w')}{\partial V^A(w)}dw' - \int_{\underline{w}}^{\bar{w}} \gamma(d)\frac{\partial h_B(w')}{\partial V^A(w)}dw' \\ = & [T(y(w)) - T(y(\check{\theta}_B(w, d))) + \gamma(d)]\frac{\partial h_A(w)}{\partial V^A(w)} + [T(y(w)) - \gamma(d) - T(y(\check{\theta}_A(w, d)))]\frac{\partial h_B(w)}{\partial V^A(w)} \\ = & -\Delta T(\check{\theta}_B(w, d), d)\frac{\partial h_A(w)}{\partial V^A(w)} + \Delta T(w, d)\frac{\partial h_B(w)}{\partial V^A(w)}\end{aligned}\tag{A2}$$

where $\Delta T(\theta_B, d) = T(y(\theta_B)) - \gamma(d) - T(y(\check{\theta}_A(\theta_B, d)))$.

A marginal increase in the indirect utility assigned to workers earning wage w , $V^A(w)$, has two opposing effects. Figure A1 illustrates these for a perturbation of $V^A(w)$ over $[w - \epsilon/2, w + \epsilon/2]$. Since only occupation-blind taxes are feasible, indirect utility for B -workers is a downward translation of that for A -workers by d . In

Figure A1: Extensive margin responses to a marginal increase in indirect utility for w -earners, $\partial V^A(w)$



this figure, the incentive compatibility constraint clearly does not hold; I disregard it for now as the objective is to explain the $S(w)$ fiscal externality. First, arrow (i) captures how marginal workers in occupation B who initially earn $\check{\theta}_B(w, d) > w$ switch to occupation A , accepting a lower wage of w . This reallocation generates a negative fiscal externality through lower income tax revenue, $-T(y(\check{\theta}_B(w, d))) + T(y(w)) < 0$, but a positive one through reduced expenditures on job quality, $\gamma(d) \geq 0$.

Conversely, arrow (ii) shows marginal A -workers initially earning $\check{\theta}_A(w, d) < w$ switching to occupation B and earning the higher wage w . This creates a positive fiscal externality through income taxes, $-T(y(\check{\theta}_A(w, d))) + T(y(w)) > 0$, but a negative one through increased expenditures on job quality, $-\gamma(d) \leq 0$.

Intuitively, the perturbed wage w exceeds what (ii)-switchers earned in A but falls short of what (i)-switchers earned in B . Thus, an increase in indirect utility for workers earning w results in some workers switching to lower-wage jobs and others to higher-wage jobs. This implies that the sign of the net fiscal externality $S(w)$ is indeterminate. However, I show next that the integral $\Xi^t(w) = \int_w^{\bar{w}} S(z) dz$, which enters the optimal tax formula, can be signed.

Rearranging (A1),

$$\eta'(w) = \frac{1}{\mu} [W'(V^A(w))h_A(w) + W'(V^A(w) - d)h_B(w)] - h(w) + S(w)$$

Integrating,

$$\eta(w) = \int_w^{\bar{w}} h(z) - \frac{1}{\mu} [W'(V^A(z))h_A(z) + W'(V^A(z) - d)h_B(z)] - S(z) dz$$

Equation (A2) shows that a small increase in $V^A(w)$ induces occupation switches with opposing fiscal implications. Integrating from w to $\bar{w}$,

$$\Xi^t(w) = \int_w^{\bar{w}} S(z) dz = \int_w^{\bar{w}} -\Delta T(\check{\theta}_B(z, d), d) \frac{\partial h_A(z)}{\partial V^A(z)} + \Delta T(z, d) \frac{\partial h_B(z)}{\partial V^A(z)} dz \quad (\text{A3})$$

Two identities simplify $\Xi^t(w)$. First, raising $V^A(z)$ and lowering $V^A(\check{\theta}_B(z, d))$ generate identical flows of workers into occupation A at wage z : since $V^B = V^A - d$, both changes improve A 's attractiveness relative

to B at the margin between wage z in A and $\check{\theta}_B(z, d)$ in B . Second, workers who switch into A at z must exit B at $\check{\theta}_B(z, d)$:

$$\frac{\partial h_A(z)}{\partial V^A(z)} = -\frac{\partial h_A(z)}{\partial V^A(\check{\theta}_B(z, d))} = \frac{\partial h_B(\check{\theta}_B(z, d))}{\partial V^A(\check{\theta}_B(z, d))}$$

For each $z \in [w, \bar{w}]$, the fiscal externality from $B \rightarrow A$ switching at z — induced by $\partial V^A(z)$ — is exactly offset by the $A \rightarrow B$ switching externality at $\check{\theta}_B(z, d)$ induced by the higher perturbation $\partial V^A(\check{\theta}_B(z, d))$. The same marginal workers are involved in both switches, and by the identities above, the fiscal effects are equal and opposite.

$$-\Delta T(\check{\theta}_B(z, d), d) \frac{\partial h_A(z)}{\partial V^A(z)} + \Delta T(\check{\theta}_B(z, d), d) \frac{\partial h_B(\check{\theta}_B(z, d))}{\partial V^A(\check{\theta}_B(z, d))} = \Delta T(\check{\theta}_B(z, d), d) \left(\frac{\partial h_B(\check{\theta}_B(z, d))}{\partial V^A(\check{\theta}_B(z, d))} - \frac{\partial h_A(z)}{\partial V^A(z)} \right) = 0$$

In $\Xi^t(w)$, these terms cancel. Changing variables from z to θ_B , (A3) reduces to

$$\begin{aligned} \Xi^t(w) &= \int_w^{\check{\theta}_B(w, d)} \Delta T(\theta_B, d) \frac{\partial h_B(\theta_B)}{\partial V^A(\theta_B)} d\theta_B \\ &= \int_w^{\check{\theta}_B(w, d)} \Delta T(\theta_B, d) f(\check{\theta}_A(\theta_B, d), \theta_B) \frac{\check{\theta}_A(\theta_B, d)}{l(\check{\theta}_A(\theta_B, d))^{1+\frac{1}{\varepsilon}}} d\theta_B \end{aligned} \quad (\text{A4})$$

where the second line uses two facts: the endogenous wage density (10) implies (A5),

$$\frac{\partial h_B(\theta_B)}{\partial V^A(\theta_B)} = f(\check{\theta}_A(\theta_B, d), \theta_B) \frac{\partial \check{\theta}_A(\theta_B, d)}{\partial V^A(\theta_B)}, \quad (\text{A5})$$

and the threshold function (9) together with the local incentive constraint (12) imply (A6):

$$\frac{\partial \check{\theta}_A(\theta_B, d)}{\partial V^A(\theta_B)} = (V^{A'}(\check{\theta}_A(\theta_B, d)))^{-1} = \frac{\check{\theta}_A(\theta_B, d)}{l(\check{\theta}_A(\theta_B, d))^{1+\frac{1}{\varepsilon}}}. \quad (\text{A6})$$

Equation (A4) shows that the occupation switching fiscal externality entering the optimal tax formula is strictly positive when $\gamma(d) = 0$, as $\Delta T(\theta_B, d) = T(y(\theta_B)) - T(y(\check{\theta}_A(\theta_B, d))) > 0$. That is, a small tax cut at w generates a positive fiscal externality since marginal workers switch from earning a wage below w in A to one above w in B . When $\gamma(d) > 0$, $\Xi^t(w)$ is no longer guaranteed to be positive, since $\gamma(d)$ may be large enough to flip the sign of $\Delta T(\theta_B, d)$.

The intuition for the limits of integration in (A4) lies in whether a worker is inframarginal to the tax change in one occupation or both. Specifically, workers with $\theta_B \in [w, \check{\theta}_B(w, d)]$ have income $y(w)$ falling between their A -income and their B -income, so a tax change at w alters the relative return to switching occupations, as represented by the dashed line in Figure 1.

In contrast, workers on the indifference locus with B -wages exceeding $\check{\theta}_B(w, d)$ have skills exceeding w and thus incomes exceeding $y(w)$ in both occupations. The tax change at w therefore affects both options equally, leaving the relative return to switching — and hence their occupation choice — unchanged.

The first-order condition with respect to $l(w)$ is

$$\mu h(w) \left(w - l^{1/\varepsilon} \right) = \mu \eta(w) \frac{l^{1/\varepsilon}}{w} \left(1 + \frac{1}{\varepsilon} \right)$$

where $\frac{\partial c}{\partial l} = MRS_{lc} = l^{1/\varepsilon}$. Since $MRS_{lc}/w = 1 - T'(y(w))$, this becomes

$$\frac{T'(y(w))}{1 - T'(y(w))} = \frac{1 + \frac{1}{\varepsilon}}{wh(w)}\eta(w)$$

The first-order condition with respect to d is

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial d} = & - \int_{\underline{w}}^{\bar{w}} (W'(V^A(\theta_B) - d) + \mu\gamma'(d)) h_B(\theta_B) d\theta_B \\ & + \mu \int_{\underline{w}}^{\bar{w}} (\theta_B l(\theta_B) - c(V^A(\theta_B), l(\theta_B))) \frac{\partial h(\theta_B)}{\partial d} d\theta_B - \mu \int_{\underline{w}}^{\bar{w}} \gamma(d) \frac{\partial h_B(\theta_B)}{\partial d} d\theta_B = 0 \end{aligned} \quad (\text{A7})$$

By (9),

$$\frac{\partial h_A(\check{\theta}_A(w, d))}{\partial d} = - \frac{\partial h_B(w)}{\partial d} = \frac{\partial h_B(w)}{\partial V^A(w)}.$$

Using the notation $\mathbb{E}_B[\cdot]$ from Proposition 1 and (A5)–(A6), (A7) simplifies to

$$\begin{aligned} -\gamma'(d) = & \frac{1}{\mu} \mathbb{E}_B [W'(V^A(\theta_B) - d)] + \frac{\int_{\underline{w}}^{\bar{w}} \Delta T(\theta_B, d) \frac{\partial h_B(\theta_B)}{\partial V^A(\theta_B)} d\theta_B}{\int_{\underline{w}}^{\bar{w}} h_B(\theta_B) d\theta_B} \\ = & \frac{1}{\mu} \mathbb{E}_B [W'(V^A(\theta_B) - d)] + \frac{\int_{\underline{w}}^{\bar{w}} \Delta T(\theta_B, d) f(\check{\theta}_A(\theta_B, d), \theta_B) \frac{\check{\theta}_A(\theta_B, d)}{l(\check{\theta}_A(\theta_B, d))^{1+\frac{1}{\varepsilon}}} d\theta_B}{\int_{\underline{w}}^{\bar{w}} h_B(\theta_B) d\theta_B} \\ = & \frac{1}{\mu} \mathbb{E}_B [W'(V^A(\theta_B) - d)] + \mathbb{E}_B \left[\Delta T(\theta_B, d) \frac{f(\check{\theta}_A(\theta_B, d), \theta_B)}{h_B(\theta_B)} \frac{\check{\theta}_A(\theta_B, d)}{l(\check{\theta}_A(\theta_B, d))^{1+\frac{1}{\varepsilon}}} \right]. \end{aligned} \quad (\text{A8})$$

The transversality conditions $\eta(\bar{w}) = \eta(\underline{w}) = 0$ imply

$$\int_{\underline{w}}^{\bar{w}} \eta(w) dw = \int_{\underline{w}}^{\bar{w}} \left(h(z) - \frac{1}{\mu} [W'(V^A(z))h^A(z) + W'(V^A(z) - d)h^B(z)] \right) dz - \Xi(\underline{w}) = 0.$$

$\Xi(\underline{w}) = 0$ because a tax perturbation at the lowest wage does not induce any occupation switching: switchers would need A -skill $\check{\theta}_A(\theta_B, d) < \underline{w}$, but no skill lies below $\underline{w}$. Thus,

$$\mu = \int_{\underline{w}}^{\bar{w}} W'(V^A(z))f^A(z) + W'(V^A(z) - d)f^B(z) dz$$

is the average social marginal welfare weight.

A.2 Proposition 2

With occupation-specific taxes, workers' occupation choice is given by $o^*(\theta_A, \theta_B, d) = \arg \max_{o \in \{A, B\}} V^o(\theta_o, d)$, where

$$V^A(\theta_A) = \max_y y - T^A(y) - \frac{(y/\theta_A)^{1+\frac{1}{\varepsilon}}}{1 + \frac{1}{\varepsilon}}, \quad V^B(\theta_B, d) = \max_y y - T^B(y) - \frac{(y/\theta_B)^{1+\frac{1}{\varepsilon}}}{1 + \frac{1}{\varepsilon}} - d. \quad (\text{A9})$$

Threshold functions are now defined by

$$V^A(\theta_A) = V^B(\check{\theta}_B(\theta_A, d))$$

Let $\tilde{V}^B(\theta_B) = V^B(\theta_B) + d$. The planner's problem is

$$\max_{\substack{l^A(\theta_A), V^A(\theta_A), \\ l^B(\theta_B), \tilde{V}^B(\theta_B), d}} \int_{\underline{w}}^{\bar{w}} W(V^A(\theta_A)) h_A(\theta_A) d\theta_A + \int_{\underline{w}}^{\bar{w}} W(\tilde{V}^B(\theta_B) - d) h_B(\theta_B) d\theta_B \quad (\text{A10})$$

subject to occupation-specific local incentive constraints,

$$\frac{\partial V^A(\theta_A)}{\partial \theta_A} - \frac{l^A(\theta_A)^{1+1/\varepsilon}}{\theta_A} = 0 \quad \forall \theta_A \in [\underline{w}, \bar{w}] \quad (\text{A11})$$

$$\frac{\partial \tilde{V}^B(\theta_B)}{\partial \theta_B} - \frac{l^B(\theta_B)^{1+1/\varepsilon}}{\theta_B} = 0 \quad \forall \theta_B \in [\underline{w}, \bar{w}] \quad (\text{A12})$$

and the resource constraint

$$\begin{aligned} \int_{\underline{w}}^{\bar{w}} [\theta_A l^A(\theta_A) - c_A(V^A(\theta_A), l^A(\theta_A))] h_A(\theta_A) d\theta_A \\ + \int_{\underline{w}}^{\bar{w}} [\theta_B l^B(\theta_B) - c_B(\tilde{V}^B(\theta_B), l^B(\theta_B)) - \gamma(d)] h_B(\theta_B) d\theta_B \geq 0. \end{aligned} \quad (\text{A13})$$

For each occupation $V^o \in \{V^A, \tilde{V}^B\}$, applying Lagrange multiplier $\eta^o \mu$ to the local incentive constraint for occupation o and μ to the resource constraint,

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial V^o(w)} &= W'(V^o(w) - \mathbb{1}(o=B)d) h_o(w) - \eta^{o'}(w) \mu - \mu h_o(w) + \mu S^o(w) = 0 \\ \frac{\partial \mathcal{L}}{\partial l^o(w)} &= \mu h_o(w) \left(w - l^{1/\varepsilon} \right) + \mu \eta^o(w) \frac{l^{1/\varepsilon}}{w} \left(1 + \frac{1}{\varepsilon} \right) = 0 \end{aligned}$$

where, using the revised wedge $\Delta T(\theta_B, d) = T^B(y^B(\theta_B)) - \gamma(d) - T^A(y^A(\check{\theta}_A(\theta_B, d)))$,

$$S^A(\theta_A) = -\Delta T(\check{\theta}_B(\theta_A, d), d) \frac{\partial h_A(\theta_A)}{\partial V^A(\theta_A)}, \quad S^B(\theta_B) = \Delta T(\theta_B, d) \frac{\partial h_B(\theta_B)}{\partial \tilde{V}^B(\theta_B)} = \Delta T(\theta_B, d) \frac{\partial h_B(\theta_B)}{\partial V^B(\theta_B)}.$$

Following the same steps as in Appendix A.1, and using

$$\begin{aligned} \frac{\partial h_A(\theta_A)}{\partial V^A(\theta_A)} &= f(\theta_A, \check{\theta}_B(\theta_A, d)) \frac{\check{\theta}_B(\theta_A, d)}{l^B(\check{\theta}_B(\theta_A, d))^{1+\frac{1}{\varepsilon}}} \\ \frac{\partial h_B(\theta_B)}{\partial V^B(\theta_B)} &= f(\check{\theta}_A(\theta_B, d), \theta_B) \frac{\check{\theta}_A(\theta_B, d)}{l^A(\check{\theta}_A(\theta_B, d))^{1+\frac{1}{\varepsilon}}}, \end{aligned}$$

we obtain the optimal tax formulas

$$\begin{aligned}\frac{T^{A'}(y^A(w))}{1 - T^{A'}(y^A(w))} &= \frac{1 + \frac{1}{\varepsilon}}{wh_A(w)} \int_w^{\bar{w}} \left[h_A(\theta_A) - \frac{1}{\mu} W'(V^A(\theta_A)) h_A(\theta_A) \right. \\ &\quad \left. + \Delta T(\check{\theta}_B(\theta_A, d), d) f(\theta_A, \check{\theta}_B(\theta_A, d)) \frac{\check{\theta}_B(\theta_A, d)}{l^B(\check{\theta}_B(\theta_A, d))^{1+\frac{1}{\varepsilon}}} \right] d\theta_A, \\ \frac{T^{B'}(y^B(w))}{1 - T^{B'}(y^B(w))} &= \frac{1 + \frac{1}{\varepsilon}}{wh_B(w)} \int_w^{\bar{w}} \left[h_B(\theta_B) - \frac{1}{\mu} W'(V^B(\theta_B)) h_B(\theta_B) \right. \\ &\quad \left. - \Delta T(\theta_B, d) f(\check{\theta}_A(\theta_B, d), \theta_B) \frac{\check{\theta}_A(\theta_B, d)}{l^A(\check{\theta}_A(\theta_B, d))^{1+\frac{1}{\varepsilon}}} \right] d\theta_B.\end{aligned}$$

The first-order condition with respect to d is

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial d} &= - \int_{\underline{w}}^{\bar{w}} \left(W'(\tilde{V}^B(\theta_B) - d) + \mu \gamma'(d) \right) h_B(\theta_B) d\theta_B \\ &\quad + \mu \int_{\underline{w}}^{\bar{w}} \left(\theta_A l^A(\theta_A) - c_A(V^A(\theta_A), l^A(\theta_A)) \right) \frac{\partial h_A(\theta_A)}{\partial d} d\theta_A \\ &\quad + \mu \int_{\underline{w}}^{\bar{w}} \left(\theta_B l^B(\theta_B) - c_B(\tilde{V}^B(\theta_B), l^B(\theta_B)) - \gamma(d) \right) \frac{\partial h_B(\theta_B)}{\partial d} d\theta_B = 0\end{aligned}$$

Using $\frac{\partial h_B(\theta_B)}{\partial d} = -\frac{\partial h_B(\theta_B)}{\partial V^B(\theta_B)}$ from (A.2), the result mirrors (A8) but with the revised wedge $\Delta T(\theta_B, d)$ reflecting the occupation-specific schedules:

$$-\gamma'(d) = \frac{1}{\mu} \mathbb{E}_B [W'(V^B(\theta_B))] + \frac{\int_{\underline{w}}^{\bar{w}} \Delta T(\theta_B, d) f(\check{\theta}_A(\theta_B, d), \theta_B) \frac{\check{\theta}_A(\theta_B, d)}{l^A(\check{\theta}_A(\theta_B, d))^{1+\frac{1}{\varepsilon}}} d\theta_B}{H^B(\bar{w})} \quad (\text{A14})$$

From the transversality condition $\int_{\underline{w}}^{\bar{w}} \eta^{B'}(\theta_B) d\theta_B = \eta^B(\bar{w}) - \eta^B(\underline{w}) = 0$,

$$\begin{aligned}\eta^B(\underline{w}) &= \int_{\underline{w}}^{\bar{w}} h_B(\theta_B) - \frac{1}{\mu} W'(V^B(\theta_B)) h_B(\theta_B) - S^B(\theta_B) d\theta_B \\ &= H^B(\bar{w}) \left[1 - \frac{1}{\mu} \mathbb{E}_B [W'(V^B(\theta_B))] - \frac{\int_{\underline{w}}^{\bar{w}} \Delta T(\theta_B, d) f(\check{\theta}_A(\theta_B, d), \theta_B) \frac{\check{\theta}_A(\theta_B, d)}{l^A(\check{\theta}_A(\theta_B, d))^{1+\frac{1}{\varepsilon}}} d\theta_B}{H^B(\bar{w})} \right] = 0\end{aligned} \quad (\text{A15})$$

Substituting (A15) into (A14),

$$\left(-\gamma'(d) - \frac{1}{\mu} \mathbb{E}_B [W'(V^B(\theta_B))] \right) - \left(1 - \frac{1}{\mu} \mathbb{E}_B [W'(V^B(\theta_B))] \right) = 0$$

or

$$-\gamma'(d) = 1.$$

A.3 Corollary 2

Suppose that workers can individually purchase job quality improvements at cost $\gamma(d)$. The government has access to occupation-specific taxes and subsidizes job quality purchases at rate s . The occupational choice rule (3, 4) is modified to reflect this additional choice of d : $o^*(\theta_A, \theta_B, d) = \arg \max_{o \in \{A, B\}} V^o(\theta_o, d)$. Substituting the budget constraint, indirect utilities are

$$V^A(\theta_A) = \max_y y - T^A(y) - \frac{(y/\theta_A)^{1+1/\varepsilon}}{1+1/\varepsilon}, \quad V^B(\theta_B, d) = \max_{y,d} y - T^B(y) - \frac{\gamma(d)}{1+s} - \frac{(y/\theta_B)^{1+1/\varepsilon}}{1+1/\varepsilon} - d. \quad (\text{A16})$$

Conditional on working in B , a worker's first-order condition with respect to d is

$$-\gamma'(d) = 1 + s \quad (\text{A17})$$

The planner's problem is again to maximize the objective in (A10) subject to local incentive constraints (A12) and the resource constraint (A13). The resource constraint is unchanged because the subsidy s is a transfer from the government to the worker: to achieve disamenity level d , the worker pays $\gamma(d)/(1+s)$ while the government pays $s\gamma(d)/(1+s)$, but in aggregate these sum to $\gamma(d)$. Thus, the first-order condition with respect to d is

$$-\gamma'(d) = 1. \quad (\text{A18})$$

Together with (A17), this implies that the optimal subsidy on job quality improvements is zero when the government can condition taxes on occupation and workers with quasilinear utility can purchase amenity improvements on their own.

A.4 Proposition 3

Building on Section 3.2, I describe properties of the pairwise threshold functions $\check{\theta}_{j\ell}(\cdot)$ for all $j \neq \ell$, generalizing $\check{\theta}_A(\cdot, d), \check{\theta}_B(\cdot, d)$ from the two-occupation case.

First, threshold functions are defined by indifference: $\theta_\ell = \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell) \iff u^\ell(\theta_\ell) = u^\ell(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) = u^j(\theta_j)$. Therefore, they implicitly depend on the income tax schedule. Second, each threshold function $\check{\theta}_{j\ell}(\cdot)$ is strictly increasing: when a worker is more skilled at occupation ℓ , they must also be more skilled at occupation j to remain indifferent between the two. Third, for any two distinct occupations $j \neq \ell$, the functions $\check{\theta}_{j\ell}(\cdot, \psi_j, \psi_\ell)$ and $\check{\theta}_{\ell j}(\cdot, \psi_\ell, \psi_j)$ are inverses of one another: for any θ_j ,

$$\check{\theta}_{\ell j}(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell), \psi_\ell, \psi_j) = \theta_j. \quad (\text{A19})$$

These properties yield (28)–(31). I generalize Proposition 1 to many occupations.

The Lagrangian is

$$\begin{aligned} \mathcal{L} = & \int_{\underline{w}}^{\bar{w}} \sum_{j=1}^J W(V(w) + \bar{\psi}_j + \chi_{j|k}) h_j(w) dw \\ & + \mu \int_{\underline{w}}^{\bar{w}} \sum_{j=1}^J h_j(w) [wl(w) - c(V(w), l(w)) - \gamma_j(\bar{\psi}_j) - \gamma_{\chi|k}(\chi) \mathbb{1}[j \leq k]] dw \\ & - \mu \int_{\underline{w}}^{\bar{w}} V(w) \eta'(w) dw - \mu \int_{\underline{w}}^{\bar{w}} \frac{l(w)^{1+1/\varepsilon}}{w} \eta(w) dw \end{aligned} \quad (\text{A20})$$

where $V(w)$ is the solution to (26) with $\psi_j = 0$.

The first-order condition with respect to $V(w)$ is

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial V(w)} &= \sum_{j=1}^J \left(W'(V(w) + \psi_j) - \mu \right) h_j(w) - \eta'(w) \mu \\ &+ \mu \int_{\underline{w}}^{\bar{w}} \sum_{j=1}^J \left(T(y(w')) - \gamma_j(\bar{\psi}_j) - \gamma_{\chi|k}(\chi) \mathbb{1}[j \leq k] \right) \frac{\partial h_j(w')}{\partial V(w)} dw' \end{aligned} \quad (\text{A21})$$

Only marginal workers earning w in one occupation switch in response to a change in $V(w)$.

$$\frac{\partial h_j(w')}{\partial V(w)} = 0 \quad \forall j \in \{1, \dots, J\}, \quad w' \in [\underline{w}, \bar{w}] \setminus \left\{ w, (\check{\theta}_{\ell j}(w, \psi_\ell, \psi_j))_{\ell \neq j} \right\} \quad (\text{A22})$$

This implies that

$$\frac{\partial h_j(w)}{\partial V(w)} = - \sum_{\ell \neq j} \frac{\partial h_\ell(\check{\theta}_{\ell j}(w, \psi_j, \psi_\ell))}{\partial V(w)} \quad (\text{A23})$$

I rewrite the occupation-switching fiscal externality in (A21) as

$$\int_{\underline{w}}^{\bar{w}} \sum_{j=1}^J \left(T(y(w')) - \gamma_j(\bar{\psi}_j) - \gamma_{\chi|k}(\chi) \mathbb{1}[j \leq k] \right) \frac{\partial h_j(w')}{\partial V(w)} dw' = - \sum_{j=1}^J \sum_{\ell \neq j} \Delta T_{j\ell}(w) \frac{\partial h_\ell(\check{\theta}_{\ell j}(w, \psi_j, \psi_\ell))}{\partial V(w)} = S(w) \quad (\text{A24})$$

where

$$\Delta T_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell, \chi) = (T(y(\theta_j)) - \gamma_j(\bar{\psi}_j) - \gamma_{\chi|k}(\chi) \mathbb{1}[j \leq k]) - (T(y(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))) - \gamma_\ell(\bar{\psi}_\ell) - \gamma_{\chi|k}(\chi) \mathbb{1}[\ell \leq k])$$

is the fiscal externality from a worker switching from earning $\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)$ in ℓ to θ_j in j . I specialize this to the coarse and granular predistribution scenarios later.

Since occupations are ordered from least amenable ($j = 1$) to most ($j = J$), these fiscal externalities can be signed in the absence of predistributive policy. When $\gamma_j(\psi_j) = 0 \quad \forall j \in \{1, \dots, J\}$ and $\gamma_{\chi|k}(\chi) = 0$, $\ell < j \iff \theta_\ell > \check{\theta}_{\ell j}(\theta_\ell, \psi_\ell, \psi_j) \iff \Delta T_{\ell j}(\theta_\ell) > 0$. When ℓ is more disamenable than j , one must earn a higher wage in ℓ than j , and the occupation-switching fiscal externality is positive.

Following the standard steps, the optimal occupation-blind tax is the same as Proposition 1 but with a redefinition of the occupation-switching fiscal externality to (A24).

$$\eta(w) = \int_{\underline{w}}^{\bar{w}} \sum_{j=1}^J h_j(z) \left(1 - \frac{1}{\mu} W'(V(z) + \psi_j) \right) - S(z) dz. \quad (\text{A25})$$

As before, the occupation-switching fiscal externality can be signed in the integral to the upper bound, implying less progressive taxation in the absence of predistribution. The goal is to sign

$$\Xi^t(w) = \int_w^{\bar{w}} S(z)dz = - \int_w^{\bar{w}} \sum_{j=1}^J \sum_{\ell \neq j} \Delta T_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell, \chi) \frac{\partial h_\ell(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))}{\partial V(\theta_j)} d\theta_j \quad (\text{A26})$$

$$= - \underbrace{\int_w^{\bar{w}} \sum_{j=1}^J \sum_{\ell=1}^{j-1} \Delta T_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell, \chi) \frac{\partial h_\ell(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))}{\partial V(\theta_j)} d\theta_j}_{(i)} \quad (\text{A27})$$

$$- \int_w^{\bar{w}} \sum_{j=1}^J \sum_{\ell=j+1}^J \Delta T_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell, \chi) \frac{\partial h_\ell(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))}{\partial V(\theta_j)} d\theta_j \quad (\text{A27})$$

$$= - \underbrace{\sum_{j=1}^J \sum_{\ell=j+1}^J \int_{\check{\theta}_{\ell j}(w, \psi_\ell, \psi_j)}^{\bar{w}} \Delta T_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell, \chi) \frac{\partial h_\ell(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))}{\partial V(\theta_j)} d\theta_j}_{(ii)} \quad (\text{A28})$$

$$- \sum_{j=1}^J \sum_{\ell=j+1}^J \int_w^{\check{\theta}_{\ell j}(w, \psi_\ell, \psi_j)} \Delta T_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell, \chi) \frac{\partial h_\ell(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))}{\partial V(\theta_j)} d\theta_j \quad (\text{A28})$$

This uses the fact that marginal workers who shift away from ℓ into j due to an increase in $V(\theta_j)$ are those who shift into ℓ from j due to an increase in $V(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))$.

$$\frac{\partial h_\ell(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))}{\partial V(\theta_j)} = - \frac{\partial h_\ell(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))}{\partial V(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))} = \frac{\partial h_j(\theta_j)}{\partial V(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))} \quad (\text{A29})$$

From the definition of $\Delta T_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell, \chi)$,

$$\Delta T_{j\ell}(\check{\theta}_{\ell j}(\theta_j, \psi_\ell, \psi_j), \bar{\psi}_j, \bar{\psi}_\ell, \chi) = -\Delta T_{\ell j}(\theta_j, \bar{\psi}_\ell, \bar{\psi}_j, \chi) \quad (\text{A30})$$

Thus

$$\Delta T_{\ell j}(\theta_j, \bar{\psi}_\ell, \bar{\psi}_j, \chi) \frac{\partial h_j(\check{\theta}_{\ell j}(\theta_j, \psi_\ell, \psi_j))}{\partial V(\theta_j)} = -\Delta T_{j\ell}(\check{\theta}_{\ell j}(\theta_j, \psi_\ell, \psi_j), \bar{\psi}_j, \bar{\psi}_\ell, \chi) \frac{\partial h_\ell(\theta_j)}{\partial V(\check{\theta}_{\ell j}(\theta_j, \psi_\ell, \psi_j))} \quad (\text{A31})$$

Fix occupation j . Consider jobs $\ell < j$, which are more disamenable than j . Given wages in job j at $\theta_j \in [w, \bar{w}]$, indifference between ℓ and j requires higher wages in ℓ : $\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell) > \theta_j \geq w$. For every such j and θ_j , there is a cancelling term of opposite sign in the integral: the incentive for a worker to switch from ℓ to j due to a marginal increase in $V(\theta_j)$, and from j to ℓ due to a marginal increase in $V(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))$, nets out. Therefore, term (i) in (A27) drops out.

Now consider jobs $\ell > j$, which are more amenable than j . Given wages in job j at $\theta_j \in [w, \bar{w}]$, indifference requires lower wages in ℓ . Cancelling ℓ terms can therefore only be found for j -wages $\theta_j \in [\check{\theta}_{\ell j}(w, \psi_\ell, \psi_j), \bar{w}]$, which ensures ℓ -wages fall within $[w, \bar{w}]$. By the same logic, term (ii) in (A28) drops out.

After cancelling all such terms,

$$\Xi^t(w) = \int_w^{\bar{w}} S(z)dz = - \sum_{j=1}^J \sum_{\ell=j+1}^J \int_w^{\check{\theta}_{\ell j}(w, \psi_\ell, \psi_j)} \Delta T_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell, \chi) \frac{\partial h_\ell(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))}{\partial V(\theta_j)} d\theta_j > 0 \quad (\text{A32})$$

This term is positive in the absence of predistribution since $\frac{\partial h_\ell(\check{\theta}_{j\ell}(w, \psi_j, \psi_\ell))}{\partial V(w)} < 0$ and, when $\gamma_j(\bar{\psi}_j) = 0 \forall j \in$

$\{1, \dots, J\}$ and $\gamma_{\chi|k}(\chi) = 0$, $\Delta T_{j\ell}(\check{\theta}_{\ell j}(\theta_j, \psi_\ell, \psi_j), \bar{\psi}_j, \bar{\psi}_\ell, \chi) > 0 \forall \ell > j$. I simplify $\frac{\partial h_\ell(\check{\theta}_{j\ell}(w, \psi_j, \psi_\ell))}{\partial V(w)}$ further to obtain a more explicit expression. The density of workers earning θ_j in j is

$$h_j(\theta_j) = \int_{\underline{w}}^{\check{\theta}_{j1}(\theta_j, \psi_j, \psi_1)} \int_{\underline{w}}^{\check{\theta}_{j2}(\theta_j, \psi_j, \psi_2)} \dots \int_{\underline{w}}^{\check{\theta}_{jJ}(\theta_j, \psi_j, \psi_J)} f(\theta_{-j}, \theta_j) d\theta_{-j} \quad (\text{A33})$$

Thus,

$$\begin{aligned} \frac{\partial h_j(\theta_j)}{\partial V(\theta_j)} &= \int_{\underline{\theta}_j}^{\check{\theta}_{j2}(\theta_j, \psi_j, \psi_2)} \dots \int_{\underline{\theta}_j}^{\check{\theta}_{jJ}(\theta_j, \psi_j, \psi_J)} f(\theta_{-(1,j)}, \check{\theta}_{j1}(\theta_j, \psi_j, \psi_1), \theta_j) d\theta_{-(1,j)} \frac{\partial \check{\theta}_{j1}(\theta_j, \psi_j, \psi_1)}{\partial V(\theta_j)} \\ &+ \int_{\underline{\theta}_j}^{\check{\theta}_{j1}(\theta_j, \psi_j, \psi_1)} \dots \int_{\underline{\theta}_j}^{\check{\theta}_{jJ}(\theta_j, \psi_j, \psi_J)} f(\theta_{-(2,j)}, \check{\theta}_{j2}(\theta_j, \psi_j, \psi_2), \theta_j) d\theta_{-(2,j)} \frac{\partial \check{\theta}_{j2}(\theta_j, \psi_j, \psi_2)}{\partial V(\theta_j)} + \dots \\ &+ \int_{\underline{\theta}_j}^{\check{\theta}_{j1}(\theta_j, \psi_j, \psi_1)} \dots \int_{\underline{\theta}_j}^{\check{\theta}_{jJ-1}(\theta_j, \psi_j, \psi_{J-1})} f(\theta_{-(j,J)}, \check{\theta}_{jJ}(\theta_j, \psi_j, \psi_J), \theta_j) d\theta_{-(j,J)} \frac{\partial \check{\theta}_{jJ}(\theta_j, \psi_j, \psi_J)}{\partial V(\theta_j)} \\ &= \sum_{\ell \neq j} f_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) \end{aligned} \quad (\text{A34})$$

$$\cdot F_{-(j,\ell)|(j,\ell)}((\check{\theta}_{ji}(\theta_j, \psi_j, \psi_i))_{i \neq j, i \neq \ell} \mid \theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) \frac{\partial \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{\partial V(\theta_j)} \quad (\text{A35})$$

$$= \sum_{\ell \neq j} h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) \frac{\partial \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{\partial V(\theta_j)} \quad (\text{A36})$$

Equivalently,

$$\frac{\partial h_\ell(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))}{\partial V(\theta_j)} = -h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) \frac{\partial \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{\partial V(\theta_j)} \quad (\text{A37})$$

$$= -h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) \frac{\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{l(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))^{1+1/\varepsilon}} \quad (\text{A38})$$

where the second line uses the local incentive constraint. Thus, I rewrite the occupation-switching fiscal externality in (A32) as

$$\Xi^t(w) = \sum_{j=1}^J \sum_{\ell=j+1}^J \int_w^{\check{\theta}_{\ell j}(w, \psi_\ell, \psi_j)} \Delta T_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell, \chi) h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) \frac{\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{l(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))^{1+1/\varepsilon}} d\theta_j > 0 \quad (\text{A39})$$

Intuitively, only workers who are indifferent between occupations *and* experience a relative income change (due to earning more than w in one occupation and less than w in the other) will switch occupations.

I first solve for optimal granular predistribution. Setting $\chi = 0$, $\gamma_{\chi|k}(\chi) = 0$, $\psi_j = \bar{\psi}_j$ and differentiating (A20) with respect to $\bar{\psi}_j$,

$$\frac{\partial \mathcal{L}}{\partial \bar{\psi}_j} = \int_{\underline{w}}^{\bar{w}} W'(V(\theta_j) + \bar{\psi}_j) h_j(\theta_j) - \mu h_j(\theta_j) \gamma'_j(\bar{\psi}_j) + \mu \sum_{\ell=1}^J (T(y(\theta_j)) - \gamma_\ell(\bar{\psi}_\ell)) \frac{\partial h_\ell(\theta_j)}{\partial \bar{\psi}_j} d\theta_j = 0. \quad (\text{A40})$$

Since

$$\frac{\partial h_j(\theta_j)}{\partial \bar{\psi}_j} = - \sum_{\ell \neq j} \frac{\partial h_\ell(\check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell))}{\partial \bar{\psi}_j} = \sum_{\ell \neq j} \frac{\partial h_\ell(\check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell))}{\partial V(\check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell))}, \quad (\text{A41})$$

and the threshold rule and (A38) imply that

$$\frac{\partial h_\ell(\check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell))}{\partial \bar{\psi}_j} = \frac{\partial h_\ell(\check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell))}{\partial V(\theta_j)} = h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell)) \frac{\check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell)}{l(\check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell))^{1+1/\varepsilon}}, \quad (\text{A42})$$

I rearrange (A40) to

$$\begin{aligned} \gamma'_j(\bar{\psi}_j) &= \frac{1}{\mu} \frac{\int_{\underline{w}}^{\bar{w}} W'(V(\theta_j) + \bar{\psi}_j) h_j(\theta_j) d\theta_j}{\int_{\underline{w}}^{\bar{w}} h_j(\theta_j) d\theta_j} \\ &\quad + \frac{1}{\int_{\underline{w}}^{\bar{w}} h_j(\theta_j) d\theta_j} \int_{\underline{w}}^{\bar{w}} \sum_{\ell \neq j} ((T(y(\theta_j)) - \gamma_j(\bar{\psi}_j)) - (T(y(\check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell))) - \gamma_\ell(\bar{\psi}_\ell))) \frac{\partial h_\ell(\check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell))}{\partial \bar{\psi}_j} d\theta_j \end{aligned} \quad (\text{A43})$$

$$\begin{aligned} &= \frac{1}{\mu} \frac{\int_{\underline{w}}^{\bar{w}} W'(V(\theta_j) + \bar{\psi}_j) h_j(\theta_j) d\theta_j}{\int_{\underline{w}}^{\bar{w}} h_j(\theta_j) d\theta_j} \\ &\quad + \frac{\int_{\underline{w}}^{\bar{w}} \sum_{\ell \neq j} \Delta T_{j\ell}^{\bar{\psi}}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell) h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell)) \frac{\check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell)}{l(\check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell))^{1+1/\varepsilon}} d\theta_j}{\int_{\underline{w}}^{\bar{w}} h_j(\theta_j) d\theta_j} \end{aligned} \quad (\text{A44})$$

$$= \frac{1}{\mu} \mathbb{E}_j [W'(V(\theta_j) + \bar{\psi}_j)] + \mathbb{E}_j \left[\sum_{\ell \neq j} \Delta T_{j\ell}^{\bar{\psi}}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell) \frac{h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell))}{h_j(\theta_j)} \frac{\check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell)}{l(\check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell))^{1+1/\varepsilon}} \right] \quad (\text{A45})$$

where $\mathbb{E}_j[x(\theta_j, \psi_j)] = \frac{\int_{\underline{w}}^{\bar{w}} x(\theta_j, \psi_j) h_j(\theta_j) d\theta_j}{\int_{\underline{w}}^{\bar{w}} h_j(\theta_j) d\theta_j}$ is the weighted average of x over workers in occupation j and $\Delta T_{j\ell}^{\bar{\psi}}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell) = T(y(\theta_j)) - \gamma_j(\bar{\psi}_j) - T(y(\check{\theta}_{j\ell}(\theta_j, \bar{\psi}_j, \bar{\psi}_\ell))) + \gamma_\ell(\bar{\psi}_\ell)$.

I now turn to coarse predistribution, where the planner uniformly reduces disamenities by χ in the k most disamenable occupations for fixed k and $\bar{\psi}_j$. Setting $\gamma_j(\bar{\psi}_j) = 0$ and differentiating (A20) with respect to χ ,

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \chi} &= \sum_{j=1}^k \int_{\underline{w}}^{\bar{w}} \left[W'(V(\theta_j) + \bar{\psi}_j) h_j(\theta_j) - \mu h_j(\theta_j) \gamma'_{\chi|k}(\chi) \right] d\theta_j \\ &\quad + \mu \int_{\underline{w}}^{\bar{w}} \sum_{j=1}^J (T(y(\theta_j)) - \gamma_{\chi|k}(\chi) \mathbb{1}[j \leq k]) \frac{\partial h_j(\theta_j)}{\partial \chi} d\theta_j = 0. \end{aligned} \quad (\text{A46})$$

By the same argument as (A36),

$$\frac{\partial h_j(\theta_j)}{\partial \chi} = \sum_{\ell \neq j} h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) \frac{\partial \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{\partial \chi} \quad (\text{A47})$$

$$= \sum_{\ell \neq j, \ell > k} h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) \frac{\partial \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{\partial \chi} + \sum_{\ell \neq j, \ell \leq k} h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) \frac{\partial \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{\partial \chi} \quad (\text{A48})$$

$$= \begin{cases} \sum_{\ell > k} h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) \frac{\partial \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{\partial \chi} & \text{if } j \leq k \\ \sum_{\ell \leq k} h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) \frac{\partial \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{\partial \chi} & \text{if } j > k \end{cases} \quad (\text{A49})$$

$$= \begin{cases} \sum_{\ell > k} h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) \frac{\partial \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{\partial V(\check{\theta}_{j\ell}(\theta_j))} & \text{if } j \leq k \\ - \sum_{\ell \leq k} h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) \frac{\partial \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{\partial V(\check{\theta}_{j\ell}(\theta_j))} & \text{if } j > k \end{cases} \quad (\text{A50})$$

where (A49) uses the fact that $\frac{\partial \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{\partial \chi} = 0$ when $j, \ell \leq k$ or $j, \ell > k$ because relative amenity levels are not changing. Alternatively,

$$\frac{\partial h_l(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))}{\partial \chi} = \begin{cases} -h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) \frac{\partial \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{\partial V(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))} & \text{if } j \leq k, \ell > k \\ h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) \frac{\partial \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{\partial V(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))} & \text{if } j > k, \ell \leq k \\ 0 & \text{else} \end{cases} \quad (\text{A51})$$

$$= \begin{cases} -h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) \frac{\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{l(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))^{1+\frac{1}{\varepsilon}}} & \text{if } j \leq k, \ell > k \\ h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) \frac{\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{l(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))^{1+\frac{1}{\varepsilon}}} & \text{if } j > k, \ell \leq k \\ 0 & \text{else} \end{cases} \quad (\text{A52})$$

The fiscal externality term in (A46) simplifies to

$$\int_{\underline{w}}^{\bar{w}} \sum_{\ell=1}^J (T(y(\theta_j)) - \gamma_{\chi|k}(\chi) \mathbb{1}[\ell \leq k]) \frac{\partial h_\ell(\theta_j)}{\partial \chi} d\theta_j = \int_{\underline{w}}^{\bar{w}} \sum_{j=1}^k (T(y(\theta_j)) - \gamma_{\chi|k}(\chi)) \frac{\partial h_j(\theta_j)}{\partial \chi} d\theta_j + \int_{\underline{w}}^{\bar{w}} \sum_{\ell=k+1}^J T(y(\theta_j)) \frac{\partial h_\ell(\theta_j)}{\partial \chi} d\theta_j \quad (\text{A53})$$

$$= \int_{\underline{w}}^{\bar{w}} \sum_{j=1}^k \sum_{\ell=k+1}^J \Delta T_{j\ell}^\chi(\theta_j, \chi) h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)) \cdot \frac{\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{l(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))^{1+\frac{1}{\varepsilon}}} d\theta_j \quad (\text{A54})$$

where $\Delta T_{j\ell}^\chi(\theta_j, \chi) = T(y(\theta_j)) - \gamma_{\chi|k}(\chi) - T(y(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)))$. Thus, (A46) can be rewritten as

$$\gamma'_{\chi|k}(\chi) = \frac{1}{\mu} \mathbb{E}_{j \leq k} [W'(V(\theta_j) + \psi_j)] + \mathbb{E}_{j \leq k} \left[\sum_{\ell > k} \Delta T_{j\ell}^\chi(\theta_j) \frac{h_{(j,\ell)}(\theta_j, \check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))}{h_j(\theta_j)} \frac{\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell)}{l(\check{\theta}_{j\ell}(\theta_j, \psi_j, \psi_\ell))^{1+1/\varepsilon}} \right]. \quad (\text{A55})$$

where $\mathbb{E}_{j \leq k}[x(\theta_j, \psi_j)] = \frac{\sum_{j \leq k} \int_{\underline{w}}^{\bar{w}} x(\theta_j, \psi_j) h_j(\theta_j) d\theta_j}{\sum_{j \leq k} \int_{\underline{w}}^{\bar{w}} h_j(\theta_j) d\theta_j}$ is the weighted average of $x(\theta_j, \psi_j)$ over workers in treated occupations $j \leq k$.

A.5 Proposition 4

Occupation choice follows

$$o^*(\theta_A, \theta_B) = \arg \max_{o \in \{A, B\}} V^o(\theta_o, d_o) \quad (\text{A56})$$

where

$$V^A(\theta_A) = \max_{y, k, d_A} y - T(y) - \frac{k}{1 + \bar{s}} - \frac{\gamma_A(d_A)}{1 + \bar{s}} + \zeta(k) - \frac{(y/\theta_A)^{1+1/\varepsilon}}{1 + 1/\varepsilon} - d_A \quad (\text{A57})$$

$$V^B(\theta_B) = \max_{y, k, d_B} y - T(y) - \frac{k}{1 + \bar{s}} - \frac{\gamma_B(d_B)}{1 + \bar{s} + s^B} + \zeta(k) - \frac{(y/\theta_B)^{1+1/\varepsilon}}{1 + 1/\varepsilon} - d_B \quad (\text{A58})$$

$$= V^A(\theta_B) + d_A^* + \gamma_A(d_A^*) - d_B^* - \gamma_B(d_B^*) \quad (\text{A59})$$

where d_A^* solves the first-order condition to (A57) with respect to d_A ,

$$-\gamma'_A(d_A^*) = 1 + \bar{s},$$

and likewise, d_B^* solves

$$-\gamma'_B(d_B^*) = 1 + \bar{s} + s^B.$$

The threshold function is again defined by indifference:

$$V^A(\theta_A) = V^B(\check{\theta}_B(\theta_A))$$

The incentive constraints do not depend on workers in the two occupations having different amenity cost functions: utility is linear in consumption, and even the lowest-skilled worker consumes the optimally chosen levels of k and d_A , so workers with the same wage face identical consumption-labor tradeoffs regardless of occupation. I therefore suppress the dependence of $c(\cdot)$ on k and d_A .

The planner's problem is to maximize

$$\max_{l(w), V^A(w), d_B} \int_{\underline{w}}^{\bar{w}} W(V^A(w)) h_A(w) + W(V^A(w) + d_A^* + \gamma_A(d_A^*) - d_B - \gamma_B(d_B)) h_B(w) dw \quad (\text{A60})$$

subject to local incentive compatibility constraints

$$V^{A'}(w) - \frac{l(w)^{1+1/\varepsilon}}{w} = 0 \quad \forall w \in [\underline{w}, \bar{w}] \quad (\text{A61})$$

and the resource constraint

$$\int_{\underline{w}}^{\bar{w}} [wl(w) - k^* - c(V^A(w), l(w))] [h_A(w) + h_B(w)] - \gamma_A(d_A) h_A(w) - \gamma_B(d_B) h_B(w) dw \geq 0. \quad (\text{A62})$$

The Lagrangian is

$$\begin{aligned}\mathcal{L} = & \int_{\underline{w}}^{\bar{w}} W(V^A(w))h_A(w) + W(V^A(w) + d_A^* + \gamma_A(d_A^*) - d_B - \gamma_B(d_B))h_B(w)dw \\ & + \mu \int_{\underline{w}}^{\bar{w}} [wl(w) - k^* - c(V^A(w), l(w))] [h_A(w) + h_B(w)] - \gamma_A(d_A^*)h_A(w) - \gamma_B(d_B)h_B(w)dw \\ & - \mu \int_{\underline{w}}^{\bar{w}} V^A(w)\eta'(w)dw - \mu \int_{\underline{w}}^{\bar{w}} \frac{l(w)^{1+1/\varepsilon}}{w}\eta(w)dw\end{aligned}\quad (\text{A63})$$

The first-order condition with respect to d_B is

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial d_B} = & - \int_{\underline{w}}^{\bar{w}} W'(V^B(\theta_B))h_B(\theta_B)d\theta_B - \mu \int_{\underline{w}}^{\bar{w}} \gamma'_B(d_B)h_B(\theta_B)d\theta_B \\ & + \mu \int_{\underline{w}}^{\bar{w}} [wl(w) - k^* - c(V^A(w), l(w)) - \gamma_A(d_A^*)] \frac{\partial h_A(w)}{\partial d_B} dw \\ & + \mu \int_{\underline{w}}^{\bar{w}} [wl(w) - k^* - c(V^A(w), l(w)) - \gamma_B(d_B^*)] \frac{\partial h_B(w)}{\partial d_B} dw = 0\end{aligned}\quad (\text{A64})$$

B -workers' first-order condition gives $-\gamma'_B(d_B) = 1 + \bar{s} + s^B$. Since marginal movers earning w in B must be earning $\check{\theta}_A(w)$ in A , we have $\frac{\partial h_B(w)}{\partial d_B} = -\frac{\partial h_A(\check{\theta}_A(w))}{\partial d_B}$. Substituting these together with (A5)–(A6), the FOC rearranges to

$$1 + \bar{s} + s^B = \frac{\int_{\underline{w}}^{\bar{w}} W'(V^B(\theta_B))h_B(\theta_B)d\theta_B}{\mu H^B(\bar{w})} - \frac{\int_{\underline{w}}^{\bar{w}} \Delta T(\theta_B, d_A, d_B) \frac{\partial h_B(\theta_B)}{\partial d_B} d\theta_B}{H^B(\bar{w})} \quad (\text{A65})$$

$$= \frac{1}{\mu} \mathbb{E}_B [W'(V^B(\theta_B))] + \frac{\int_{\underline{w}}^{\bar{w}} \Delta T(\theta_B, d_A, d_B) \frac{f(\check{\theta}_A(\theta_B), \theta_B)}{h_B(\theta_B)} \frac{\check{\theta}_A(\theta_B)}{l(\check{\theta}_A(\theta_B))^{1+\frac{1}{\varepsilon}}} h_B(\theta_B) d\theta_B}{H^B(\bar{w})} \quad (\text{A66})$$

$$= \frac{1}{\mu} \mathbb{E}_B [W'(V^B(\theta_B))] + \mathbb{E}_B \left[\Delta T(\theta_B, d_A, d_B) \frac{f(\check{\theta}_A(\theta_B), \theta_B)}{h_B(\theta_B)} \frac{\check{\theta}_A(\theta_B)}{l(\check{\theta}_A(\theta_B))^{1+\frac{1}{\varepsilon}}} \right] \quad (\text{A67})$$

where the fiscal externality $\Delta T(\theta_B, d_A, d_B)$ is the net fiscal gain from a switch from $\check{\theta}_A(\theta_B)$ in A to θ_B in B : increased income tax revenue minus the differential amenity subsidy cost,

$$\Delta T(\theta_B, d_A, d_B) = [\theta_B l(\theta_B) - k^* - c(V^A(\theta_B), l(\theta_B)) - \gamma_B(d_B^*)] \quad (\text{A68})$$

$$- [\check{\theta}_A(\theta_B) l(\check{\theta}_A(\theta_B)) - k^* - c(V^A(\check{\theta}_A(\theta_B)), l(\check{\theta}_A(\theta_B))) - \gamma_A(d_A^*)] \quad (\text{A69})$$

$$= \left(T(y(\theta_B)) - \frac{(\bar{s} + s^B)\gamma_B(d_B)}{1 + \bar{s} + s^B} \right) - \left(T(y(\check{\theta}_A(\theta_B))) - \frac{\bar{s}\gamma_A(d_A)}{1 + \bar{s}} \right) \quad (\text{A70})$$

$$= (T(y(\theta_B)) - T(y(\check{\theta}_A(\theta_B)))) - \left(\frac{(\bar{s} + s^B)\gamma_B(d_B)}{1 + \bar{s} + s^B} - \frac{\bar{s}\gamma_A(d_A)}{1 + \bar{s}} \right). \quad (\text{A71})$$

A.6 General equilibrium wage effects

Proofs in this subsection are almost identical to [Rothschild and Scheuer \(2013\)](#). It is helpful to define and derive some expressions for use later. Given x , the bottom wage $\underline{w}_x$ and top wage $\bar{w}_x$ are

$$\underline{w}_x = Y_A(x) \max \{ \underline{\theta}_A, \check{\theta}_{A,x}(\underline{\theta}_B, d) \}, \quad \bar{w}_x = \max \{ Y_A(x)\bar{\theta}_A, Y_B(x)\bar{\theta}_B \}. \quad (\text{A72})$$

Changing variables using $Y_A(x)\theta_A = w$ and differentiating (41) with respect to w ,

$$\check{\theta}'_{B,x} \left(\frac{w}{Y_A(x)}, d \right) = \frac{V'(w)}{V' \left(Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right)} \frac{Y_A(x)}{Y_B(x)} = \left(\frac{l(w)}{l \left(Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right)} \right)^{1+\frac{1}{\varepsilon}} \frac{\check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right)}{\frac{w}{Y_A(x)}} \quad (\text{A73})$$

$$\check{\theta}'_{A,x} \left(\frac{w}{Y_B(x)}, d \right) = \left(\frac{l(w)}{l \left(Y_A(x) \check{\theta}_{A,x} \left(\frac{w}{Y_B(x)}, d \right) \right)} \right)^{1+\frac{1}{\varepsilon}} \frac{\check{\theta}_{A,x} \left(\frac{w}{Y_B(x)}, d \right)}{\frac{w}{Y_B(x)}} \quad (\text{A74})$$

where $\check{\theta}'_{o,x}$ denotes the partial derivative of $\check{\theta}_{o,x}(\theta_o, d)$ with respect to its first argument.

Changing variables using $Y_A(x)\theta_A = w$ and differentiating (41) with respect to $V(w)$,

$$\frac{\partial \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right)}{\partial V(w)} = \frac{1}{V' \left(Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right)} \quad (\text{A75})$$

$$\frac{\partial \check{\theta}_{A,x} \left(\frac{w}{Y_B(x)}, d \right)}{\partial V(w)} = \frac{1}{V' \left(Y_A(x) \check{\theta}_{A,x} \left(\frac{w}{Y_B(x)}, d \right) \right)} \quad (\text{A76})$$

Differentiating (46),

$$\begin{aligned} \frac{dh_x^A}{dw} &= \frac{1}{Y_A(x)^2} \left[\int_{\underline{\theta}_B}^{\check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right)} f_1 \left(\frac{w}{Y_A(x)}, \theta_B \right) d\theta_B \right. \\ &\quad \left. + f \left(\frac{w}{Y_A(x)}, \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right) \left(\frac{l(w)}{l \left(Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right)} \right)^{1+\frac{1}{\varepsilon}} \frac{\check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right)}{\frac{w}{Y_A(x)}} \right] \end{aligned} \quad (\text{A77})$$

$$\begin{aligned} \frac{dh_x^B}{dw} &= \frac{1}{Y_B(x)^2} \left[\int_{\underline{\theta}_A}^{\check{\theta}_{A,x} \left(\frac{w}{Y_B(x)}, d \right)} f_2 \left(\theta_A, \frac{w}{Y_B(x)} \right) d\theta_A \right. \\ &\quad \left. + f \left(\check{\theta}_{A,x} \left(\frac{w}{Y_B(x)}, d \right), \frac{w}{Y_B(x)} \right) \left(\frac{l(w)}{l \left(Y_A(x) \check{\theta}_{A,x} \left(\frac{w}{Y_B(x)}, d \right) \right)} \right)^{1+\frac{1}{\varepsilon}} \frac{\check{\theta}_{A,x} \left(\frac{w}{Y_B(x)}, d \right)}{\frac{w}{Y_B(x)}} \right] \end{aligned} \quad (\text{A78})$$

Differentiating (41) with respect to x ,

$$\frac{\partial V(Y_A(x)\theta_A)}{\partial Y_A(x)\theta_A} \theta_A Y_A'(x) = \frac{\partial V(Y_B(x)\check{\theta}_{B,x}(\theta_A, d))}{\partial Y_B(x)\check{\theta}_{B,x}(\theta_A, d)} \left(Y_B'(x)\check{\theta}_{B,x}(\theta_A, d) + Y_B(x) \frac{\partial \check{\theta}_{B,x}(\theta_A, d)}{\partial x} \right) \quad (\text{A79})$$

$$\frac{\partial \check{\theta}_{B,x}(\theta_A, d)}{\partial x} = \left(\frac{l(Y_A(x)\theta_A)}{l(Y_B(x)\check{\theta}_{B,x}(\theta_A, d))} \right)^{1+\frac{1}{\varepsilon}} \frac{Y_A'(x)}{Y_A(x)} \check{\theta}_{B,x}(\theta_A, d) - \frac{Y_B'(x)}{Y_B(x)} \check{\theta}_{B,x}(\theta_A, d) \quad (\text{A80})$$

$$\frac{d\check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right)}{dx} = \frac{\partial\check{\theta}_{B,x}(\theta_A, d)}{\partial\theta_A}\Big|_{\theta_A=\frac{w}{Y_A(x)}} \frac{\partial\frac{w}{Y_A(x)}}{\partial x} + \frac{\partial\check{\theta}_{B,x}(\theta_A, d)}{\partial x}\Big|_{\theta_A=\frac{w}{Y_A(x)}} \quad (\text{A81})$$

$$= - \left(\frac{l(w)}{l\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right)\right)} \right)^{1+\frac{1}{\varepsilon}} \frac{\check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right)}{\frac{w}{Y_A(x)}} \frac{wY'_A(x)}{Y_A(x)^2} \quad (\text{A82})$$

$$+ \left(\frac{l(w)}{l(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right))} \right)^{1+\frac{1}{\varepsilon}} \frac{Y'_A(x)}{Y_A(x)} \check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right) - \frac{Y'_B(x)}{Y_B(x)} \check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right) \quad (\text{A83})$$

$$= - \frac{Y'_B(x)}{Y_B(x)} \check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right) \quad (\text{A84})$$

By symmetry,

$$\frac{\partial\check{\theta}_{A,x}(\theta_B, d)}{\partial x} = \check{\theta}_{A,x}(\theta_B, d) \left[\frac{Y'_B(x)}{Y_B(x)} \left(\frac{l(Y_B(x)\theta_B)}{l(Y_A(x)\check{\theta}_{A,x}(\theta_B, d))} \right)^{1+\frac{1}{\varepsilon}} - \frac{Y'_A(x)}{Y_A(x)} \right] \quad (\text{A85})$$

$$\frac{d\check{\theta}_{A,x}\left(\frac{w}{Y_B(x)}, d\right)}{dx} = - \frac{Y'_A(x)}{Y_A(x)} \check{\theta}_{A,x}\left(\frac{w}{Y_B(x)}, d\right) \quad (\text{A86})$$

A.6.1 Analog of **Rothschild and Scheuer (2013)** Lemma 4

$$H_x^A(w) = \frac{1}{Y_A(x)} \int_{\underline{w}_x}^w \int_{\underline{\theta}_B}^{\check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)} f\left(\frac{w'}{Y_A(x)}, \theta_B\right) d\theta_B dw' \quad (\text{A87})$$

$$\begin{aligned} \frac{dH_x^A(w)}{dx} &= - \frac{Y'_A(x)}{Y_A(x)^2} \int_{\underline{w}_x}^w \int_{\underline{\theta}_B}^{\check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)} f\left(\frac{w'}{Y_A(x)}, \theta_B\right) d\theta_B dw' \\ &\quad + \frac{1}{Y_A(x)} \int_{\underline{w}_x}^w f\left(\frac{w'}{Y_A(x)}, \check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)\right) \frac{d\check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)}{dx} dw' \\ &\quad - \frac{Y'_A(x)}{Y_A(x)^3} \int_{\underline{w}_x}^w \int_{\underline{\theta}_B}^{\check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)} w' f_a\left(\frac{w'}{Y_A(x)}, \theta_B\right) d\theta_B dw' \end{aligned} \quad (\text{A88})$$

Substituting (A84) and collecting terms,

$$\begin{aligned} \frac{dH_x^A(w)}{dx} &= - \frac{Y'_A(x)}{Y_A(x)} \int_{\underline{w}_x}^w \int_{\underline{\theta}_B}^{\check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)} f\left(\frac{w'}{Y_A(x)}, \theta_B\right) d\theta_B dw' \\ &\quad - \frac{1}{Y_A(x)} \frac{Y'_B(x)}{Y_B(x)} \int_{\underline{w}_x}^w f\left(\frac{w'}{Y_A(x)}, \check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)\right) \check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right) dw' \\ &\quad - \frac{Y'_A(x)}{Y_A(x)^3} \int_{\underline{w}_x}^w \int_{\underline{\theta}_B}^{\check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)} w' f_a\left(\frac{w'}{Y_A(x)}, \theta_B\right) d\theta_B dw' \end{aligned} \quad (\text{A89})$$

Define

$$\Omega_x^A(w) = \frac{1}{Y_A(x)Y_B(x)} \int_{\underline{w}_x}^w f\left(\frac{w'}{Y_A(x)}, \check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)\right) Y_B(x) \check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right) \cdot \left[\frac{Y'_A(x)}{Y_A(x)} \frac{l(w')^{1+\frac{1}{\varepsilon}}}{l\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)\right)^{1+\frac{1}{\varepsilon}}} - \frac{Y'_B(x)}{Y_B(x)} \right] dw' \quad (\text{A90})$$

$$= \int_{\check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right)}^{\check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right)} f\left(\check{\theta}_{A,x}(\theta_B, d), \theta_B\right) \check{\theta}_{A,x}(\theta_B, d) \cdot \left[\frac{Y'_A(x)}{Y_A(x)} - \frac{Y'_B(x)}{Y_B(x)} \frac{l(\theta_B Y_B(x))^{1+\frac{1}{\varepsilon}}}{l(\check{\theta}_{A,x}(\theta_B, d) Y_A(x))^{1+\frac{1}{\varepsilon}}} \right] d\theta_B \quad (\text{A91})$$

where the second line follows from a change of variables: $\theta_B = \check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)$ and (A74). Collecting terms,

$$\frac{dH_x^A(w)}{dx} = - \frac{Y'_A(x)}{Y_A(x)} \left(\frac{1}{Y_A(x)} \int_{\underline{w}_x}^w \int_{\underline{\theta}_B}^{\check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)} f\left(\frac{w'}{Y_A(x)}, \theta_B\right) d\theta_B dw' \right. \quad (\text{A92})$$

$$+ \int_{\underline{w}_x}^w \frac{w'}{Y_A(x)} \left[f\left(\frac{w'}{Y_A(x)}, \check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)\right) \frac{\check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)}{\frac{w'}{Y_A(x)}} \cdot \left(\frac{l(w')}{l\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)\right)} \right)^{1+\frac{1}{\varepsilon}} \frac{1}{Y_A(x)} \right. \quad (\text{A93})$$

$$\left. + \int_{\underline{\theta}_B}^{\check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)} \frac{1}{Y_A(x)} f_a\left(\frac{w'}{Y_A(x)}, \theta_B\right) d\theta_B \right] dw' \Big) + \Omega_x^A(w) \quad (\text{A94})$$

$$= - \frac{Y'_A(x)}{Y_A(x)} \left[\frac{w'}{Y_A(x)} \int_{\underline{\theta}_B}^{\check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)} f\left(\frac{w'}{Y_A(x)}, \theta_B\right) d\theta_B \right]_{\underline{w}}^w + \Omega_x^A(w) \quad (\text{A95})$$

$$= - \frac{Y'_A(x)}{Y_A(x)} w h_{A,x}(w) + \Omega_x^A(w) \quad (\text{A96})$$

where lines (A93)-(A94) are integrated by parts by defining

$$u = \frac{w'}{Y_A(x)} \quad (\text{A97})$$

$$dv = \left[f\left(\frac{w'}{Y_A(x)}, \check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)\right) \frac{\check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)}{\frac{w'}{Y_A(x)}} \left(\frac{l(w')}{l\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)\right)} \right)^{1+\frac{1}{\varepsilon}} \frac{1}{Y_A(x)} + \int_{\underline{\theta}_B}^{\check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)} \frac{1}{Y_A(x)} f_a\left(\frac{w'}{Y_A(x)}, \theta_B\right) d\theta_B \right] dw' \quad (\text{A98})$$

$$du = \frac{dw'}{Y_A(x)} \quad (\text{A99})$$

$$v = \int_{\underline{\theta}_B}^{\check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)} f\left(\frac{w'}{Y_A(x)}, \theta_B\right) d\theta_B. \quad (\text{A100})$$

$$\int_{\underline{w}}^w u dv = \left[\frac{w'}{Y_A(x)} \int_{\underline{\theta}_B}^{\check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)} f\left(\frac{w'}{Y_A(x)}, \theta_B\right) d\theta_B \right]_{\underline{w}}^w - \frac{1}{Y_A(x)} \int_{\underline{w}}^w \int_{\underline{\theta}_B}^{\check{\theta}_{B,x}\left(\frac{w'}{Y_A(x)}, d\right)} f\left(\frac{w'}{Y_A(x)}, \theta_B\right) d\theta_B dw' \quad (\text{A101})$$

Substituting (A101) into (A93)-(A94), Now, we do the same for $H_x^B(w)$.

$$H_x^B(w) = \frac{1}{Y_B(x)} \int_{\underline{w}_x}^w \int_{\underline{\theta}_A}^{\check{\theta}_{A,x}\left(\frac{w'}{Y_B(x)}, d\right)} f\left(\theta_A, \frac{w'}{Y_B(x)}\right) d\theta_A dw' \quad (\text{A102})$$

$$\begin{aligned} \frac{dH_x^B(w)}{dx} &= -\frac{Y'_B(x)}{Y_B(x)^2} \int_{\underline{w}_x}^w \int_{\underline{\theta}_A}^{\check{\theta}_{A,x}\left(\frac{w'}{Y_B(x)}, d\right)} f\left(\theta_A, \frac{w'}{Y_B(x)}\right) d\theta_A dw' \\ &\quad + \frac{1}{Y_B(x)} \int_{\underline{w}_x}^w f\left(\check{\theta}_{A,x}\left(\frac{w'}{Y_B(x)}, d\right), \frac{w'}{Y_B(x)}\right) \frac{d\check{\theta}_{A,x}\left(\frac{w'}{Y_B(x)}, d\right)}{dx} dw' \\ &\quad - \frac{Y'_B(x)}{Y_B(x)^3} \int_{\underline{w}_x}^w \int_{\underline{\theta}_A}^{\check{\theta}_{A,x}\left(\frac{w'}{Y_B(x)}, d\right)} w' f_b\left(\theta_A, \frac{w'}{Y_B(x)}\right) d\theta_A dw' \end{aligned} \quad (\text{A103})$$

Substituting (A86),

$$\begin{aligned} \frac{dH_x^B(w)}{dx} &= -\frac{Y'_B(x)}{Y_B(x)^2} \int_{\underline{w}_x}^w \int_{\underline{\theta}_A}^{\check{\theta}_{A,x}\left(\frac{w'}{Y_B(x)}, d\right)} f\left(\theta_A, \frac{w'}{Y_B(x)}\right) d\theta_A dw' \\ &\quad - \frac{1}{Y_B(x)} \frac{Y'_A(x)}{Y_A(x)} \int_{\underline{w}_x}^w f\left(\check{\theta}_{A,x}\left(\frac{w'}{Y_B(x)}, d\right), \frac{w'}{Y_B(x)}\right) \check{\theta}_{A,x}\left(\frac{w}{Y_B(x)}, d\right) dw' \\ &\quad - \frac{Y'_B(x)}{Y_B(x)^3} \int_{\underline{w}_x}^w \int_{\underline{\theta}_A}^{\check{\theta}_{A,x}\left(\frac{w'}{Y_B(x)}, d\right)} w' f_b\left(\theta_A, \frac{w'}{Y_B(x)}\right) d\theta_A dw' \end{aligned} \quad (\text{A104})$$

$$\begin{aligned} &= -\frac{Y'_B(x)}{Y_B(x)} \left[\frac{1}{Y_B(x)} \int_{\underline{w}_x}^w \int_{\underline{\theta}_A}^{\check{\theta}_{A,x}\left(\frac{w'}{Y_B(x)}, d\right)} f\left(\theta_A, \frac{w'}{Y_B(x)}\right) d\theta_A dw' \right. \\ &\quad + \int_{\underline{w}_x}^w \frac{w'}{Y_B(x)} \left\{ f\left(\check{\theta}_{A,x}\left(\frac{w'}{Y_B(x)}, d\right), \frac{w'}{Y_B(x)}\right) \frac{\check{\theta}_{A,x}\left(\frac{w}{Y_B(x)}, d\right)}{\frac{w'}{Y_B(x)}} \right. \\ &\quad \cdot \left(\frac{l(w)}{l\left(Y_A(x)\check{\theta}_{A,x}\left(\frac{w}{Y_B(x)}, d\right)\right)} \right)^{1+\frac{1}{\varepsilon}} \frac{1}{Y_B(x)} \\ &\quad \left. \left. + \int_{\underline{\theta}_A}^{\check{\theta}_{A,x}\left(\frac{w'}{Y_B(x)}, d\right)} \frac{1}{Y_B(x)} f_b\left(\theta_A, \frac{w'}{Y_B(x)}\right) d\theta_A \right\} dw' \right] - \Omega_x^B(w) \end{aligned} \quad (\text{A105})$$

where

$$\begin{aligned} \Omega_x^B(w) &= \frac{1}{Y_A(x)Y_B(x)} \int_{\underline{w}_x}^w f\left(\check{\theta}_{A,x}\left(\frac{w'}{Y_B(x)}, d\right), \frac{w'}{Y_B(x)}\right) Y_A(x)\check{\theta}_{A,x}\left(\frac{w'}{Y_B(x)}, d\right) \\ &\quad \cdot \left[\frac{Y'_A(x)}{Y_A(x)} - \frac{Y'_B(x)}{Y_B(x)} \frac{l(w')^{1+\frac{1}{\varepsilon}}}{l\left(Y_A(x)\check{\theta}_{A,x}\left(\frac{w'}{Y_B(x)}, d\right)\right)^{1+\frac{1}{\varepsilon}}} \right] dw' \end{aligned} \quad (\text{A106})$$

$$= \int_{\frac{w}{Y_B(x)}}^{\frac{w}{Y_B(x)}} f(\check{\theta}_{A,x}(\theta_B, d), \theta_B) \check{\theta}_{A,x}(\theta_B, d) \left[\frac{Y'_A(x)}{Y_A(x)} - \frac{Y'_B(x)}{Y_B(x)} \frac{l(\theta_B Y_B(x))^{1+\frac{1}{\varepsilon}}}{l(\check{\theta}_{A,x}(\theta_B, d) Y_A(x))^{1+\frac{1}{\varepsilon}}} \right] d\theta_B \quad (\text{A107})$$

where the second line follows from a change of variables $\theta_B = \frac{w}{Y_B(x)}$ and (A73). Integration by parts again yields

$$\frac{dH_x^B(w)}{dx} = -\frac{Y_B'(x)}{Y_B(x)}wh_{B,x}(w) - \Omega_x^B(w) \quad (\text{A108})$$

A.6.2 Proposition 5

The planner's problem the same as in Proposition 1, but with the addition of the consistency constraint (47), which ensures labor market clearing. The Lagrangian is

$$\begin{aligned} \mathcal{L} = & \int_{\underline{w}_x}^{\bar{w}_x} W(V(w))h_{A,x}(w) + W(V(w) - d)h_{B,x}(w)dw \\ & + \mu \int_{\underline{w}_x}^{\bar{w}_x} [wl(w) - c(V(w), l(w))] [h_{A,x}(w) + h_{B,x}(w)] - \gamma(d)h_{B,x}(w)dw \\ & - \mu \int_{\underline{w}_x}^{\bar{w}_x} V(w)\eta'(w)dw - \mu \int_{\underline{w}_x}^{\bar{w}_x} \frac{l(w)^{1+1/\varepsilon}}{w}\eta(w)dw \\ & + \mu\xi \left[(1 - \alpha(x)) \int_{\underline{w}_x}^{\bar{w}_x} wl(w)h_{A,x}(w)dw - \alpha(x) \int_{\underline{w}_x}^{\bar{w}_x} wl(w)h_{B,x}(w)dw \right] \end{aligned} \quad (\text{A109})$$

The first-order condition with respect to $V(w)$ is

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial V(w)} = & W'(V(w))h_{A,x}(w) + W'(V(w) - d)h_{B,x}(w) - \eta'(w)\mu - \mu h_x(w) \\ & + \mu \left[-\Delta T \left(Y_B(x)\check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right) \frac{\partial h_{A,x}(w)}{\partial V(w)} + \Delta T(w) \frac{\partial h_{B,x}(w)}{\partial V(w)} \right] \\ & - \mu\xi \left[-\Delta C \left(Y_B(x)\check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right) \frac{\partial h_{A,x}(w)}{\partial V(w)} + \Delta C(w) \frac{\partial h_{B,x}(w)}{\partial V(w)} \right] = 0 \end{aligned} \quad (\text{A110})$$

where

$$\Delta T(w) = T(y(w)) - \gamma(d) - T \left(y \left(Y_A(x)\check{\theta}_{A,x} \left(\frac{w}{Y_B(x)}, d \right) \right) \right), \quad (\text{A111})$$

$$\Delta C(w) = \alpha(x)wl(w) + (1 - \alpha(x))Y_A(x)\check{\theta}_{A,x} \left(\frac{w}{Y_B(x)}, d \right) l \left(Y_A(x)\check{\theta}_{A,x} \left(\frac{w}{Y_B(x)}, d \right) \right). \quad (\text{A112})$$

Making $\eta'(w)$ the subject and integrating,

$$\begin{aligned} \eta(w) = & \int_w^{\bar{w}_x} h_x(z) - \frac{1}{\mu} [W'(V(z))h_{A,x}(z) + W'(V(z) - d)h_{B,x}(z)] dz \\ & + \int_w^{\bar{w}_x} \left[-\Delta T \left(Y_B(x)\check{\theta}_{B,x} \left(\frac{z}{Y_A(x)}, d \right) \right) \frac{\partial h_{A,x}(z)}{\partial V(z)} + \Delta T(z) \frac{\partial h_{B,x}(z)}{\partial V(z)} \right] dz \end{aligned} \quad (\text{A113})$$

$$\begin{aligned} & - \xi \int_w^{\bar{w}_x} \left[-\Delta C \left(Y_B(x)\check{\theta}_{B,x} \left(\frac{z}{Y_A(x)}, d \right) \right) \frac{\partial h_{A,x}(z)}{\partial V(z)} + \Delta C(z) \frac{\partial h_{B,x}(z)}{\partial V(z)} \right] dz \\ = & \int_w^{\bar{w}_x} h_x(z) - \frac{1}{\mu} [W'(V(z))h_{A,x}(z) + W'(V(z) - d)h_{B,x}(z)] dz \\ & + \int_w^{Y_B(x)\check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right)} (\Delta T(z) - \xi \Delta C(z)) f \left(\check{\theta}_{A,x} \left(\frac{z}{Y_B(x)}, d \right), \frac{z}{Y_B(x)} \right) \frac{Y_A(x)\check{\theta}_{A,x} \left(\frac{z}{Y_B(x)}, d \right)}{l \left(Y_A(x)\check{\theta}_{A,x} \left(\frac{z}{Y_B(x)}, d \right) \right)^{1+\frac{1}{\varepsilon}}} dz \end{aligned} \quad (\text{A114})$$

The first-order condition with respect to $l(w)$ is

$$\mu h_x(w) \left(w - l^{1/\varepsilon} \right) + \mu \xi [(1 - \alpha(x)) w h_{A,x}(w) - \alpha(x) w h_{B,x}(w)] = \mu \eta(w) \frac{l^{1/\varepsilon}}{w} \left(1 + \frac{1}{\varepsilon} \right) \quad (\text{A115})$$

Rearranging and using $\frac{l(w)^{1/\varepsilon}}{w} = \frac{MRS(w)}{w} = 1 - T'(y(w))$,

$$\begin{aligned} \frac{w - l^{1/\varepsilon}}{l^{1/\varepsilon}} &= \frac{1 + \frac{1}{\varepsilon}}{w h_x(w)} \eta(w) - \frac{\xi w}{l^{1/\varepsilon}} \left(\frac{h_{A,x}(w)}{h_x(w)} - \alpha(x) \right) \\ \frac{T'(y(w))}{1 - T'(y(w))} &= \frac{1 + \frac{1}{\varepsilon}}{w h_x(w)} \eta(w) - \frac{\xi}{1 - T'(y(w))} \left(\frac{h_{A,x}(w)}{h_x(w)} - \alpha(x) \right) \end{aligned} \quad (\text{A116})$$

The optimal tax formula (A116) contains an extra term that depends on the difference between sector A's local income share $\frac{h_{A,x}(w)}{h_x(w)}$ and its aggregate income share $\alpha(x)$. When sector A's local share exceeds its aggregate share, a tax cut at $y(w)$ disproportionately encourages effort by A-workers, raising B-wages relative to A; if $\xi > 0$, this is welfare-improving. [Rothschild and Scheuer \(2013\)](#) express optimal policy in terms of the keep rate $1 - T'(y(w))$ instead. Rearranging (A116) yields

$$1 - T'(y(w)) = \left(1 + \xi \left(\frac{h_{A,x}(w)}{h_x(w)} - \alpha(x) \right) \right) \left(1 + \frac{1 + \frac{1}{\varepsilon}}{w h_x(w)} \eta(w) \right)^{-1} \quad (\text{A117})$$

The first-order condition with respect to d is, after much cancelling,

$$\begin{aligned} \int_{\underline{w}_x}^{\bar{w}_x} W'(V(w) - d) h_{B,x}(w) dw + \mu \int_{\underline{w}_x}^{\bar{w}_x} \gamma'(d) h_{B,x}(w) dw \\ + \mu \int_{\underline{w}_x}^{\bar{w}_x} (\Delta T(z) - \xi \Delta C(z)) f(\check{\theta}_{A,x}(z, d), z) \frac{Y_A(x) \check{\theta}_{A,x}(z, d)}{l(Y_A(x) \check{\theta}_{A,x}(z, d))^{1 + \frac{1}{\varepsilon}}} dz = 0 \end{aligned} \quad (\text{A118})$$

$$-\gamma'(d) = \frac{1}{\mu} \mathbb{E}_B [W'(V(w) - d)] + \mathbb{E}_B \left[(\Delta T(w) - \xi \Delta C(w)) \frac{f(\check{\theta}_{A,x}(w, d), w)}{h_{B,x}(w)} \frac{Y_A(x) \check{\theta}_{A,x}(w, d)}{l(Y_A(x) \check{\theta}_{A,x}(w, d))^{1 + \frac{1}{\varepsilon}}} \right] \quad (\text{A119})$$

where $\mathbb{E}_B[x(w)] = \frac{\int_{\underline{w}}^{\bar{w}} x(w) h_{B,x}(w) dw}{\int_{\underline{w}}^{\bar{w}} h_{B,x}(w) dw}$ is the weighted average of $x(w)$ among B-workers.

A.6.3 Lemma 1

Lemma 1 *At an optimal x , the multiplier on the consistency constraint is given by*

$$\xi = - \frac{I + \frac{1}{\mu} (R + Q_1) + A_1 + D_T}{\Lambda(x) \sigma(x) \alpha(x) Y_B(x) E_B + \hat{C} + S + A_2 - D_C + Q_2} \quad (\text{A120})$$

where

$$\hat{C} = \int_{\underline{w}_x}^{\bar{w}_x} w^2 l'(w) \Lambda(x) \frac{h_{A,x}(w) h_{B,x}(w)}{h_x(w)} dw \quad (\text{A121})$$

$$S = \int_{\underline{w}_x}^{\bar{w}_x} w l(w) \frac{d\bar{\Omega}_x(w)}{dw} dw \quad (\text{A122})$$

$$R = \int_{\underline{w}_x}^{\bar{w}_x} V'(w) \left(\Lambda(x) w \frac{h_{A,x}(w) h_{B,x}(w)}{h_x(w)} - \bar{\Omega}_x(w) \right) (W'(V(w)) - W'(V(w) - d)) dw \quad (\text{A123})$$

$$I = - \int_{\underline{w}_x}^{\bar{w}_x} \eta(w) l(w)^{1+\frac{1}{\varepsilon}} \frac{d}{dw} \left[\frac{1}{w h_x(w)} \frac{dH_x(w)}{dx} \right] dw \quad (\text{A124})$$

$$A_1 = \int_{\underline{w}_x}^{\bar{w}_x} l(w) \Delta \Omega_x(w) dw \quad (\text{A125})$$

$$A_2 = \int_{\underline{w}_x}^{\bar{w}_x} \left(\frac{h_{A,x}(w)}{h_x(w)} - \alpha(x) \right) l(w) \Delta \Omega_x(w) dw \quad (\text{A126})$$

$$\begin{aligned} D_y = & \int_{\underline{w}_x}^{\bar{w}_x} \left(\frac{l(w)}{l \left(Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right)} \right)^{1+\frac{1}{\varepsilon}} \frac{Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right)}{w} \frac{1}{Y_A(x)} f \left(\frac{w}{Y_A(x)}, \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right) \\ & \cdot \Delta y \left(Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right) \\ & \cdot \left[\frac{1}{h_x \left(Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right)} \frac{dH_x \left(Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right)}{dx} - \frac{1}{h_x(w)} \frac{dH_x(w)}{dx} \right] dw, \text{ with } y \in \{C, T\} \end{aligned} \quad (\text{A127})$$

$$Q_1 = - \left[W(V(\bar{w}_x)) - W(V(\bar{w}_x) - d) + \mu \gamma(d) \right] \left(\Lambda(x) \bar{w}_x \frac{h_{A,x}(\bar{w}_x) h_{B,x}(\bar{w}_x)}{h_x(\bar{w}_x)} - \bar{\Omega}_x(\bar{w}_x) \right) \quad (\text{A128})$$

$$Q_2 = - \xi \mu \bar{w}_x l(\bar{w}_x) \Lambda(x) \bar{w}_x \frac{h_{A,x}(\bar{w}_x) h_{B,x}(\bar{w}_x)}{h_x(\bar{w}_x)} \quad (\text{A129})$$

This generalizes [Rothschild and Scheuer \(2013\)](#)'s results on optimal policy in the presence of general equilibrium effects and extensive margin occupation-switching to account for compensating differentials. The expression consequently includes several additional terms relative to their model, though it collapses to their formulation when amenities and compensating differentials are absent. We briefly discuss these additional terms below; for further details, see [Rothschild and Scheuer \(2013\)](#).

Terms in the denominator of (A120) characterize the impact of a marginal change in x on the consistency constraint. $\Lambda(x) \sigma(x) \alpha(x) Y_B(x) E_B$ represents the direct effect of x on the consistency constraint, holding behavior fixed.

$\hat{C}$ accounts for intensive margin behavioral responses: as relative wages change, workers adjust their labor supply along the $l(w)$ schedule. This reallocation of labor supply among workers initially earning identical wages induces changes in aggregate income that affect the consistency constraint.

S reflects the impact of occupation switching on the consistency constraint: general equilibrium wage changes induce workers to switch occupations, altering aggregate income in both occupations. Its full decomposition is in (A131)-(A133). In the absence of compensating differentials, aggregate income changes in the two occupations are equal and opposite because marginal workers earn the same wage and income in both occupations. With compensating differentials, however, the picture is richer. At any given wage w , two

distinct groups may be indifferent between occupations: workers with skill vector $\left(\frac{w}{Y_A(x)}, \check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right)\right)$ who would earn w in occupation A , and workers with skill vector $\left(\check{\theta}_{A,x}\left(\frac{w}{Y_B(x)}, d\right), \frac{w}{Y_B(x)}\right)$ who would earn w in occupation B . The occupation-switching effect at wage w thus combines: (i) a weighted average of these two groups' responses, where weights equal the respective occupation shares at that wage, and (ii) a composition term reflecting how the occupation shares themselves adjust.

$$S = \int_{\underline{w}_x}^{\bar{w}_x} w l(w) \frac{d\bar{\Omega}_x(w)}{dw} dw \quad (\text{A130})$$

$$= \int_{\underline{w}_x}^{\bar{w}_x} w l(w) \left(\underbrace{\frac{h_{A,x}(w)}{h_x(w)} \frac{d\Omega_x^B(w)}{dw}}_{\text{switchers earning } w \text{ in } A} + \underbrace{\frac{h_{B,x}(w)}{h_x(w)} \frac{d\Omega_x^A(w)}{dw}}_{\text{switchers earning } w \text{ in } B} + \underbrace{\frac{d}{dw} \left[\frac{h_{A,x}(w)}{h_x(w)} \right] \Delta\Omega_x(w)}_{\text{composition}} \right) dw \quad (\text{A131})$$

$$\begin{aligned} \frac{d\Omega_x^A(w)}{dw} &= \frac{1}{Y_A(x)Y_B(x)} Y_B(x) \check{\theta}_B\left(\frac{w}{Y_A(x)}\right) f\left(\frac{w}{Y_A(x)}, \check{\theta}_B\left(\frac{w}{Y_A(x)}\right)\right) \\ &\quad \cdot \left[\frac{Y'_A(x)}{Y_A(x)} \left(\frac{l(w)}{l\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right)\right)} \right)^{1+\frac{1}{\varepsilon}} - \frac{Y'_B(x)}{Y_B(x)} \right] \end{aligned} \quad (\text{A132})$$

$$\begin{aligned} \frac{d\Omega_x^B(w)}{dw} &= \frac{1}{Y_A(x)Y_B(x)} Y_A(x) \check{\theta}_A\left(\frac{w}{Y_B(x)}\right) f\left(\check{\theta}_A\left(\frac{w}{Y_B(x)}\right), \frac{w}{Y_B(x)}\right) \\ &\quad \cdot \left[\frac{Y'_A(x)}{Y_A(x)} - \frac{Y'_B(x)}{Y_B(x)} \left(\frac{l(w)}{l\left(Y_A(x)\check{\theta}_{A,x}\left(\frac{w}{Y_B(x)}, d\right)\right)} \right)^{1+\frac{1}{\varepsilon}} \right] \end{aligned} \quad (\text{A133})$$

A_2 captures how changes in x affect the wage distribution $F(w)$ through compensating differentials, as summarized by $\Delta\Omega_x(w)$, which in turn affects the consistency constraint. This channel is only operative when one occupation is locally overrepresented at wage w , that is, when $\frac{h_A(w)}{h(w)} \neq \alpha(x)$. In this case, an increase in the density of workers earning wage w raises aggregate income in one occupation disproportionately relative to the other. By contrast, when $\frac{h_A(w)}{h(w)} = \theta$, the occupational composition at wage w mirrors the aggregate share, so the induced shift in $F(w)$ is neutral and has no effect on the consistency constraint.

D_C captures how changes in x amplify or dampen the occupation-switching effects on the consistency constraint that are already embedded in the optimal tax system. Consider that the planner chose taxes knowing they would induce occupation switching, which has effects on the consistency constraint that are valued at shadow cost ξ . D_C thus quantifies how a change in x interacts with the optimal tax to affect the consistency constraint.

While S, A_2 , and D_C all arise from occupation switching, they represent distinct channels. S directly measures occupation switching induced by changes in x affect aggregate income across occupations. A_2 captures how that switching affects the wage distribution. D_C is more subtle: it captures how the change in x alters the impact that *tax-induced* occupation switching has on the consistency constraint—an impact that was already internalized in the optimal tax schedule.

Q_2 is a boundary term for the consistency constraint, but it is generically zero. The numerator contains the product $h_{A,x}(\bar{w}_x)h_{B,x}(\bar{w}_x)$, which equals zero whenever one occupation has no mass at the top wage $\bar{w}_x$. This occurs whenever one occupation strictly dominates at the top—that is, whenever $Y_A(x)\bar{\theta}_A \neq Y_B(x)\bar{\theta}_B$. This is the case for almost all values of x .

The terms in the numerator capture how changes in x affect the planner's objective, the incentive constraints,

and the resource constraint. I is what [Rothschild and Scheuer \(2013\)](#) call a “generalized [Stiglitz \(1982\)](#) effect”: it captures how changes in x affect incentive constraints. It can be decomposed into two distinct channels through which changes in x affect incentive constraints.

$$I = \int_{\bar{w}_x}^{\bar{w}_x} \eta(w) l(w)^{1+\frac{1}{\varepsilon}} \left(\frac{1}{\Lambda(x)} \frac{d}{dw} \left[\frac{h_{A,x}(w)}{h_x(w)} \right] + \frac{d}{dw} \left[\frac{\Delta\Omega_x(w)}{wh_x(w)} \right] \right) dw \quad (\text{A134})$$

The first term in brackets operates through the skill composition of occupations. When $\frac{h_{A,x}(w)}{h_x(w)}$ is increasing in w (i.e., $\frac{d}{dw} \frac{h_{A,x}(w)}{h_x(w)} > 0$), occupation A is high-skill, in the sense that the share of workers employed in A increases with the wage level. An increase in x then lowers wages for occupation A workers while raising them for occupation B workers. When A is the high-skill occupation, this compression of the wage distribution relaxes incentive constraints.

The second term in brackets captures how changes in x affect the wage distribution through induced occupation switching. In [Rothschild and Scheuer \(2013\)](#), this term vanishes because marginal workers earn identical wages in both occupations: changes in x affect the wage distribution solely through the direct effect on relative wages within each occupation, holding occupation choices fixed. In the presence of compensating differentials, however, marginal workers earn different wages across occupations. Consequently, occupation switching induced by changes in x affects the wage distribution, captured by $\Delta\Omega_x(w)$. This second component of I is therefore essential for measuring the total effect of changes in x on the wage distribution and, through this channel, on incentive constraints. Expressing I as in (A124) combines these two channels, revealing that the overall compression or decompression of the wage distribution is the sufficient statistic for how x affects incentive constraints.

R is the redistributive effect. Under a concave social welfare function, the social planner places a higher social welfare weight on B -workers than on A -workers earning the same wage, so an increase in x that raises wages for B -workers relative to A -workers has a redistributive benefit.

A_1 measures how a change in x affects the resource constraint through occupation switching. A change in x leads to a change in the composition of workers across occupations and wages, altering aggregate resources since workers supply different quantities of labor at different wages.

D_T is analogous to D_C , but for the resource constraint: it captures how changes in x interact with the fiscal externality from compensating differentials that results from tax-induced occupation switching.

Finally, Q_1 is a boundary correction term that captures the welfare and fiscal effects of occupation switching at the top of the wage distribution. Since generically only one occupation dominates at the top wage, Q_1 simplifies to

$$Q_1 = \begin{cases} [W(V(Y_A(x)\bar{\theta}_A)) - W(V(Y_A(x)\bar{\theta}_A) - d) + \mu\gamma(d)] \Omega_x^B(Y_A(x)\bar{\theta}_A) & \text{if } \bar{w}_x = Y_A(x)\bar{\theta}_A > Y_B(x)\bar{\theta}_B \\ [W(V(Y_B(x)\bar{\theta}_B)) - W(V(Y_B(x)\bar{\theta}_B) - d) + \mu\gamma(d)] \Omega_x^A(Y_B(x)\bar{\theta}_B) & \text{if } \bar{w}_x = Y_B(x)\bar{\theta}_B > Y_A(x)\bar{\theta}_A. \end{cases} \quad (\text{A135})$$

A.6.4 Proof of Lemma 1

The first-order condition with respect to x is

$$\begin{aligned} & \int_{\underline{w}_x}^{\bar{w}_x} W(V(w)) \frac{dh_{A,x}(w)}{dx} + W(V(w) - d) \frac{dh_{B,x}(w)}{dx} dw \\ & + \mu \int_{\underline{w}_x}^{\bar{w}_x} [wl(w) - c(V(w), l(w))] \frac{dh_x(w)}{dx} - \gamma(d) \frac{dh_{B,x}(w)}{dx} dw \\ & - \mu \xi \theta'(x) \int_{\underline{w}_x}^{\bar{w}_x} wl(w) h_x(w) dw + \mu \xi \int_{\underline{w}_x}^{\bar{w}_x} wl(w) \left((1 - \alpha(x)) \frac{dh_{A,x}(w)}{dx} - \alpha(x) \frac{dh_{B,x}(w)}{dx} \right) dw + B_1 = 0 \end{aligned} \quad (\text{A136})$$

where

$$\begin{aligned} B_1 = & \frac{d\bar{w}_x}{dx} \left[W(V(\bar{w}_x)) h_{A,x}(\bar{w}_x) + W(V(\bar{w}_x) - d) h_{B,x}(\bar{w}_x) + \mu(\bar{w}_x l(\bar{w}_x) - c(V(\bar{w}_x), l(\bar{w}_x))) h_x(\bar{w}_x) \right. \\ & \left. - \mu \gamma(d) h_{B,x}(\bar{w}_x) + \mu \xi ((1 - \alpha(x)) \bar{w}_x l(\bar{w}_x) h_{A,x}(\bar{w}_x) - \alpha(x) \bar{w}_x l(\bar{w}_x) h_{B,x}(\bar{w}_x)) \right] \\ & - \frac{d\underline{w}_x}{dx} \left[W(V(\underline{w}_x)) h_{A,x}(\underline{w}_x) + W(V(\underline{w}_x) - d) h_{B,x}(\underline{w}_x) + \mu(\underline{w}_x l(\underline{w}_x) - c(V(\underline{w}_x), l(\underline{w}_x))) h_x(\underline{w}_x) \right. \\ & \left. - \mu \gamma(d) h_{B,x}(\underline{w}_x) + \mu \xi ((1 - \alpha(x)) \underline{w}_x l(\underline{w}_x) h_{A,x}(\underline{w}_x) - \alpha(x) \underline{w}_x l(\underline{w}_x) h_{B,x}(\underline{w}_x)) \right] \end{aligned} \quad (\text{A137})$$

Integrating by parts and using $\int_{\underline{w}_x}^{\bar{w}_x} wl(w) h_x(w) dw = Y(L_A, L_B)$, we can rewrite the FOC (A136) as

$$\begin{aligned} & - \int_{\underline{w}_x}^{\bar{w}_x} W'(V(w)) V'(w) \frac{dH_x^A(w)}{dx} + W'(V(w) - d) V'(w) \frac{dH_x^B(w)}{dx} dw \\ & - \mu \int_{\underline{w}_x}^{\bar{w}_x} \left(wl'(w) + l(w) - V'(w) - l(w)^{1/\varepsilon} l'(w) \right) \frac{dH_x(w)}{dx} dw \\ & + \mu \xi \int_{\underline{w}_x}^{\bar{w}_x} (wl'(w) + l(w)) \left(\alpha(x) \frac{dH_x^B(w)}{dx} - (1 - \alpha(x)) \frac{dH_x^A(w)}{dx} \right) dw \\ & - \mu \xi \theta'(x) Y(L_A, L_B) + B_1 + B_2 = 0 \end{aligned} \quad (\text{A138})$$

where

$$\begin{aligned} B_2 = & \left[W(V(w)) \frac{dH_x^A(w)}{dx} + W(V(w) - d) \frac{dH_x^B(w)}{dx} + \mu (wl(w) - c(V(w), l(w))) \frac{dH_x(w)}{dx} \right. \\ & \left. - \mu \gamma(d) \frac{dH_x^B(w)}{dx} + \mu \xi wl(w) \left((1 - \alpha(x)) \frac{dH_x^A(w)}{dx} - \alpha(x) \frac{dH_x^B(w)}{dx} \right) \right]_{\underline{w}_x}^{\bar{w}_x} \end{aligned} \quad (\text{A139})$$

From the inner problem FOCs (A110) and (A115),

$$\begin{aligned} & \mu \int_{\underline{w}_x}^{\bar{w}_x} \frac{V'(w)}{h_x(w)} \left(\frac{1}{\mu} (W'(V(w)) h_{A,x}(w) + W'(V(w) - d) h_{B,x}(w)) - \eta'(w) - h_x(w) \right. \\ & \left. + \left[\Delta \hat{T}(w) \frac{\partial h_{B,x}(w)}{\partial V(w)} - \Delta \hat{T} \left(Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right) \frac{\partial h_{A,x}(w)}{\partial V(w)} \right] \right) \frac{dH_x(w)}{dx} dw = 0 \end{aligned} \quad (\text{A140})$$

where $\Delta\hat{T}(w) = \Delta T(w) - \xi\Delta C(w)$, and

$$\mu \int_{\underline{w}_x}^{\bar{w}_x} \frac{l'(w)}{h_x(w)} \left[h_x(w) \left(w - l^{1/\varepsilon} \right) + \xi w [(1 - \alpha(x))h_{A,x}(w) - \alpha(x)h_{B,x}(w)] - \eta(w) \frac{l^{1/\varepsilon}}{w} \left(1 + \frac{1}{\varepsilon} \right) \right] \frac{dH_x(w)}{dx} dw = 0 \quad (\text{A141})$$

Adding these integrals to (A138) and cancelling terms,

$$\int_{\underline{w}_x}^{\bar{w}_x} V'(w) \left(W'(V(w)) \left(\frac{h_{A,x}(w)}{h_x(w)} \frac{dH_x(w)}{dx} - \frac{dH_x^A(w)}{dx} \right) + W'(V(w) - d) \left(\frac{h_{B,x}(w)}{h_x(w)} \frac{dH_x(w)}{dx} - \frac{dH_x^B(w)}{dx} \right) \right) dw \quad (\text{A142})$$

$$- \mu \int_{\underline{w}}^{\bar{w}} l(w) \frac{dH_x(w)}{dx} dw - \mu \int_{\underline{w}_x}^{\bar{w}_x} \left[\eta(w) l'(w) l^{1/\varepsilon} \left(1 + \frac{1}{\varepsilon} \right) + w V'(w) \eta'(w) \right] \frac{1}{w h_x(w)} \frac{dH_x(w)}{dx} dw \quad (\text{A143})$$

$$+ \mu \xi \int_{\underline{w}_x}^{\bar{w}_x} \left[(w l'(w) + l(w)) \left(\alpha(x) \frac{dH_x^B(w)}{dx} - (1 - \alpha(x)) \frac{dH_x^A(w)}{dx} \right) + w l'(w) \left((1 - \alpha(x)) \frac{h_{A,x}(w)}{h_x(w)} - \alpha(x) \frac{h_{B,x}(w)}{h_x(w)} \right) \frac{dH_x(w)}{dx} \right] dw \quad (\text{A144})$$

$$+ \mu \int_{\underline{w}_x}^{\bar{w}_x} \frac{V'(w)}{h_x(w)} \left[\Delta\hat{T}(w) \frac{\partial h_{B,x}(w)}{\partial V(w)} - \Delta\hat{T} \left(Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right) \frac{\partial h_{A,x}(w)}{\partial V(w)} \right] \frac{dH_x(w)}{dx} dw \quad (\text{A145})$$

$$- \mu \xi \theta'(x) Y(L_A, L_B) + B_1 + B_2 = 0 \quad (\text{A146})$$

Performing a change of variables $w = Y_B(x)\check{\theta}_B\left(\frac{q}{Y_A}\right)$ for the term containing $\Delta\hat{T}(w)\frac{\partial h_{B,x}(w)}{\partial V(w)}$ in (A145),

$$\begin{aligned} & \mu \int_{\underline{w}_x}^{\bar{w}_x} \frac{V'\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{q}{Y_A}, d\right)\right)}{h_x\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{q}{Y_A}, d\right)\right)} \Delta\hat{T}\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{q}{Y_A}, d\right)\right) \\ & \cdot \frac{\partial h_{B,x}\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{q}{Y_A}, d\right)\right)}{\partial V\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{q}{Y_A}, d\right)\right)} \frac{dH_x\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{q}{Y_A}, d\right)\right)}{dx} \frac{Y_B(x)}{Y_A(x)} \check{\theta}'_{B,x}\left(\frac{q}{Y_A(x)}\right) dq \end{aligned} \quad (\text{A147})$$

$$\begin{aligned} & - \mu \int_{\underline{w}_x}^{\bar{w}_x} \frac{V'(w)}{h_x(w)} \Delta\hat{T}\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right)\right) \frac{\partial h_{A,x}(w)}{\partial V(w)} \frac{dH_x(w)}{dx} dw \\ & = \mu \int_{\underline{w}_x}^{\bar{w}_x} V'(w) \Delta\hat{T}\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A}, d\right)\right) \frac{\partial h_{A,x}(w)}{\partial V(w)} \\ & \cdot \left[\frac{1}{h_x\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A}, d\right)\right)} \frac{dH_x\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A}, d\right)\right)}{dx} - \frac{1}{h_x(w)} \frac{dH_x(w)}{dx} \right] dw \end{aligned} \quad (\text{A148})$$

$$\begin{aligned} & = \mu \int_{\underline{w}_x}^{\bar{w}_x} \frac{V'(w)}{V'\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right)\right)} \frac{1}{Y_A(x)} f\left(\frac{w}{Y_A(x)}, \check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right)\right) \Delta\hat{T}\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A}, d\right)\right) \\ & \cdot \left[\frac{1}{h_x\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A}, d\right)\right)} \frac{dH_x\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A}, d\right)\right)}{dx} - \frac{1}{h_x(w)} \frac{dH_x(w)}{dx} \right] dw \end{aligned} \quad (\text{A149})$$

$$\begin{aligned} & \cdot \left[\frac{1}{h_x\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A}, d\right)\right)} \frac{dH_x\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A}, d\right)\right)}{dx} - \frac{1}{h_x(w)} \frac{dH_x(w)}{dx} \right] dw \\ & = \mu \int_{\underline{w}_x}^{\bar{w}_x} \left(\frac{l(w)}{l\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right)\right)} \right)^{1+\frac{1}{\varepsilon}} \frac{Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right)}{w} \frac{1}{Y_A(x)} f\left(\frac{w}{Y_A(x)}, \check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right)\right) \\ & \cdot \Delta\hat{T}\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A}, d\right)\right) \\ & \cdot \left[\frac{1}{h_x\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A}, d\right)\right)} \frac{dH_x\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A}, d\right)\right)}{dx} - \frac{1}{h_x(w)} \frac{dH_x(w)}{dx} \right] dw \end{aligned} \quad (\text{A150})$$

where changing the integral limits is not necessary as we are still integrating over income. Line (A148) uses (A73) and

$$\frac{\partial h_{A,x}(w)}{\partial V(w)} = - \frac{\partial h_{A,x}(w)}{\partial V(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right))} = \frac{\partial h_{B,x}(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right))}{\partial V(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right))}, \quad (\text{A151})$$

line (A149) uses

$$\frac{\partial h_{A,x}(w)}{\partial V(w)} = \frac{1}{Y_A(x)} f\left(\frac{w}{Y_A(x)}, \check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right)\right) \frac{1}{V'\left(Y_B(x)\check{\theta}_{B,x}\left(\frac{w}{Y_A(x)}, d\right)\right)} \quad (\text{A152})$$

and line (A150) uses the local incentive constraint.

Rearranging (A142)-(A146),

$$\int_{\underline{w}_x}^{\bar{w}_x} V'(w) \left(W'(V(w)) \left(\frac{h_{A,x}(w)}{h_x(w)} \frac{dH_x(w)}{dx} - \frac{dH_x^A(w)}{dx} \right) + W'(V(w) - d) \left(\frac{h_{B,x}(w)}{h_x(w)} \frac{dH_x(w)}{dx} - \frac{dH_x^B(w)}{dx} \right) \right) dw \quad (\text{A153})$$

$$- \mu \int_{\underline{w}_x}^{\bar{w}_x} \left[\eta(w) l'(w) l^{1/\varepsilon} \left(1 + \frac{1}{\varepsilon} \right) + w V'(w) \eta'(w) \right] \frac{1}{w h_x(w)} \frac{dH_x(w)}{dx} dw \quad (\text{A154})$$

$$+ \mu \xi \int_{\underline{w}_x}^{\bar{w}_x} w l'(w) \left[(1 - \alpha(x)) \left(\frac{h_{A,x}(w)}{h_x(w)} \frac{dH_x(w)}{dx} - \frac{dH_x^A(w)}{dx} \right) - \alpha(x) \left(\frac{h_{B,x}(w)}{h_x(w)} \frac{dH_x(w)}{dx} - \frac{dH_x^B(w)}{dx} \right) \right] dw \quad (\text{A155})$$

$$+ \mu \int_{\underline{w}_x}^{\bar{w}_x} l(w) \left[\xi \left(\alpha(x) \frac{dH_x^B(w)}{dx} - (1 - \alpha(x)) \frac{dH_x^A(w)}{dx} \right) - \frac{dH_x(w)}{dx} \right] dw \quad (\text{A156})$$

$$+ \mu \int_{\underline{w}_x}^{\bar{w}_x} \left(\frac{l(w)}{l \left(Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right)} \right)^{1+\frac{1}{\varepsilon}} \frac{Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right)}{w} \frac{1}{Y_A(x)} f \left(\frac{w}{Y_A(x)}, \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right) \cdot \Delta \hat{T} \left(Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right) \quad (\text{A157})$$

$$\cdot \left[\frac{1}{h_x \left(Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right)} \frac{dH_x \left(Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right)}{dx} - \frac{1}{h_x(w)} \frac{dH_x(w)}{dx} \right] dw \quad (\text{A158})$$

$$- \mu \xi \theta'(x) Y(L_A, L_B) + B_1 + B_2 = 0 \quad (\text{A159})$$

From (A96) and (A108),

$$\frac{h_{A,x}(w)}{h_x(w)} \frac{dH_x(w)}{dx} - \frac{dH_x^A(w)}{dx} = \frac{h_{A,x}(w)}{h_x(w)} \left[- \frac{Y'_A(x)}{Y_A(x)} w h_{A,x}(w) + \Omega_x^A(w) - \frac{Y'_B(x)}{Y_B(x)} w h_{B,x}(w) - \Omega_x^B(w) \right] \quad (\text{A160})$$

$$+ \frac{Y'_A(x)}{Y_A(x)} w h_{A,x}(w) - \Omega_x^A(w) \quad (\text{A161})$$

$$= w \Lambda(x) \frac{h_{A,x}(w) h_{B,x}(w)}{h_x(w)} - \bar{\Omega}_x(w) \quad (\text{A162})$$

$$\frac{h_{B,x}(w)}{h_x(w)} \frac{dH_x(w)}{dx} - \frac{dH_x^B(w)}{dx} = -w \Lambda(x) \frac{h_{A,x}(w) h_{B,x}(w)}{h_x(w)} + \bar{\Omega}_x(w) \quad (\text{A163})$$

where $\Lambda(x) = \frac{Y'_A(x)}{Y_A(x)} - \frac{Y'_B(x)}{Y_B(x)}$ and $\bar{\Omega}_x(w) = \frac{h_{A,x}(w) \Omega_x^B(w) + h_{B,x}(w) \Omega_x^A(w)}{h_x(w)}$.

Thus (A153) simplifies to

$$\int_{\underline{w}_x}^{\bar{w}_x} V'(w) \left(\Lambda(x) w \frac{h_{A,x}(w) h_{B,x}(w)}{h_x(w)} - \bar{\Omega}_x(w) \right) (W'(V(w)) - W'(V(w) - d)) dw, \quad (\text{A164})$$

(A155) to

$$\mu \xi \int_{\underline{w}_x}^{\bar{w}_x} w l'(w) \left(\Lambda(x) w \frac{h_{A,x}(w) h_{B,x}(w)}{h_x(w)} - \bar{\Omega}_x(w) \right) dw, \quad (\text{A165})$$

and (A156) to

$$\mu \int_{\underline{w}_x}^{\bar{w}_x} l(w) \left[\xi \alpha(x) \left(-\frac{Y'_B(x)}{Y_B(x)} w h_{B,x}(w) - \Omega_x^B(w) \right) - \xi(1 - \alpha(x)) \left(-\frac{Y'_A(x)}{Y_A(x)} w h_{A,x}(w) + \Omega_x^A(w) \right) \right] \quad (\text{A166})$$

$$+ \frac{Y'_A(x)}{Y_A(x)} w h_{A,x}(w) + \frac{Y'_B(x)}{Y_B(x)} w h_{B,x}(w) + \Omega_x^B(w) - \Omega_x^A(w) \Big] dw$$

$$= \mu \left(\frac{Y'_A(x)}{Y_A(x)} \int_{\underline{w}_x}^{\bar{w}_x} w l(w) h_{A,x}(w) dw + \frac{Y'_B(x)}{Y_B(x)} \int_{\underline{w}_x}^{\bar{w}_x} w l(w) h_{B,x}(w) dw \right) \quad (\text{A167})$$

$$+ \xi \mu \int_{\underline{w}_x}^{\bar{w}_x} w l(w) \left(-\alpha(x) \frac{Y'_B(x)}{Y_B(x)} h_{B,x}(w) + (1 - \alpha(x)) \frac{Y'_A(x)}{Y_A(x)} h_{A,x}(w) \right) dw$$

$$+ \xi \mu \int_{\underline{w}_x}^{\bar{w}_x} l(w) [-\alpha(x) \Omega_x^B(w) - (1 - \alpha(x)) \Omega_x^A(w)] dw$$

$$+ \mu \int_{\underline{w}_x}^{\bar{w}_x} l(w) [\Omega_x^B(w) - \Omega_x^A(w)] dw$$

$$= \mu \underbrace{[Y'_A(x) E_A + Y'_B(x) E_B]}_{=0} + \xi \mu \left(Y'_A(x) E_A - \alpha(x) \underbrace{[Y'_A(x) E_A + Y'_B(x) E_B]}_{=0} \right) \quad (\text{A168})$$

$$- \xi \mu \int_{\underline{w}_x}^{\bar{w}_x} l(w) \Omega_x^A(w) dw + \mu(1 - \xi \alpha(x)) \int_{\underline{w}_x}^{\bar{w}_x} l(w) [\Omega_x^B(w) - \Omega_x^A(w)] dw \quad (\text{A169})$$

using the identity $Y'_A(x) E_A + Y'_B(x) E_B \equiv 0$. Since $Y'_A(x) E_A - \theta'(x) Y(L_A, L_B) = -\frac{\alpha(x) Y_B(x) E_B}{x}$ and $-\frac{1}{x} = \sigma(x) \Lambda(x)$ where $\sigma(x) \equiv \frac{dx}{x} \frac{\frac{Y_A(x)}{Y_B(x)}}{d\left(\frac{Y_A(x)}{Y_B(x)}\right)}$ is the substitution elasticity (Rothschild and Scheuer Lemma 3), we can combine $\xi \mu Y'_A(x) E_A$ in (A168) with (A159) to obtain

$$\xi \mu Y'_A(x) E_A - \xi \mu \theta'(x) Y(L_A, L_B) = \xi \mu \Lambda(x) \sigma(x) \alpha(x) Y_B(x) E_B. \quad (\text{A170})$$

Thus, combining the Ω_x terms in (A165) and (A169),

$$- \mu \xi \left(\int_{\underline{w}_x}^{\bar{w}_x} w l'(w) \bar{\Omega}_x(w) dw + \int_{\underline{w}_x}^{\bar{w}_x} l(w) (\Omega_x^A(w) - \bar{\Omega}_x(w) + \bar{\Omega}_x(w)) dw \right) \quad (\text{A171})$$

$$= -\mu \xi \int_{\underline{w}_x}^{\bar{w}_x} (w l'(w) + l(w)) \bar{\Omega}_x(w) dw + \mu \xi \int_{\underline{w}_x}^{\bar{w}_x} \frac{h_{A,x}(w)}{h_x(w)} l(w) \Delta \Omega_x(w) dw \quad (\text{A172})$$

Integrating by parts yields

$$- \xi \mu \bar{w}_x l(\bar{w}_x) \bar{\Omega}_x(\bar{w}_x) + \xi \mu \int_{\underline{w}_x}^{\bar{w}_x} w l(w) \frac{d\bar{\Omega}_x(w)}{dw} dw + \mu \xi \int_{\underline{w}_x}^{\bar{w}_x} \frac{h_{A,x}(w)}{h_x(w)} l(w) \Delta \Omega_x(w) dw \quad (\text{A173})$$

where we use the fact that $\bar{\Omega}_x(\underline{w}_x) = 0$.

Noticing that $\eta(w) w V'(w) = \eta(w) l(w)^{1+\frac{1}{\varepsilon}}$, and so

$$\frac{d}{dw} \eta(w) w V'(w) = \eta'(w) l(w)^{1+\frac{1}{\varepsilon}} + \eta(w) \left(1 + \frac{1}{\varepsilon} \right) l(w)^{1+\frac{1}{\varepsilon}} l'(w),$$

we can integrate (A154) by parts. Choosing

$$dv = \left[\eta(w)l'(w)l^{1/\varepsilon} \left(1 + \frac{1}{\varepsilon} \right) + wV'(w)\eta'(w) \right] dw \quad (\text{A174})$$

$$u = -\frac{1}{wh_x(w)} \frac{dH_x(w)}{dx} \quad (\text{A175})$$

$$v = \eta(w)l(w)^{1+\frac{1}{\varepsilon}} \quad (\text{A176})$$

$$du = -\frac{d}{dw} \left[\frac{1}{wh_x(w)} \frac{dH_x(w)}{dx} \right] dw \quad (\text{A177})$$

and using the transversality conditions $\eta(\underline{w}_x) = \eta(\bar{w}_x) = 0$, (A154) simplifies to

$$-\mu \int_{\underline{w}_x}^{\bar{w}_x} \eta(w)l(w)^{1+\frac{1}{\varepsilon}} \frac{d}{dw} \left[\frac{1}{wh_x(w)} \frac{dH_x(w)}{dx} \right] dw \quad (\text{A178})$$

Define $\tilde{H}(w, x) \equiv H_x(w)$. Analogous definitions hold for $\tilde{H}^A(w, x)$ and $\tilde{H}^B(w, x)$. Since $\tilde{H}(\bar{w}_x, x) \equiv 1$ for all x ,

$$\frac{d\tilde{H}(\bar{w}_x, x)}{dx} = \frac{\partial \tilde{H}(\bar{w}_x, x)}{\partial x} + \frac{\partial \tilde{H}(\bar{w}_x, x)}{\partial w} \frac{d\bar{w}_x}{dx} = \frac{dH(\bar{w}_x)}{dx} + h_x(\bar{w}_x) \frac{d\bar{w}_x}{dx} = 0 \quad (\text{A179})$$

Using (A179),

$$\frac{d\tilde{H}_x^A(\bar{w}_x, x)}{dx} = \frac{dH_x^A(\bar{w}_x)}{dx} + h_{A,x}(\bar{w}_x) \frac{d\bar{w}_x}{dx} \quad (\text{A180})$$

$$= \frac{dH_x^A(\bar{w}_x)}{dx} - \frac{h_{A,x}(\bar{w}_x)}{h_x(\bar{w}_x)} \frac{dH_x(\bar{w}_x)}{dx} \quad (\text{A181})$$

$$= -\frac{h_{A,x}(\bar{w}_x)h_{B,x}(\bar{w}_x)}{h_x(\bar{w}_x)} \bar{w}_x \Lambda(x) + \bar{\Omega}_x(\bar{w}_x) \quad (\text{A182})$$

This and the fact that $\Omega_x^A(\underline{w}_x) = 0$ allows us to simplify B_1 and B_2 to

$$B_1 + B_2 = \left[\xi \mu \bar{w}_x l(\bar{w}_x) + W(V(\bar{w}_x)) - W(V(\bar{w}_x) - d) + \mu \gamma(d) \right] \left(-\frac{h_{A,x}(\bar{w}_x)h_{B,x}(\bar{w}_x)}{h_x(\bar{w}_x)} \bar{w}_x \Lambda(x) + \bar{\Omega}_x(\bar{w}_x) \right) \quad (\text{A183})$$

Substituting (A164), (A165), (A170), (A173), (A178) and (A183) into (A153)-(A159), the first-order condi-

tion with respect to x is

$$\begin{aligned}
& \int_{\underline{w}_x}^{\bar{w}_x} V'(w) \left(\Lambda(x) w \frac{h_{A,x}(w) h_{B,x}(w)}{h_x(w)} - \bar{\Omega}_x(w) \right) (W'(V(w)) - W'(V(w) - d)) dw \\
& - \mu \int_{\underline{w}_x}^{\bar{w}_x} \eta(w) l(w)^{1+\frac{1}{\varepsilon}} \frac{d}{dw} \left[\frac{1}{w h_x(w)} \frac{dH_x(w)}{dx} \right] dw \\
& + \mu \xi \int_{\underline{w}_x}^{\bar{w}_x} w^2 l'(w) \Lambda(x) \frac{h_{A,x}(w) h_{B,x}(w)}{h_x(w)} dw \\
& + \xi \mu \int_{\underline{w}_x}^{\bar{w}_x} w l(w) \frac{d\bar{\Omega}_x(w)}{dw} dw + \mu \xi \int_{\underline{w}_x}^{\bar{w}_x} \left(\frac{h_{A,x}(w)}{h_x(w)} - \alpha(x) \right) l(w) \Delta \Omega_x(w) dw + \mu \int_{\underline{w}_x}^{\bar{w}_x} l(w) \Delta \Omega_x(w) dw \quad (\text{A184}) \\
& + \mu \int_{\underline{w}_x}^{\bar{w}_x} \left(\frac{l(w)}{l \left(Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right)} \right)^{1+\frac{1}{\varepsilon}} \frac{Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right)}{w} \frac{1}{Y_A(x)} f \left(\frac{w}{Y_A(x)}, \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right) \\
& \cdot \Delta \hat{T} \left(Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right) \\
& \cdot \left[\frac{1}{h_x \left(Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right)} \frac{dH_x \left(Y_B(x) \check{\theta}_{B,x} \left(\frac{w}{Y_A(x)}, d \right) \right)}{dx} - \frac{1}{h_x(w)} \frac{dH_x(w)}{dx} \right] dw \\
& + \left[W(V(\bar{w}_x)) - W(V(\bar{w}_x) - d) + \mu \gamma(d) \right] \left(-\Lambda(x) \bar{w}_x \frac{h_{A,x}(\bar{w}_x) h_{B,x}(\bar{w}_x)}{h_x(\bar{w}_x)} + \bar{\Omega}_x(\bar{w}_x) \right) \\
& + \xi \mu \Lambda(x) \sigma(x) \alpha(x) Y_B(x) E_B - \xi \mu \bar{w}_x l(\bar{w}_x) \Lambda(x) \bar{w}_x \frac{h_{A,x}(\bar{w}_x) h_{B,x}(\bar{w}_x)}{h_x(\bar{w}_x)} = 0
\end{aligned}$$

This can be rewritten as

$$\mu (A_1 + D_T + I) + Q_1 + R + \xi \mu \left(A_2 + Q_2 + \hat{C} - D_C + S + \Lambda(x) \sigma(x) \alpha(x) Y_B(x) E_B \right) = 0 \quad (\text{A185})$$

Rearranging yields Lemma 1.

B Empirical Appendix

B.1 Inferring policy-invariant primitives

Under the logit model of occupational choice (50), the probability that a worker chooses occupation j given skill vector θ is

$$P(j|\theta) = \frac{\exp \left(\frac{\beta}{\bar{\theta}(\theta)} V(\theta_j) + \tilde{\psi}_j + \chi_{\ell|k} \right)}{\sum_{\ell=1}^K \exp \left(\frac{\beta}{\bar{\theta}(\theta)} V(\theta_\ell) + \tilde{\psi}_\ell + \chi_{\ell|k} \right)}, \quad (186)$$

where I suppress the dependence of choice probabilities on amenities $\tilde{\psi}$ for brevity.

Under the Gaussian copula assumption (49), the conditional distribution of θ_{-j} given θ_j has a Gaussian copula structure: for $\ell \neq j$, the $\Phi^{-1}(F_\ell^\theta(\theta_\ell))$ are multivariate normal with mean vector $\Phi^{-1}(F_j^\theta(\theta_j))\varrho$ and covariance matrix $\Sigma_{-1} - \varrho'\varrho$, where Σ_{-1} is the upper-left $(n-1) \times (n-1)$ submatrix of Σ , and $\varrho = (\rho, \dots, \rho)$ has dimension $1 \times (n-1)$. Given empirical occupation shares $s_1, \dots, s_J$ and observed wage distributions $\tilde{f}_1^\theta, \dots, \tilde{f}_J^\theta$, the underlying productivity distributions $f_1^\theta, \dots, f_J^\theta$ and baseline occupational amenities $\psi_1^0, \dots, \psi_J^0$

solve the J functional equations

$$s_j \tilde{f}_j^\theta(\theta_j) = f_j^\theta(\theta_j) \int_{[\underline{w}, \bar{w}]^{J-1}} P(j|\theta) f_{\Phi^{-1}(F_j^\theta(\theta_j))_{\ell}, \Sigma_{-1-\ell'}_{\ell}}^{\mathcal{N}}(\Phi^{-1}(F_1^\theta(\theta_1)), \dots, \Phi^{-1}(F_n^\theta(\theta_n))) d\theta_{-j}, \quad (187)$$

where $s_j \tilde{f}_j^\theta(\theta_j)$ are the empirical analog of the endogenous, occupation-specific wage densities

$$h_j(\theta_j) = \int_{[\underline{w}, \bar{w}]^{J-1}} P(j|\theta_{-j}, \theta_j) f(\theta_{-j}, \theta_j) d\theta_{-j}. \quad (188)$$

These J equations pin down the productivity distributions f_j^θ and the amenities ψ_j^0 (up to an additive constant) conditional on β . As β scales the income component of utility relative to the fixed amenities ψ_j^0 , the two are not separately identified from the cross-sectional wage and occupation data alone: changes in β can be offset by shifts in ψ_j^0 without altering the equilibrium sorting pattern at a fixed tax schedule.

I calibrate β using the occupation-switching elasticity from [Powell and Shan \(2012\)](#), who estimate that wages rise by 0.33% for a 10% increase in the net-of-tax rate. Within the model, this equals

$$\frac{d \int \ln(\theta) \tilde{f}^\theta(\theta) d\theta}{d \ln(1-\tau)} = \int \ln(\theta) \frac{d \tilde{f}^\theta(\theta)}{d \ln(1-\tau)} d\theta, \quad (189)$$

where, since only $P(j|\theta)$ depends on τ ,

$$\begin{aligned} \frac{d \tilde{f}^\theta(\theta)}{d \ln(1-\tau)} &= \sum_j f_j^\theta(\theta_j) \int_{[\underline{w}, \bar{w}]^{J-1}} \frac{dP(j|\theta)}{d \ln(1-\tau)} \\ &\quad \cdot f_{\Phi^{-1}(F_j^\theta(\theta_j))_{\ell}, \Sigma_{-1-\ell'}_{\ell}}^{\mathcal{N}}(\Phi^{-1}(F_1^\theta(\theta_1)), \dots, \Phi^{-1}(F_n^\theta(\theta_n))) d\theta_{-j}, \end{aligned} \quad (190)$$

with

$$\frac{dP(j|\theta)}{d \ln(1-\tau)} = \frac{\beta}{\theta(\theta)} (1-\tau)^{1+\varepsilon} P(j|\theta) \left[\theta_j^{1+\varepsilon} - \sum_{\ell=1}^J P(\ell|\theta) \theta_\ell^{1+\varepsilon} \right]. \quad (191)$$

The parameters $(f_j^\theta, \psi_j^0, \beta)$ are estimated jointly by solving the augmented system of $J+1$ equations (187) and (189), imposing $\rho = 1$ and a constant marginal tax rate of $\tau = 0.30$ in place of the actual nonlinear U.S. schedule. This simplification prevents the discontinuous marginal rates of the actual schedule from generating bunching and is standard in the literature ([Rothschild and Scheuer, 2013](#); [Lockwood et al., 2017](#)).

B.2 Optimal policy

The Lagrangian is

$$\begin{aligned} \mathcal{L} &= \int_{[\underline{w}, \bar{w}]^J} (IV(\theta) + \gamma) g(\theta) d\theta \\ &\quad + \mu \int_{[\underline{w}, \bar{w}]} \sum_{j=1}^J h_j(w) [wl(w) - c(V(w), l(w)) - \gamma_{\chi|k}(\chi) \mathbb{1}[j \leq k]] dw \\ &\quad - \mu \int_{[\underline{w}, \bar{w}]} V(w) \eta'(w) dw - \mu \int_{[\underline{w}, \bar{w}]} \frac{l(w)^{1+1/\varepsilon}}{w} \eta(w) dw. \end{aligned} \quad (192)$$

The derivation follows Proposition 3 almost identically. Under coarse predistribution, optimal marginal tax rates are characterized by

$$\frac{T'(y(w))}{1 - T'(y(w))} = \frac{1 + \frac{1}{\varepsilon}}{wh(w)} \left(\sum_{j=1}^J \int_w^{\bar{w}} h_j(\theta_j) \left(1 - \frac{1}{\mu} g_j^T(\theta_j) \right) d\theta_j - \Xi(w) \right), \quad (193)$$

where

$$\begin{aligned} \Xi(w) &= \sum_{\{j,\ell\}} \left[\int_0^w \int_w^{\bar{w}} \Upsilon_{j\ell}(\theta_j, \theta_\ell) d\theta_\ell d\theta_j - \int_w^{\bar{w}} \int_0^w \Upsilon_{j\ell}(\theta_j, \theta_\ell) d\theta_\ell d\theta_j \right] \\ \Delta T_{j\ell}^X(\theta_j, \theta_\ell, \chi) &= T(y(\theta_j)) - \gamma_{\chi|k}(\chi) \mathbb{1}[j \leq k] - (T(y(\theta_\ell)) - \gamma_{\chi|k}(\chi) \mathbb{1}[\ell \leq k]) \\ \Upsilon_{j\ell}(\theta_j, \theta_\ell) &= \Delta T_{j\ell}^X(\theta_j, \theta_\ell, \chi) h_{(j,\ell)}^T(\theta_j, \theta_\ell) \\ g_j^T(\theta_j) &= \frac{\int_{[\underline{w}, \bar{w}]^{J-1}} \beta \bar{\theta}(\theta_{-j}, \theta_j)^{-1} g(\theta_{-j}, \theta_j) P(j|\theta_{-j}, \theta_j) d\theta_{-j}}{h_j(\theta_j)} \\ h_{(j,\ell)}^T(\theta_j, \theta_\ell) &= \int_{[\underline{w}, \bar{w}]^{J-2}} \beta \bar{\theta}(\theta_{-\{j,\ell\}}, \theta_\ell, \theta_j)^{-1} P(j|\theta_{-\{j,\ell\}}, \theta_\ell, \theta_j) \\ &\quad \cdot P(\ell|\theta_{-\{j,\ell\}}, \theta_\ell, \theta_j) f(\theta_{-\{j,\ell\}}, \theta_\ell, \theta_j) d\theta_{-\{j,\ell\}}. \end{aligned}$$

Optimal coarse predistribution is characterized by

$$\gamma'_{\chi|k}(\chi) = \frac{1}{\mu} \mathbb{E}_{j \leq k} [g_j^p(\theta_j)] + \mathbb{E}_{j \leq k} \left[\sum_{\ell > k} \int_{[\underline{w}, \bar{w}]} \Delta T_{j\ell}^X(\theta_j, \theta_\ell, \chi) \frac{h_{(j,\ell)}^p(\theta_j, \theta_\ell)}{h_j(\theta_j)} d\theta_\ell \right], \quad (194)$$

where

$$\begin{aligned} g_j^p(\theta_j) &= \frac{\int_{[\underline{w}, \bar{w}]^{J-1}} g(\theta_{-j}, \theta_j) P(j|\theta_{-j}, \theta_j) d\theta_{-j}}{h_j(\theta_j)} \\ h_{(j,\ell)}^p(\theta_j, \theta_\ell) &= \int_{[\underline{w}, \bar{w}]^{J-2}} P(\ell|\theta_{-\{j,\ell\}}, \theta_\ell, \theta_j) P(j|\theta_{-\{j,\ell\}}, \theta_\ell, \theta_j) f(\theta_{-\{j,\ell\}}, \theta_\ell, \theta_j) d\theta_{-\{j,\ell\}}. \end{aligned}$$

More generally, for an arbitrary set $\mathcal{S} \subseteq \{1, \dots, J\}$ of target occupations for predistribution (not necessarily the k most disamenable), redefine

$$\chi_{j|\mathcal{S}} = \begin{cases} \chi & \text{if } j \in \mathcal{S} \\ 0 & \text{else,} \end{cases} \quad (195)$$

and let $\gamma_{\chi|\mathcal{S}}(\chi)$ denote the optimal predistribution cost per worker for $\mathcal{S}$. The characterization above continues to hold with $\mathbb{E}_{j \leq k}$ replaced by $\mathbb{E}_{j \in \mathcal{S}}$ and $\sum_{\ell > k}$ replaced by $\sum_{\ell \notin \mathcal{S}}$:

$$\gamma'_{\chi|\mathcal{S}}(\chi) = \frac{1}{\mu} \mathbb{E}_{j \in \mathcal{S}} [g_j^p(\theta_j)] + \mathbb{E}_{j \in \mathcal{S}} \left[\sum_{\ell \notin \mathcal{S}} \int_{[\underline{w}, \bar{w}]} \Delta T_{j\ell}^X(\theta_j, \theta_\ell, \chi) \frac{h_{(j,\ell)}^p(\theta_j, \theta_\ell)}{h_j(\theta_j)} d\theta_\ell \right]. \quad (196)$$

B.3 Additional figures and tables

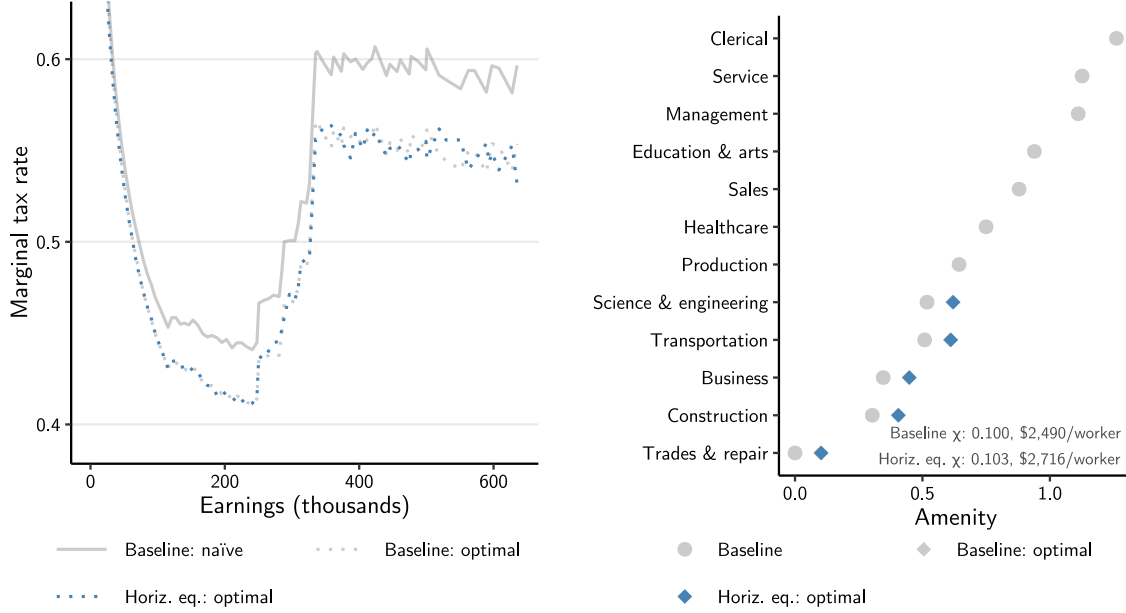
Table B1: Occupation group definitions

Occupation group	Census codes
Management	0010–0430
Business	0500–0950
Science & engineering	1000–1980
Education & arts	2000–2960
Healthcare	3000–3650
Service	3700–4650
Sales	4700–4965
Clerical	5000–5940
Construction	6200–6940
Trades & repair	7000–7630
Production	7700–8965
Transportation	9000–9750

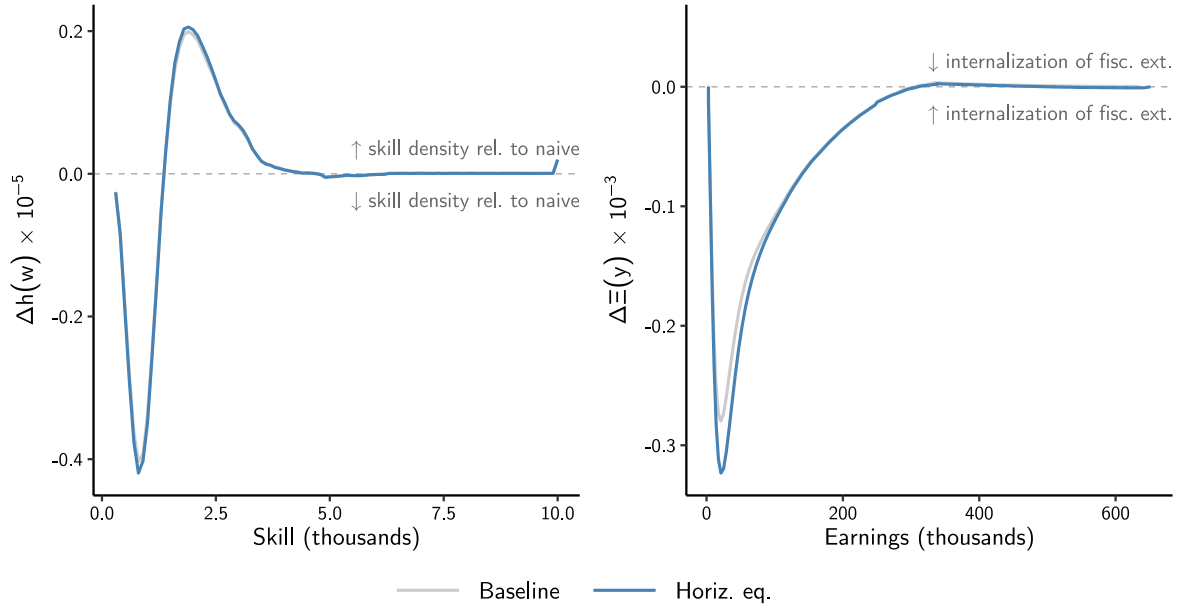
Note: 2010 Census occupation codes.

Figure B2: Robustness to horizontal equity motive, λ

(a) Optimal MTRs and predistribution



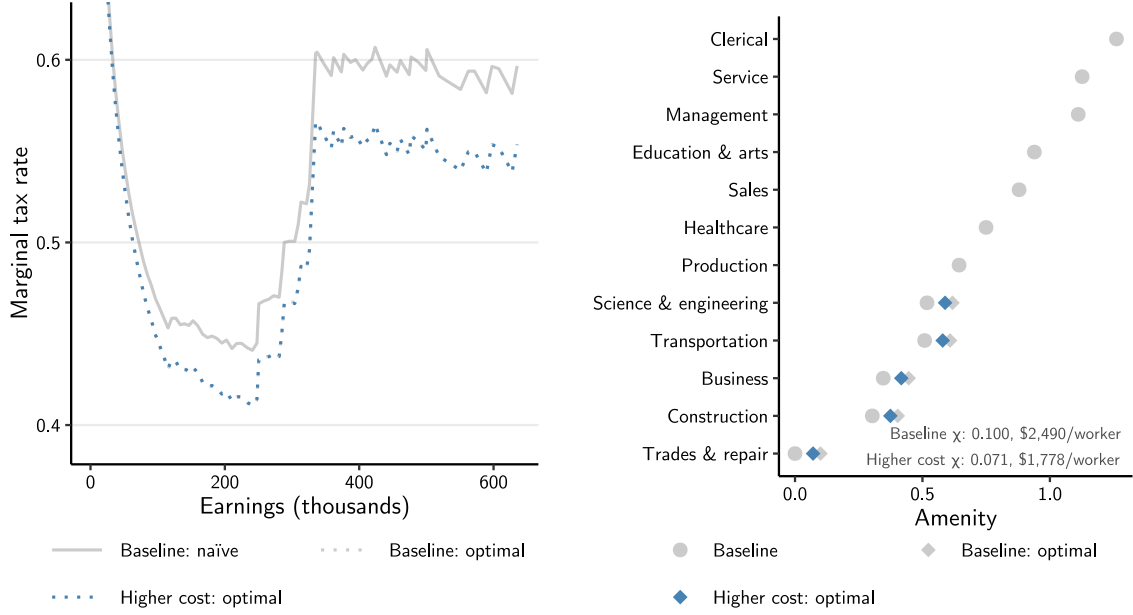
(b) Changes in skill density $h(w)$ and fiscal externality $\Xi(y)$ relative to naïve case



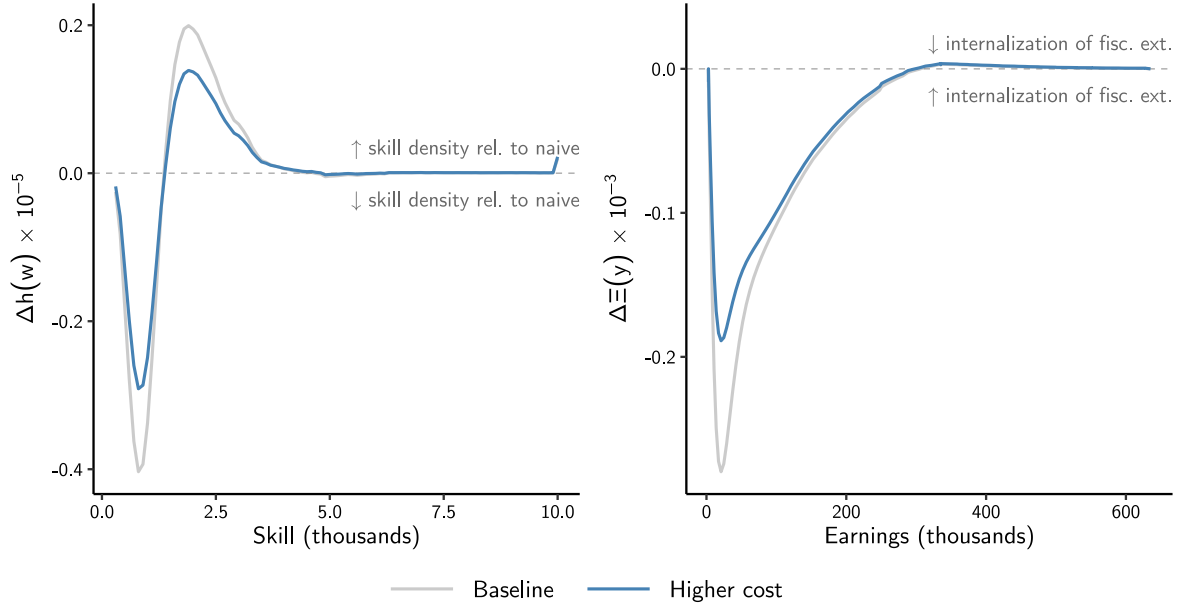
Note: The parameter λ governs the extent to which the planner assigns higher welfare weight to workers with comparative advantage in disamenable occupations. The baseline specification, corresponding to Figures 2–3, sets $\lambda = 0$, placing no weight on comparative advantage. The “Horiz. eq.” specification sets $\lambda = 5.0$, with primitives β, ρ, f^θ , and ψ^0 held at their baseline values. The naïve and optimal cases follow the naïve and “Fisc. ext. correction + predistribution” specifications in Figure 2, respectively. Panel (b) follows the variable definitions therein.

Figure B3: Robustness to higher predistribution cost, c_p

(a) Optimal MTRs and predistribution



(b) Changes in skill density $h(w)$ and fiscal externality $\Xi(y)$ relative to naïve case



Note: The predistribution cost parameter c_p scales the per-worker cost of amenity improvements in the cost function $\gamma(\chi)$. The “Higher cost” specification doubles c_p relative to the baseline, holding all primitives β, ρ, f^θ , and ψ^0 fixed. The baseline specification corresponds to Figures 2–3. The naïve and optimal cases follow the naïve and “Fisc. ext. correction + predistribution” specifications in Figure 2, respectively. Panel (b) follows the variable definitions therein.